Open For Debate: Governance, Power, and the Limits of Internet Openness

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I. Open Internet: Key Issues and Evolution of the Concept

The concept of an "open internet" has evolved alongside technological, political, and economic transformations. Once defined primarily in technical terms—interoperability, universal access, and decentralised governance—it now intersects with complex debates over data governance, sovereignty, and digital rights. This report invites readers to interrogate what openness means in a fragmented and contested digital order. It explores normative, institutional, and empirical dimensions of openness, and whether traditional definitions remain adequate for understanding today's internet ecosystem.

Patryk Pawlak opens the volume with an introduction that frames openness as a layered and contested condition—shaped by design choices, market structures, and governance arrangements at the infrastructure, logical, and policy layers—and sets out the structural forces driving its transformation. Wolfgang Kleinwächter traces the duality at the heart of the internet: one world at the transport layer, 193 national jurisdictions at the application layer. His analysis examines how the multistakeholder governance model that emerged from the WSIS process remains both necessary and under increasing pressure. Arturo Filastò unpacks the vocabulary of "open," "closed," and "splinternet," showing how fragmentation can serve either authoritarian control or civic liberation depending on who deploys it. Ceren Ünal shifts the frame toward political economy, arguing that connectivity alone no longer produces shared economic benefits when control over computational capacity, data infrastructure, and semiconductor supply chains has become as consequential as access to shared protocols.

Chinasa T. Okolo closes the chapter by examining the contradictions facing the Global Majority, where the promise of an open internet has collided with structural barriers to meaningful participation. She identifies three interconnected dimensions of an emerging AI divide: linguistic exclusion; the marginalisation of cultures and knowledge systems in AI models due to severe data scarcity; and the concentration of computational power in a small number of countries. The chapter as a whole establishes that openness, at each layer of the internet and for each segment of the global population, is more contingent and more unevenly distributed than standard accounts may suggest.

Reclaiming Internet Openness: Dynamic, Layered, and Contested

Patryk Pawlak

Layered openness of the internet

The open internet is widely regarded as one of the defining institutional and technological developments of the contemporary era. Its significance lies not only in its global reach but in the way it historically enabled innovation, communication, and participation without requiring prior authorisation from a central gatekeeper. Scholars have long associated this openness with the internet's end-to-end design, interoperable standards, and relatively low barriers to entry, all of which allowed new services to emerge at the network's edges and facilitated the transnational circulation of information, ideas, and economic activity (Lessig, 2001; Zittrain, 2008). However, as a substantial body of scholarship has shown, the openness of the internet has never been absolute, nor has it remained stable over time. Rather, it has evolved unevenly across different layers of technical architecture and governance, shaped by design choices, market structures, public policy, and geopolitical contestation (DeNardis, 2014; Mueller, 2017; Radu et al., 2024; De Gregorio, 2024).

Understanding the future of the open internet, therefore, requires moving beyond any singular principle or slogan. Openness is not a unitary condition; it is produced and constrained through the interaction of infrastructure, logical resources, and institutional frameworks. This layered perspective reflects a longstanding insight in internet governance scholarship: that the internet is best understood not as a monolithic system, but as an assemblage of physical networks, logical coordination mechanisms, and governance arrangements that together make global connectivity possible (Clark et al., 2005; DeNardis, 2014). Recent works also show that digital sovereignty is increasingly being "infrastructured" through technical, economic, and regulatory practices rather than simply asserted in law or policy rhetoric (Musiani, 2025; Fratini et al., 2024).

At the infrastructure layer, openness depends on the physical and economic conditions of connectivity, including access to broadband networks, interconnection arrangements, submarine cables, data centres, cloud systems, and the competitive structure of telecommunications markets. At the logical layer, it is shaped by standards, protocols, naming and numbering systems, software architectures, and platform dependencies that determine whether networks and services can interoperate. At the policy layer, openness is conditioned by legal rules, regulatory choices, and governance institutions, as well as by the balance struck among security, competition, innovation, and fundamental rights.

The layered approach matters because the openness of the internet has not changed in a simple linear direction. In some respects, openness has expanded. Internet access has extended to billions of users; open standards have enabled communication at a global scale; and digital tools have lowered the costs of publishing, organising, and transacting online. In other respects, openness has narrowed. Control over key infrastructures has become more concentrated; dependence on a small number of platforms and cloud providers has intensified; and governments have become increasingly willing to intervene in traffic flows, content governance, data localisation, and cross-border digital exchange. Recent scholarship on internet fragmentation and global digital constitutionalism underscores that these shifts are not only technical or commercial, but also constitutional and geopolitical, as states and private actors contest the governance of digital spaces (Nocetti, 2024; De Gregorio, 2024; Radu et al., 2024).

Internet openness in flux

The central question, therefore, is not whether the internet is open or closed in the abstract, but how openness is being redefined, by whom, and with what consequences for innovation, competition, rights, and political power (Komaitis, 2026).

At the infrastructure layer, early accounts of the internet emphasised decentralisation, redundancy, and distributed design. Packet-switched networking and the interconnection of autonomous systems enabled a network architecture that could scale without a single point of control, while a heterogeneous mix of public institutions, research networks, civil society actors, and private firms supported its expansion (Abbate, 1999). Over time, however, this physical network became embedded in commercial markets and strategic supply chains. Broadband connectivity increasingly depended on a limited number of fixed and mobile network operators. In contrast, international connectivity became tied to large-scale investment in submarine cable systems, internet exchange points, data centres, and content delivery networks. More recently, hyperscale cloud providers and large content firms have expanded vertically into backbone infrastructure, regional computing facilities, edge caching, and private interconnection. Recent work on infrastructuring digital sovereignty suggests that these infrastructural shifts are not merely commercial adjustments but part of a wider renegotiation of control, autonomy, and dependency in digital systems (Musiani, 2025; Fratini et al., 2024). As scholarship on infrastructural power suggests, such developments improve efficiency and performance, but they also create new dependencies by concentrating control over the material conditions of connectivity in a relatively small number of firms (Plantin et al., 2018; Musiani et al., 2016).

At the logical layer, the open internet was built on shared protocols and interoperable standards. The adoption of TCP/IP, the development of the Domain Name System, and the role of open standards bodies created a common technical environment in which independently operated networks and applications could communicate. This architecture supported “generativity”: the capacity of a system to enable unanticipated innovation by diverse and distributed actors (Zittrain, 2008). It also underpinned the idea of permissionless innovation, which holds that new applications can be launched without prior negotiation with network operators or device manufacturers. Yet here too, openness has become more ambivalent. The internet’s logical layer increasingly relies on systems that are formally interoperable but functionally concentrated. A limited number of operating systems, browsers, cloud platforms, app stores, and identity providers now mediate access to users and digital markets. In this sense, control points have shifted upward from the transport layer toward application ecosystems and service environments. Recent policy work on interoperability, especially in social networking services, indicates that interoperability is increasingly being treated as a governance instrument for contestability rather than only as a technical property of the network (CERRE 2026). As DeNardis has argued, technical architecture itself functions as a site of governance, shaping who can participate, under what terms, and with what visibility (DeNardis, 2009; 2014).

The policy layer has undergone the most visible transformation. Early internet governance was often characterised by relatively light-touch regulation, private ordering, and transnational coordination, with many core functions managed through technical communities, industry bodies, and multistakeholder institutions rather than through traditional intergovernmental control. This arrangement was closely linked to the normative assumption that openness would best be preserved through interoperability, limited barriers to entry, and restrained ex ante regulation (Mueller, 2002; Brousseau et al., 2012). That model, however, has come under sustained pressure as the internet has become central to economic life, democratic participation, public administration, and national security. Recent work on digital sovereignty and global digital constitutionalism shows that the policy layer is now shaped by competing efforts to reclaim authority over digital infrastructure, regulate powerful intermediaries, and align online governance with constitutional values and institutional legitimacy (Fratini et al., 2024; De Gregorio, 2024; Musiani, 2025). States have expanded their role through rules governing data protection, cybersecurity, platform accountability, competition, child safety, harmful content, taxation, and critical infrastructure resilience. Many of these interventions respond to genuine failures, including concentration, online abuse, disinformation, cyber insecurity, and inadequate consumer protection. At the same time, they generate new tensions, since measures intended to enhance safety, sovereignty, or accountability may also fragment the network, raise compliance costs, entrench incumbent firms, or facilitate censorship and surveillance (Goldsmith & Wu, 2006; Suzor, 2019; Bradford, 2023).

The end of openness?

Several broad structural forces help explain these changes across all three layers. The first is commercialisation. The transformation of the internet from a research and communications network into a global commercial system brought scale, investment, and mass adoption, but also incentives for vertical integration, enclosure, and concentration. The second is securitisation. Cyberattacks, infrastructure sabotage, foreign interference, and concerns over strategic dependency have pushed both governments and firms to prioritise resilience, trusted suppliers, and greater control over networks and services. The third is platformisation: an increasing share of online activity now occurs within managed ecosystems rather than on the open web, shifting authority from shared protocols toward proprietary interfaces, algorithmic ranking systems, and private rule-making (Gillespie, 2018; Helmond, 2015). A fourth is geopolitical competition. The internet is no longer understood solely as a global communications medium or market infrastructure; it is increasingly treated as a domain of strategic rivalry, affecting supply chains, standards-setting, cross-border data flows, semiconductor access, and digital regulation (Farrell & Newman, 2019; 2023; Nocetti, 2024). A fifth is the growing recognition that openness alone does not necessarily produce socially desirable outcomes. Concerns over privacy, manipulation, bias, online violence, and labour exploitation have weakened the once-dominant presumption that fewer constraints automatically yield better social and democratic results.

These developments suggest that the core challenge for contemporary internet governance is not simply to defend openness in the abstract, but to specify what forms of openness remain normatively and institutionally valuable under changed conditions. In this context, interoperability is becoming increasingly important not merely as a technical feature, but as a policy objective linked to contestability, innovation, and the reduction of dependency on dominant gatekeepers. Recent debates over data portability, fair interconnection, open standards, and interoperability mandates reflect a shift away from idealised narratives of complete decentralisation toward more practical efforts to preserve openness within concentrated and highly managed digital environments (Crémer et al., 2019; CERRE, 2026). At the same time, resilience has emerged as a parallel concern. The open internet of the future will be judged not only by whether users can connect and publish, but also by whether networks can withstand disruption, whether governance remains transparent and inclusive, and whether technical and regulatory systems preserve trust without unnecessary centralisation.

About this report

This report proceeds from the premise that openness remains a foundational value of the internet, but one that must be understood as dynamic, layered, and contested. The relevant question is no longer whether the internet should be open in principle. Rather, it is which forms of openness should be protected, where new safeguards are required, and how governance can respond to changing risks without undermining the generative and participatory features that made the internet so consequential in the first place.

This report examines how the idea and practice of an "open internet" have evolved under changing technological, economic, and geopolitical conditions. It brings together contributions from scholars and practitioners across law, economics, political science, and technical communities to examine how openness is being redefined—and contested—at different layers of the digital ecosystem.

The report starts with a discussion of the **concept of the open internet.** It traces its evolution from a primarily technical notion—rooted in interoperable protocols, end-to-end connectivity, and decentralised governance—to a multidimensional concept entangled with questions of digital rights, security, and state and corporate power. It asks whether traditional understandings of openness still hold in an era characterised by fragmentation, national "segments" of the network, and a growing reliance on platform-mediated environments.

It then explores **sovereignty, interoperability, and standards** as key fault lines in contemporary internet governance. Chapters in this section analyse how different actors pursue digital sovereignty through regulation, industrial policy, and infrastructural choices; how technical standards and interoperability requirements can both enable and constrain openness; and how multistakeholder institutions and standards-setting bodies navigate rising geopolitical tensions while maintaining a coherent, globally interoperable network.

The report proceeds to examine **data and digital trade policies to demonstrate** how divergent regulatory regimes for data protection, national security, and digital trade shape the practical conditions of openness. It highlights how measures such as data localisation, adequacy decisions, and trade commitments can simultaneously protect important public interests and contribute to a more fragmented network of partially interoperable legal and technical regimes.

The subsequent contributions look at the relationship among **open internet principles, innovation, and economic development**. Authors analyse how openness affects competition, investment, and inclusion, and how governments and international organisations are re-evaluating earlier "light-touch" approaches in the face of platform power, cyber threats, and systemic risks to democratic processes and public services.

Finally, the report turns to **companies, markets, and business models**, assessing how commercial strategies and industry structures influence the conditions of internet openness. It analyses the role of hyperscale cloud providers, app stores, content platforms, and emerging AI-related intermediaries as new infrastructural gatekeepers; explores how advertising-driven and data-extractive business models shape incentives for enclosure or openness; and examines new proposals for interoperability, portability, and alternative governance arrangements that might rebalance power in digital markets.

Across these sections, the report adopts a **layered and systemic perspective**: it treats the internet as an interconnected set of infrastructure, logical, and policy layers in which decisions at one level reverberate through the others. Rather than asking simply whether the internet is still open, the contributions collectively investigate *how* openness is being redefined, where it is being narrowed or expanded, and which institutional and technical arrangements might sustain an internet that remains interoperable, accessible, and accountable under new conditions.

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Open Internet and a Holistic Approach to Governance in the Digital Age

Wolfgang Kleinwächter

The concept of an open internet is in the DNA of the internet architecture, which allowed it to blossom from a research project in its early days into the foundation of today's economic, political and social life. It has evolved alongside technological, political, and economic transformations. Once defined primarily in technical terms—interoperability, universal access, and decentralised governance—it now intersects with complex debates over data governance, sovereignty, and digital rights. Did the new challenges of the digital transformation change the understanding of an open internet in a growing, fragmented world?

If we take the internet as a layered system, the answer is: yes and no. No, because at the transport layer, the universal codes, protocols and standards (e.g., TCP/IP, DNS, SMTP, HTTP, BGP, IPv4 and IPv6; DNSSec) guarantee that everyone can communicate with everyone, anytime and anywhere. Organisations such as ICANN, IETF, RIRs, and W3C, which are managed by the technical community and non-governmental stakeholders, help preserve the vision of "one world, one internet". Without the universality of those protocols, there is no open internet.

Concurrently, in the application layer, national political strategies and commercial economic interests have fragmented the internet into "national internet segments" and "walled gardens". The open internet and free communication are under growing pressure from practices such as internet shutdowns, censorship, mass surveillance and closed shops, where internet users must pay an entrance fee. Here, the internet, shaped by transnational tech corporations and sovereign nation states, represents "one world, 193 national jurisdictions".

Interconnected layers and internet fragmentation

The two worlds are inseparably interconnected. The challenge is to navigate a friction between the realities of the "*virtual world", where* old barriers of time and space do not exist anymore, and legal and political borders that still exist in the "*real world"*. This distinction didn't matter much as long as the internet was a playground for geeks. But now that more than six billion people are online, the distinction between "real places" (as states) and "virtual spaces" (as networks) has become a global political issue.

In the 1990s, many internet pioneers dreamed that the philosophy of the transport layer would spill over to the application layer and would lead to an "unified internet world." Today's reality is that many governments would like to see spillover from the application layer to the transport layer, which will inevitably lead to a fragmented internet. From a public-interest perspective, splintering the global, unified internet is a bad idea. Metcalfe's Law states that the value of a network is proportional to the square of the number of connected users. The greater the number of users, the more valuable the service becomes to the community. A network with five billion users has far more options than one with five million. In other words, cutting out parts of the internet reduces its value. Adding more networks and services increases its value.

The risk of internet fragmentation is not new (Cerf et al., 2015). In the 1990s, the discussion about alternative root signalled this risk. Before ICANN introduced internationalized domain names (iDNS), there was a fear that the internet would fall apart along language lines. In the 2000s, a discussion started on an Object Naming System (ONS) for the Internet of Things (IoT), separated from the DNS. With the creation of fully moderated "app space" and social networks, the fear of internet fragmentation returned, with many new "walled gardens": from blockchain, .eth, Web3, to the metaverse and AI (Kleinwächter and Klimburg, 2023). Now, the BRICS states put forward a politically motivated alternative root, the "MeltNet*" (*Görgen, 2026*)*.

The UN World Summit on the Information Society (WSIS)

Globally, this duality was addressed during the two *UN World Summits on the Information Society* (WSIS) in Geneva (2003) and Tunis (2005), followed by two *UNCSTD Working Groups on Enhanced Cooperation (*WGEC, 2018) between 2013 and 2018, and returned to focus more recently during the *WSIS+20 Review Conference* in New York (2025). The WSIS process tried to build bridges between governments and non-governmental stakeholders, with limited success. It recognised that the non-state actors played an essential role in building the internet.[1] The UN Resolution 56/183, which started WSIS in 2001, invited not only governments but also *"non-governmental organisations, civil society and the private sector to contribute to, and actively participate*" in the intergovernmental negotiations (UN General Assembly, 2002).

It was Kofi Annan, then UN Secretary-General, who understood this new challenge. He invited policymakers to get inspired by the fathers and mothers of the internet and called for "*political innovation"*. In his 2004 address to the *UN Working Group on internet Governance (WGIG),* he called for development of an "*inclusive and participatory models of governance. The medium must be made accessible and responsive to the needs of all the world's people"*. *He also stressed that while there is a need for governance, this "does not necessarily mean that it has to be done in the traditional way for something that is so very different" (*Annan, 2004*)*. The WGIG took Annan's proposal for more political creativity seriously. The WGIG Final Report (2005) stated that governance in the information age must move from the intergovernmental *"one stakeholder leadership model"* to a "*multistakeholder collaboration model."*

The question that emerged was not whether governmental top-down regulation should be extended to the "technical world" or replaced by private-sector self-regulation

[1] The "dot-com-boom" started after the invention of the World Wide Web in 1989. The Internet Society (ISOC) was established in 1992, ICANN in 1998. Manuel Castell published his "The Rise of the Network Society" in 1996, where he argued that "borderless spaces" will become more important than "bordered places." It was a time when code makers were much faster than lawmakers.

altogether. Rather, the challenge is to bring these two worlds into constructive coexistence. WGIG differentiated between the "*development of the internet*" (the technical layer) and the "*use of the internet*" (the political layer). It did not recommend *replacing* the existing systems, but rather *enhan*cing communication, coordination and collaboration among all concerned stakeholders (WGIG, 2005). The reasoning behind this was that the "*United Nations*" of governments need the "*United Constituencies*" of non-governmental stakeholders and vice versa. Furthermore, the argument was that governance in the digital age will function only on the basis of co-regulatory models that take into account both the sovereignty of the nation-state and the universality of global networks. Regulation by states and self-regulatory mechanisms by non-governmental networks must be complementary. The weakness of one partner in one area can be compensated for by the strength of the other, and vice versa.

Finding the right balance

There is no "one size fits all" solution to strengthening multistakeholder internet governance for a more open internet. Addressing cybersecurity problems requires a different governance model than managing critical internet resources, such as domain names or IP addresses. As internet policy and regulation become more issue-specific, each internet-related governance challenge requires a distinct governance model to adequately address the problems at hand.

At the same time, to avoid governance conflicts and not undermine the internet's global nature, there is a clear need for a holistic and more collaborative approach. Such an approach should promote constructive coexistence among the different stakeholders and support the development of new and innovative models of co-governance, where each stakeholder understands its respective role within a decentralised, multilayer, multiplayer mechanism of communication, coordination, and collaboration.

This distinction between the governance *of* the internet and the governance *on* the internet allows stakeholders to adopt a broader approach to internet Governance and to accommodate different interpretations and perspectives. In that respect, the multistakeholder internet Governance approach offers general guidance for all forms of digital policymaking in areas such as cybersecurity, data, or artificial intelligence. This is a continued open invitation for political innovation. There is no need to reinvent the wheel. But there is a clear need to strengthen the wheel.

Two decades after the adoption of the Tunis Agenda, the WSIS+20 Review conference in 2025 reiterated this approach. Members of the UN reaffirmed the working definition of internet governance *as "the development and application by Governments, the private sector and civil society, in their respective roles, of shared principles, norms, rules, decision-making procedures and programmes that shape the evolution and use of the internet"* and confirmed the multistakeholder approach as *"the best way to govern digital policy issues"* (Kleinwächter, 2025).

Finding the right balance in internet governance across the different yet interlinked layers is a complex challenge that is becoming increasingly difficult amid today's geopolitical battles. This difficulty, however, should not be a pretext for abandoning multistakeholder approaches to governing digital policies, but rather an invitation to learning and more innovative modes of engagement.

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Open, Closed, Splinternet

Arturo Filasto

The early notions of an Open internet can be traced in the concept of the internet being "a loosely-organised international collaboration of autonomous, internet-connected networks [...] through voluntary adherence to open protocols and procedures defined by internet standards" (Huizer & Crocker, 1994). This concept is often simplified to the notion of the internet as a "network of networks", where open and interoperable protocols define the rules of engagement and where no single actor can unilaterally change the rules. Rather, decisions about protocols are made by consensus, yet no one needs permission to build tools or services that use these protocols. These principles ultimately made the internet resilient by allowing each part to operate autonomously and ensuring power is distributed throughout the network.

The splinternet duality

As internet penetration increased globally, so did the interest in its stronger regulation. As the principles of the internet were ultimately connected to a libertarian ethos, regulatory interference was seen as an attack by governments and corporations on the very foundation of the internet (Barlow, 1996).

This debate gave rise to the concept of the splinternet (Crews, 2001), initially used to promote "parallel internets" as a libertarian response to state regulation, in which different communities and business models could flourish. The splinternet was imagined as a healthy response to state control, a way to break free from centralised governance and create autonomous online spaces. Today, references to the splinternet mean something quite different. Generally, they describe the increase of state control over the internet to enact digital sovereignty and create national spaces, often heavily monitored and censored. We see this in projects such as national intranets in Russia and Iran, and in pervasive filtering regimes that restrict citizens' ability to access information, organise, and communicate freely.

What is perhaps even more interesting is how the solution to the splinternet of the repressive-regime variety might indeed be a splinternet of the decentralised, libertarian flavour. On the one hand, we have "repressive" splinternets (national intranets, pervasive censorship, state-controlled routing) that weaponise fragmentation for control. On the other hand, we have "liberatory" splinternets, where decentralised technical interventions, such as Tor onion services, VPN-based overlay networks, and other private autonomous zones, create separate safe spaces for people to communicate while being shielded from overreaching governments and corporations. These parallel networks are technically a form of splinternet in the original sense of the term, but they are tools for liberation, not control.

Perhaps the open internet, as a network of networks, is "meant" to be a splinternet, yet one that empowers individuals and non-governmental entities to create their own autonomous spaces, rather than one in which governments carve the network into isolated, heavily policed domains.

Blurred lines between open and closed internets

The line between "open" and "closed" internets has become particularly difficult to navigate for democracies, where interventions in transnational information flows have become part of a broader toolbox of foreign and security policy instruments.

In response to Russia's war of aggression against Ukraine, the European Union imposed a package of sanctions, a component of which was dedicated to preventing the (online) broadcast of Russian state-backed media in the EU territory (tenOever et al., 2024). Some of these measures illustrate the friction between the technical/operational and application layers of the internet, as well as important policy dilemmas.[2] Unlike classic authoritarian filtering regimes, these measures did not involve seizing domains or building a single centralised "Great Firewall". Instead, they relied on a patchwork of technical implementations, the most common of which has been DNS-based blocking, with striking variations across countries and internet Service Providers (Kristoff et al., 2024). Some networks return NXDOMAIN or SERVFAIL for sanctioned domains, which, to the end user, may appear as the result of connectivity issues. Others serve block pages that transparently explain the reason for the unavailability of the service. There is also a group that barely implements the sanctions. These methods curtail the reach of pro-Kremlin propaganda machinery. At the same time, they institutionalise new forms of geographically and politically bounded access that depend on opaque decisions by regulators, courts, and intermediaries.

The resulting picture is of an internet increasingly shaped by overlapping regimes of control and resistance: authoritarian splinternets where openness is constrained by firewalls in Moscow or Tehran, decentralised circumvention infrastructures delivered by digital rights and resilience groups, and segments of the internet "sanctioned" by Brussels.

The internet as a complex system

The internet is no longer a single, stable object with fixed properties, but rather a complex, evolving system shaped by a multitude of forces: from technical design choices to economic incentives, political projects, and social practices. Its foundational idea, a loosely organised "network of networks" built on open protocols and distributed decision-making, was never neutral. It carried a particular set of power relations about autonomy, interoperability and permissionless innovation. Over time, those expectations have been both reinforced and challenged by new actors, events and regulatory imaginations.

[2] See contribution by Kleinwachter in this volume.

The terms used to describe this system—open, closed or splinternet—reflect that complexity rather than resolve it. Each of them carries a duality of sorts. Open can denote both a technical standard and a political ideal. Closed can describe both authoritarian firewalls and regulatory boundaries. splinternet can mean both liberatory decentralised enclaves and repressive, state-mandated partitions. As the balance of power shifts between states, platforms, civil society and users, these terms evolve, acquiring new meanings and losing old ones. What once may have sounded like a libertarian fantasy of parallel internets now serves as shorthand for the fragmentation of a global communications space under competing sovereignties.

The complex and adaptive nature of the internet makes it difficult to predict with confidence the direction of its evolution. It is impossible to anticipate how "openness" will be redefined, how "splinternet" will be redeployed, or what new categories will emerge. What is certain, however, is that the future trajectory of the internet will be determined by various stakeholder groups that shape the norms governing this shared infrastructure and define what they consider legitimate interventions. In that sense, the future of the internet will be written not only in code and regulation, but also in the way these actors continue to argue about what openness, closure and fragmentation mean and what kind of networked world they wish to establish.

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The Governance and Geopolitics of Openness in a Fragmented Digital Order

Ceren Ünal

Introduction: openness as an economic variable

For much of the internet's history, openness operated as both a normative commitment and an economic premise. Digital policy debates relied on a relatively stable relationship between openness, innovation and growth. Open architectures were expected to lower barriers to entry, enable experimentation and generate productivity gains across borders.

The belief that openness would reliably generate innovation did not emerge from optimism alone. It followed from how the internet was understood to function. Its architecture limited central control and allowed users to modify behaviour at the edges, which made experimentation inexpensive and decentralised (Lessig, 1999; 2006; Zittrain, 2008). Distributed participation could therefore translate into economic production rather than mere communication (Benkler, 2006). Openness mattered because the technical structure aligned economic participation with innovation.

Economic theory reinforced this expectation. Innovation tends to increase where entry remains contestable (Aghion et al., 2005), long-term investment depends on predictable institutional environments (North, 1990), and cross-border interaction spreads knowledge even under distributional tension (Rodrik, 2011). Applied to digital networks, these mechanisms justified interoperable systems, cross-border data flows, and multistakeholder governance as growth-enhancing arrangements rather than political preferences.

The assumption that connectivity automatically produces shared economic benefits, however, has become difficult to sustain. The network remains globally reachable, yet its economic effects are increasingly uneven. Earlier analyses help explain why this shift was predictable. Pressures toward network segmentation had already been identified (Mueller, 2017). Interdependence was shown to operate as a source of strategic leverage rather than mutual vulnerability (Farrell & Newman, 2019). Data governance had begun to migrate from coordination into trade and security policy (Aaronson, 2018). More recent work describes the outcome as a single technical network operating across multiple incompatible governance environments (O'Hara & Hall, 2021). Taken together, these developments suggest that openness has not disappeared but has changed function. Connectivity persists, yet the economic consequences of participation no longer distribute automatically across participants.

Openness, therefore, remains relevant but no longer serves as an automatic driver of growth. It needs to be analysed as an economic variable shaped by governance structures rather than architecture alone. The key question is no longer whether openness is desirable, but under what conditions it continues to generate economic and political value. In this essay, openness is not treated as a single commodity but as a layered condition: architectural accessibility, economic contestability and institutional participation.

From architecture to capability

The early internet distributed control across the network. The end-to-end principle located intelligence at the edges rather than in the core, allowing users and developers to innovate without permission from central operators (Saltzer et al., 1984; Cerf & Kahn, 1974; Abbate, 1999). Interoperable protocols created a layered system in which applications could scale without owning infrastructure. Innovation depended primarily on software and connectivity rather than physical capacity.

That structure has changed. Innovation increasingly depends on resources that exist below the application layer. Large-scale computation, specialised hardware and proprietary datasets now determine the frontier of technological development. The constraints are no longer primarily architectural but material.

Platform economics illustrates this transition. Firms compete within ecosystems defined by control over data aggregation, cloud infrastructure and integrated service environments (Srnicek, 2017; Crémer et al., 2019). Value creation increasingly depends on operating environments rather than individual products. Competitive advantage arises from maintaining persistent access to training data, compute resources and distribution channels. Artificial intelligence development intensifies this shift. Training frontier models requires massive computational capacity and capital investment, producing high fixed costs and strong economies of scale (Stanford AI Index, 2025; Sevilla et al., 2022; OECD, 2024). As model performance scales with data and compute, access to infrastructure becomes a precondition for innovation rather than a complement.

Semiconductor supply chains reinforce the same pattern. Advanced chip design and fabrication concentrate in a limited set of specialised actors and facilities (Miller, 2022; Semiconductor Industry Association, 2024). The production of high-performance processors involves technical expertise, manufacturing capacity and supply chain coordination that cannot easily be replicated through software-level competition.

These developments create layered bottlenecks. Protocol interoperability still enables communication across networks, yet meaningful participation in advanced digital markets depends on access to compute, data storage and hardware production. The locus of technological power has therefore shifted from shared architecture to controlled capability.

Openness at the protocol layer persists, but it no longer determines competitive conditions. The capacity to build, train and operate complex digital systems now shapes innovation outcomes more than the ability to connect to the network itself.

Digital sovereignty

As technological capability becomes unevenly distributed, sovereignty is increasingly defined by control over critical digital functions rather than territorial jurisdiction alone.

The issue is no longer whether networks cross borders but whether participation in them exposes domestic economies to decisions taken elsewhere. Dependence on foreign cloud infrastructure, AI providers and semiconductor supply chains transforms openness into a question of bargaining power.

In Europe, this response takes a regulatory form, seeking influence through rule-setting and using legal standards to shape market behaviour beyond its borders (Bradford, 2020). The GDPR, the Digital Services Act (DSA), and the Digital Markets Act (DMA) illustrate this strategy. Rather than competing directly in infrastructure, the EU structures the conditions under which infrastructure can be used. The competitiveness debate reflects this concern: limited scale, fragmented capital markets and slow coordination risk leaving Europe structurally behind in advanced digital industries, prompting a shift toward domestic compute capacity, industrial capability and accelerated regulation (Draghi, 2024).

Transatlantic data transfers demonstrate both the reach and the limits of this model. The Schrems II judgment invalidated the Privacy Shield because foreign surveillance law undermined equivalent protection (CJEU, 2020). The subsequent EU–US Data Privacy Framework restored legal transfers but remains contested because legal guarantees depend on institutional trust across jurisdictions. The dispute concerns authority over data once it crosses jurisdictional boundaries, not connectivity itself.

Alongside legal disputes over data transfers, a harder form of digital sovereignty has emerged through technological containment policies. The United States increasingly treats advanced computing capacity as a security asset, restricting semiconductor exports, investment and certain digital platforms. Export controls on advanced chips (U.S. Department of Commerce, 2022; 2023) and legislation targeting foreign-controlled platforms such as TikTok (U.S. Congress, 2024) shifted policy from regulating conduct to controlling capability. Recent executive actions extend this logic to the application and regulatory layer, linking market access to security and competitiveness considerations (Executive Office of the President, 2025; 2026). After contract negotiations collapsed over Anthropic's refusal to remove restrictions on its model's use for autonomous weapons and mass domestic surveillance, the Pentagon designated the company a supply-chain risk under statutes historically reserved for foreign adversaries (Anthropic PBC v. U.S. Department of War et al., 2026). Anthropic sued the Department of Defence, and employees of OpenAI and Google filed amicus briefs in a personal capacity (Amici Curiae Employees of OpenAI and Google, 2026), while Microsoft submitted a separate institutional brief (Microsoft Corporation, 2026), both arguing that the designation threatened the competitiveness of the entire American AI sector. Participation in digital markets, therefore, depends not only on compliance but on political alignment, turning interdependence into an instrument of power (Farrell & Newman, 2019).

These tensions reveal a structural asymmetry between regulatory and technological power. Regulatory systems can condition market access but cannot independently supply the capabilities on which markets rely. Sovereignty, therefore, becomes a bargaining position, not full autonomy. States negotiate acceptable levels of dependence rather than eliminate it. The shift from jurisdiction to dependency management marks a change in the function of sovereignty. It no longer guarantees control over infrastructure but structures the terms under which reliance on external systems remains politically and economically acceptable.

Fragmentation producing hierarchy

Fragmentation in the digital economy reorganises participation and produces a structured dependence. Global digital production increasingly resembles a value chain in which different actors occupy distinct functional tiers rather than equivalent positions (Gereffi, 2018). Connectivity remains global, yet authority over standards, infrastructure and innovation is unevenly distributed. Cross-country indicators increasingly show diverging digital productivity and AI capability despite universal connectivity (OECD, 2024).

In earlier phases of globalisation, fragmentation referred primarily to geographically dispersed manufacturing. In the digital economy, it refers to functional separation across layers of the system (Baldwin, 2016). Some actors operate core infrastructure, others build services on top of it, and many rely on platforms governed elsewhere. Access to the network, therefore, does not imply influence over its operation.

This layered structure creates gradual participation. States and firms positioned at core technological nodes shape protocols, technical standards and operational practices. Others adapt to these conditions. Market access does not translate into rule-shaping capacity, and integration does not guarantee value capture. Economic inclusion coexists with institutional dependence.

Emerging industrial strategies illustrate this adjustment. Gulf countries, for example, invest heavily in compute infrastructure and national AI programmes to move up the digital value chain. Initiatives under the UAE AI Strategy 2031 and the Saudi Vision 2030 aim to transform consumption into capability by securing bargaining power in technological ecosystems. The objective is not to exit the system but to reposition within the hierarchy.

The global digital system, therefore, functions less as an open network and more as a layered order. Fragmentation differentiates roles rather than isolating actors. Some define operational conditions while others operate within them. The outcome is neither full decoupling nor seamless integration but a structured distribution of authority. Openness persists at the level of access, yet influence depends on position within the hierarchy. Connectivity alone no longer determines participation in economic governance.

Incentive failure

If openness once functioned as a self-reinforcing equilibrium, today it lacks an equivalent incentive structure. The closest modern example of restoring such alignment is privacy by design. Data protection became operational not only through legal obligation but because compliance reduced liability exposure and enabled market trust (Cavoukian, 2011; GDPR, Art. 25). Firms incorporated it into routine risk management. The rule changed behaviour because the payoff structure supported adoption.

Openness lacks such incentive logic. Interoperability imposes direct costs on dominant actors by reducing switching barriers and facilitating entry. Maintaining proprietary interfaces, vertically integrated services, and controlled data environments remains economically rational. The private payoff from closure exceeds that from openness even when the collective outcome is less efficient.

This divergence explains the persistence of gated ecosystems. Firms optimise for user retention, data accumulation and ecosystem lock-in. Open standards, shared data access or portability measures often benefit competitors more than incumbents. The market, therefore, underproduces openness even where it would improve aggregate welfare.

Regulation attempts to correct this imbalance. The Digital Markets Act explicitly targets structural gatekeeping positions by imposing interoperability, data access and portability obligations on designated platforms. Its economic rationale is to restore market contestability rather than to punish, and to lower entry barriers where network effects entrench incumbency.

The challenge is therefore one of incentive alignment, not normative commitment. Openness cannot rely on voluntary adoption because its benefits are systemic while its costs are concentrated. Without institutional mechanisms that redistribute these incentives, rational actors will continue to prefer controlled ecosystems over interoperable ones. Correcting this requires not stronger rules alone but a deliberate restructuring of the conditions under which openness becomes economically rational.

Openness by design

Principles shape outcomes only when embedded in institutions that organise incentives and expectations; their absence explains why openness persists rhetorically while declining in practice. If openness no longer follows from network architecture and market interaction, it becomes a governance problem. The preceding analysis shows that actors behave rationally in response to incentives that now favour control, integration, and dependency management. The issue is therefore not commitment to openness but the lack of mechanisms that make it economically viable. This leads to a practical question: can openness be deliberately constructed rather than assumed?

Institutional economics offers a useful lens on why such construction is necessary. Stable economic coordination depends on rules that reduce uncertainty and monitoring mechanisms that discourage opportunistic behaviour (North, 1990; Ostrom, 1990). More recent work on regulation by design shows how behavioural constraints can be embedded directly into technical and organisational systems rather than imposing sanctions only after violations occur (Yeung, 2018; Brownsword, 2019). Openness, therefore, requires operational embedding rather than policy aspiration.

Standards form the first layer of coordination and translate abstract goals into operational practices. Interoperability protocols, auditability requirements and model evaluation benchmarks define predictable expectations across markets. International initiatives, including OECD AI principles (OECD, 2019) and ISO/IEC AI governance standards (ISO/IEC, 2023), stabilise expectations across jurisdictions without imposing uniform regulation. They structure competition while preventing incompatible technical environments.

The second layer involves policy instruments. Regulation alone cannot generate openness if compliance remains prohibitively costly or strategically irrational. Governments, therefore, rely on tools that alter incentives rather than only impose obligations. Public procurement can reward interoperable solutions; certification schemes can reduce information asymmetry, and reporting frameworks can distribute compliance burdens across value chains, especially where unequal digital capabilities otherwise prevent effective market participation (OECD, 2024). These mechanisms reshape payoffs so that participation in open systems becomes economically viable rather than merely legally required.

Competition policy operates within the same logic. Interoperability mandates and data access obligations aim to restore contestability where network effects entrench incumbency, targeting structural lock-in rather than uniformity of business models. When openness becomes a condition for market participation rather than an optional feature, firms incorporate it into their strategy.

Openness by design is a governance methodology that makes cooperative behaviour rational for self-interested actors through institutional rules, standards, and economic incentives. The relevant question has shifted from whether openness is desirable to whether governance arrangements can make it economically sustainable in a competitive and fragmented digital environment.

Conclusion

Early internet governance treated openness as self-sustaining. The architecture appeared to align economic incentives with decentralisation, and policy largely assumed that preserving connectivity would also preserve its benefits. In retrospect, this rested on favourable technical and market conditions rather than on stable institutional foundations. Openness worked, but partly by accident.

The current geopolitical environment encourages the opposite intuition. Control over infrastructure, data and computation can support industrial policy and short-term growth. States, therefore, have rational reasons to prefer managed dependence over unrestricted participation, and the growing assumption that openness has ceased to be economically relevant is the real risk. However, openness still matters, not because networks naturally reward it, but because complex, interdependent systems require predictable interaction beyond political alignment. The question has shifted from how to preserve openness from interference to how to justify it under conditions where closure can appear advantageous.

Openness by design responds to that shift. It does not attempt to restore the past or deny geopolitical competition. It recognises that openness now survives only where institutions deliberately sustain it. The task of governance is therefore not to defend openness as a principle, but to organise incentives so that choosing openness remains rational even when power is concentrated, and trust is limited.

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The Paradox of an "Open" Internet and AI Realities in the Global Majority

Chinasa T. Okolo

Introduction: the illusion of an open internet

The internet is often described as an open infrastructure: anyone with connectivity can theoretically access its networks and services. In practice, however, openness is deeply uneven. For billions of people across the Global Majority3, the experience of an open internet is fundamentally different and shaped by structural barriers that shape what information they can access, how they communicate, and whether they can meaningfully participate online. AI systems are now systematically encoding and amplifying these inequalities. Understanding how this transformation is occurring requires moving beyond the assumption that technical access alone guarantees openness. Instead, the emerging AI divide reveals how linguistic exclusion, unequal data representation, and the concentration of computational resources are reshaping who can participate in the digital ecosystem.

From the digital divide to the AI divide

The digital divide has traditionally been defined by disparities in internet access (Sambuli, 2016), high mobile data costs that are, on average, three times higher than in economically developed countries (GSMA, 2025), and limited internet infrastructure penetration (Okolo, 2023). While there has been significant progress over the past decade, African countries continue to struggle with reliable and affordable internet access (ITU & UNESCO, 2025). At the same time, governments, telecommunications companies, tech giants, and international development institutions have accelerated investments in digital infrastructures. Examples include the World Bank's financing of fibre-optic expansion in Nigeria (World Bank Group, 2025) and Meta's 2Africa subsea internet cable, constructed in collaboration with a consortium of leading telecom companies like Orange, MTN, and Vodafone (2Africa, 2022).

Increasingly, the digital divide is evolving into what policymakers describe as the AI divide (United Nations & ILO, 2024). Inequalities in internet access are now compounded by disparities in AI development, research capacity, and governance. This new divide and crisis in openness manifests in three interconnected dimensions. First, linguistic exclusion limits access to information in the languages spoken by much of the world's population. Second, epistemic erasure of culture and values due to data scarcity compounds challenges to linguistic inclusion. This leads to the marginalisation of

[3] Global Majority includes communities in low and middle-income countries across Africa, Asia, Latin America, the Caribbean, and the Pacific Islands.

cultures and knowledge systems in AI models. Finally, the concentration of computational power in a small number of countries limits how countries, particularly those in the Global Majority, can contribute to the development of truly global AI systems.

Linguistic access and meaningful participation

Language remains one of the most significant barriers to meaningful participation online. When speakers of non-English and non-Romance languages attempt to access information or communicate online in their respective languages, they often encounter systematic barriers. These impact their ability to receive effective content moderation (Elswah et al., 2025), achieve meaningful dissemination of their content (Graham, 2014), or maintain control over their intellectual property (Bhuiyan, 2014; Story et al., 2006).

AI systems amplify these inequities exponentially. Large language models perform significantly worse in low-resource languages (Li et al., 2025). Content moderation algorithms struggle to identify hate speech or misinformation in languages outside their training datasets while simultaneously over-flagging benign content because they lack cultural context (De Gregorio & Stremlau, 2023; Shahid & Vashistha, 2023). These issues cause recommendation systems to systematically deprioritise non-English content, creating algorithmic invisibility for billions of people who become more vulnerable online. In some cases, shortcomings in content moderation for low-resource languages amplify harmful content, further exacerbating internet challenges for millions of users (Nigatu & Raji, 2024).

More broadly, these challenges represent a fundamental barrier to Global Majority internet users' participation in the digital economy, access to knowledge, and the ability to shape public discourse. When openness is measured solely by the ability to connect to a platform rather than the ability to use it meaningfully, technical accessibility becomes a misleading indicator of meaningful participation.

Data colonialism and the homogenization of knowledge

A second challenge concerns whose knowledge, culture, and values are encoded in AI systems. Research from the Data Provenance Initiative suggests that training data from regions such as Africa and South America accounts for only approximately 0.2% of the datasets used to develop major machine-learning models (Longpre et al., 2024).

This lack of representative datasets constitutes a profound epistemic injustice. AI systems, particularly large language models and generative image systems, are gradually becoming authoritative sources of information (ARTICLE19, 2025) and redefining cultural narratives (Symons & Abumusab, 2024). Yet they homogenise thought, culture, and values into a mainstream ideal (Sourati et al., 2025). This homogenization predominantly reflects Western, and frequently American, perspectives, opinions, and aesthetics, further deprioritising the vast array of worldviews.

This dynamic mirrors historical patterns of colonial knowledge extraction and erasure, but with a crucial difference: it is being automated, scaled, and embedded into infrastructure that increasingly mediates how people and communities access information and understand the world. The open internet promised democratisation of knowledge, but AI-driven concentration threatens to narrow whose knowledge counts as legitimate, whose culture deserves representation, and whose values shape the collective digital future. As AI tools become integrated into search engines, productivity platforms, and educational resources, they risk reinforcing a narrow set of cultural assumptions while sidelining diverse intellectual traditions.

The geography of computational power

A third and more structural challenge to openness lies in the geography of computational resources. The infrastructure required to train advanced AI systems—data centres, specialised graphics processors, and large-scale cloud platforms—is disproportionately concentrated in North America and Europe (Lehdonvirta et al., 2024). Research from the Oxford internet Institute describes this AI compute inequality in three categories: the Compute North, Compute South and, Compute Desert. Wealthy countries in the Compute North host data centres equipped with the latest GPUs capable of training frontier AI models. These countries can participate in cutting-edge AI development, iterate on novel architectures, and maintain technological sovereignty. Countries in the "Compute South" are mainly middle-income countries with access to older GPU infrastructure, which is sufficient for deploying and fine-tuning existing models but insufficient for training frontier systems. These nations are relegated to being consumers and adapters of AI technology rather than originators. Meanwhile, countries in the Compute Desert have little or no access to AI computing infrastructure. These nations have no meaningful participation in the AI economy beyond basic usage of consumer applications.

These disparities create a structural hierarchy in the AI ecosystem. Notable compute shortages in Africa, Latin America, the Caribbean, and Oceania mean that these regions are systematically relegated to the bottom of the AI development hierarchy. This is not due to lack of talent, ambition, or potential applications, but rather structural inequalities in access to the fundamental infrastructure required for participation.

The compute divide between regions creates a vicious cycle, compounding challenges of both the AI and digital divides. Without access to computational resources, researchers and developers in these regions cannot train models on their own data or build systems tailored to local challenges, in their own languages. They become dependent on models developed elsewhere, trained on others' data, optimised for others' use cases (Chege, 2025). The promise of openness rings hollow when the basic infrastructure required for meaningful participation is so unevenly distributed.

Governing AI for genuine openness

The challenges posed by linguistic exclusion, data scarcity, and the consolidation of computational power require governance frameworks that move beyond expanding connectivity alone. Ensuring meaningful openness, transparency, and accountability in the age of AI means addressing substantive participation throughout the entire value chain of AI development: who can build, train, and shape these systems.

First, research communities should advocate for linguistic equity, requiring AI systems operating globally to demonstrate meaningful performance across diverse language communities. This could trigger investments in low-resource language technologies, supporting community-driven dataset development, and establishing performance benchmarks that account for linguistic diversity.

Second, policymakers and non-governmental stakeholders should support frameworks for data sovereignty and epistemic justice that recognise the rights of different communities to shape how their knowledge, culture, and contexts are represented in training datasets. This might include requirements for geographic and cultural representation in training data, community-consent mechanisms for data collection, and support for locally developed models that reflect local contexts.

Third, governments, research institutions, and civil society organisations must invest in computational infrastructure as a global public good. These investments must also be accompanied by initiatives that democratise access to the computing power needed for AI development. This could involve distributed compute networks, subsidised access for researchers in under-resourced regions, and support for regional AI research centres.

Conclusion: reclaiming openness

The open internet was once envisioned as a democratizing force capable of expanding access to information and enabling new forms of global participation. In the age of artificial intelligence, however, openness cannot be defined by connectivity alone. True openness now depends on linguistic inclusion, equitable data representation, and access to the computational infrastructure required to build digital technologies.

Fortunately, many of the approaches described in the contribution are already underway. The newly launched Africa Green Compute Coalition (AGCC), led by UNDP, aims to democratise access to sustainable, affordable AI computing power across the African continent (UNDP, 2025). A growing number of national governments and regional initiatives across the Global Majority are developing open-source AI models. These include the development of LatamGPT led by Chile's National Center for Artificial Intelligence and a consortium of government, academic, and civil society partners (CENIA, 2026), the launch of the SEA-LION family of open-source LLMs that better understands Southeast Asia's diverse contexts (Ng et al., 2025), and Nigeria's N-ATLAS LLM, developed for 3 Nigerian languages and Nigerian pidgin English to increase access to government services (FMCIDE, 2025).

These efforts demonstrate that through coordinated action, meaningful inclusion, and local collaboration, the internet can begin to fulfil its long-deferred promise as a genuinely equitable resource shaped by and for the communities it has historically overlooked. By evolving into a more just and inclusive ecosystem, AI systems may one day reflect the full diversity of human experience rather than perpetuating the compounding barriers that have long defined access for the Global Majority.

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II. Sovereignty, Interoperability, Standards

The pursuit of digital sovereignty and strategic autonomy has become a central objective for many states seeking to assert control over data, infrastructure, and standards. Yet these ambitions often coexist uneasily with the global internet's dependence on openness and interoperability. This section of the report examines how governments, regional blocs, and standards-setting bodies negotiate the balance between sovereign control and global interdependence. Drawing on perspectives from political economy, international law, and technology governance, it will explore whether strategic autonomy can be achieved without deepening internet fragmentation, and how standardisation processes shape power asymmetries in the digital order.

Several contributions focus on the European Union, where this tension is particularly acute. Julia Carver and Dennis Broeders examine the competing demands of cloud sovereignty and AI sovereignty within EU policy, arguing that reducing dependency on US infrastructure may simultaneously constrain the EU's ability to develop competitive AI systems. Anushka Kaushik and Sara Čela each approach interoperability not as a technical given but as a deliberate economic and regulatory strategy—one that can serve both to reduce dependency and to create new forms of market leverage. Arno Lodder traces how the Union's understanding of digital sovereignty has developed in relation to its territorial jurisdiction and the legal tensions this produces.

Panagiotis Delimatsis examines how standardisation processes have become strategic instruments through which states and firms compete for influence over digital markets, with analysis spanning multilateral bodies and the increasingly contested politics of standards-setting. Yoshiko Naiki extends that analysis to data interoperability in cross-border supply chains, focusing on international trust frameworks developing at UNECE and ISO, with implications well beyond any single jurisdiction. Rebeca Rabêlo situates comparable dynamics within the BRICS context, where member states pursue hybrid strategies of regulatory assertion, infrastructural investment, and open-source development to negotiate structural asymmetries in an AI economy concentrated in two dominant poles. Across these pieces, sovereignty and interoperability emerge less as opposing values and more as instruments deployed to advance different positions within a shared but unequal digital order.

Operationalising Digital Sovereignty, Governing Interdependence: The Role of Standardisation

Panagiotis Delimatsis

Introduction

Standards are the foundation of any innovation system; they constitute the invisible infrastructure of innovation (Delimatsis, 2015). They are the conditions under which technologies interoperate, innovations diffuse, markets scale, and infrastructures become durable. In the current geopolitical turn, standards have become strategic assets that determine market access and shape innovation. The ability of a country to influence standardisation processes or outcomes, especially internationally, allows it to shape the future of digital technologies and decrease strategic dependencies, or even reverse them (World Bank, 2025).

This makes standards central to contemporary debates on digital sovereignty and strategic autonomy (European Commission, 2022; ANSI, 2026). States increasingly seek greater control over the infrastructures and technological dependencies on which their societies rely. Even the EU, whose economic project is based on openness, trade, and international cooperation, is pursuing an ambitious plan to reduce digital vulnerabilities and develop key technologies and inputs, such as semiconductors, cloud services, and AI tools, domestically (European Commission, 2020).

Yet such ambitions coexist uneasily with the fact that the internet, digital trade, transnational information systems and innovation depend on openness and interoperability. Experience shows that standardisation can support fair competition in identifying the best available technologies, ensure fair returns for innovators, and distribute the benefits of knowledge globally (Spulber, 2013). Participation and cooperation in standardisation can therefore help manage interdependence while enhancing strategic autonomy.

The core argument of this contribution is that active participation and leadership in standardisation processes, together with like-minded partners, enable a country to operationalise digital sovereignty and govern interdependence globally. Technical standards are often described as neutral and highly specialised instruments: background rules that enable devices to connect, networks to function, and services to interoperate. That description is increasingly misleading. In the digital environment, standards are not merely technical artefacts. They are strategic instruments that shape market access, lock in technological pathways, distribute advantages across value chains, and influence who exercises effective control over data, infrastructure and digital services.

Why standards matter politically

Standards structure the conditions under which technologies scale. They enable interoperability, facilitate legal compliance, reduce uncertainty, and create common vocabularies across markets. But their role is not exhausted by efficiency. Technical standards undergird the digital tools used every day and, depending on their design and implementation, can facilitate empowerment and market opportunities or deepen inequalities and widen digital divides. Standards can support openness and the exercise of rights, but they can also entrench asymmetries and exclusions (United Nations, 2023).

Once this is acknowledged, standard-setting can no longer be understood as free of politics. Standards reflect the interests, values and assumptions of those who participate in their development (Mattli & Seddon, 2015). They can harden power asymmetries just as much as they can reduce transaction costs and address market failures. In emerging fields such as AI, cybersecurity, cloud computing, and digital identity, standards increasingly determine whose technical architectures become the default infrastructure for others. Standard-setting is therefore not a secondary site of governance. It is one of the prime global arenas where digital governance is constructed and interdependence managed.

Deep-pocketed companies benefit from investing in standardisation. Governments, too, can accrue strategic and political benefits if they recognise early the importance of investing in standardisation and have the resources to do so. But digital sovereignty can accentuate power asymmetries in standardisation because developed economies now have broader security-related incentives to invest heavily. Such an investment race will not necessarily lead to better technologies being adopted; it may intensify zero-sum dynamics, alienate innovators, and produce suboptimal or illegitimate standards.

The EU's standards turn: from market integration to strategic autonomy

The EU offers a particularly revealing case study of the tension between fragmentation and openness. It has moved from using standardisation as a largely technical co-regulatory tool for deepening the internal market to treating it as a strategic field where industrial policy, market integration, values, fundamental rights, and external action converge (European Commission, 2022; Delimatsis, 2026). Yet the EU continues to present itself as committed to openness, multilateralism and interoperable digital governance, as Article 21 TEU requires. The result is a distinctive but delicate project: to achieve "open strategic autonomy" by shaping standards, preferably with trusted partners, rather than by retreating from interdependence (European Commission, 2025b).

This marks a major conceptual shift. In the classic internal market narrative, standards removed barriers and facilitated free movement. In the newer strategic autonomy narrative, they also become instruments through which the EU seeks to defend agency in a world marked by geopolitical rivalry and asymmetric dependencies (European Commission, 2025a). The International Digital Strategy projects this logic externally. It treats digital policy as a core element of EU external action. It commits the Union to promoting open standards and common specifications to secure interoperability, transparency and cross-border digital integration, including in digital public infrastructure, digital identity, e-signatures and e-invoicing (European Commission, 2025b).

For the EU, standards now sit at the intersection of four projects. First, they support the internal market and conformity assessment, as seen in the AI Act and Cyber Resilience Act. Second, they serve industrial and geopolitical goals by reducing dependencies and strengthening European influence in key technologies. Third, they carry a normative charge because the EU increasingly treats standard-setting as part of a constitutional and values-based digital order. Fourth, they can project and protect a distinct European way of living, innovating and doing business.

Digital sovereignty *through* (not *despite*) interdependence

The EU operationalises strategic autonomy and digital sovereignty through a mix of regulatory, industrial, international and funding instruments that aim to strengthen its capacity to shape its digital future while remaining globally engaged. This increasingly blurs the line between domestic regulation and international digital and commercial policy. The concept of digital sovereignty that best fits the EU approach emphasises preserving agency and securing access to critical technologies without creating one-sided structural dependence or loss of capacity to act (Blind, 2025; JRC, 2025). This matters because it avoids a false binary between openness and sovereignty. If sovereignty is understood as agency, interdependence becomes problematic only when it is asymmetric or weaponised (Farrell & Newman, 2019).

In this respect, a standards strategy can be sovereignty-enhancing because standardisation fora are spaces where the terms of interdependence are contested and stabilised. A key concept that standardisation can shape is interoperability. The EU digital strategy focuses on partnering with trusted countries with complementary strengths in the digital value chain to co-shape global standards and promote regulatory convergence toward a human-centric, trustworthy digital transformation. Its toolbox includes digital partnerships, digital trade agreements, trade and technology councils, MoUs, digital alliances and older e-commerce chapters. These instruments aim to foster co-creation and the adoption of international standards in areas such as semiconductors, quantum and AI, and to ensure coordination among standardisation bodies (see Table 1 regarding the EU's digital partnerships). Through them, the EU seeks to pool digital sovereignty with trusted partners.

Table 1. Standardisation in the EU's Digital Partnerships

Korea	Japan	Singapore	Canada
• Focus on AI, cyber, chips, HPC, quantum, 5G/6G, digital standardization • Cooperation on int'l standards (eg trusted chips/chip security) • Safety, fairness, contestability • Interoperability of trust services (public sector)	• Focus on privacy, chips supply chain, 5G/6G, HPC, quantum, AI, digital connectivity, platforms, data, digital trade, SMEs, blockchain, int'l standardisation, digital education, trust services • Coordination in int'l standards fora, achieve common goals, work on early stages of R&I	• Focus on digital trade and connectivity, 5G/6G, data, chips supply chain, AI, platforms, trust services, FinTech, SMEs, digital education • Cooperation on int'l standards, notably on cyber, 5G/6G and AI • Coordination on early stages of R&I	• Focus on connectivity, digital identity and trust services, data governance, AI, chips supply chains, research, quantum, Earth system modelling, language technologies, cyber, platforms, FIMI, digital skills • Cooperation on int'l standards (eg in human-centric digital identity, cyber, AI standards coordination)

Proceduralisation and digital sovereignty

The plurality of formal bodies, such as ISO, IEC, and ITU, and informal consortia, such as IETF, IEEE, 3GPP, and W3C, creates a complex landscape in constant pursuit of relevance, coherence, and legitimacy. While most bodies formally adhere to WTO/TBT principles such as openness, transparency, consensus and coherence (Kanevskaia, 2023), the current geopolitical moment introduces a different logic: speed, strategic alignment and institutional control. Complaints about delays and the lack of openness as causes of reduced leverage are increasingly common (Kanevskaia et al., 2024).

This creates a crucial normative tension in the European approach. The EU promotes standards that are agile, geopolitically effective and aligned with EU interests, but it also wants standard-setting to remain open, balanced and inclusive (European Commission, 2022). This raises the question of whether proceduralisation is a building block or a stumbling block for digital sovereignty. Arguably, it is both. Procedural guarantees may slow standard-setting in the short term, but without them, standards lose legitimacy and become vulnerable to capture.

The EU's approach is also explicitly normative. It seeks to embed privacy, accessibility, accountability, resilience and democratic values into technical standards, digital infrastructures and international agreements. Still, the more standards are required to carry substantive value choices, the harder it becomes to distinguish technical coordination from regulatory intervention. A more proactive industrial policy may nevertheless allow the EU to support infrastructures and standards that reflect those values, whether through open-source components, federated structures such as Gaia-X, or interoperability rules that enable accountability.

The gamble and the limits of the EU approach

An EU that regulates pervasively but builds insufficiently risks exercising normative influence without corresponding infrastructural control (Draghi, 2024). Standards cannot compensate for industrial weakness or long-term dependence. Effective influence in standard-setting is strengthened by technological capacity, investment, market presence and active company participation.

Moreover, despite the achievements of the internal market, internal fragmentation and implementation gaps continue to undermine EU capabilities (Letta, 2024). Without addressing institutional and market realities, including barriers in intra-EU trade in goods and services (Adilbish et al., 2025), strategic ambition will be difficult to realise.

Finally, the EU is not alone in seeking to instrumentalise standards geopolitically. This makes it harder to sustain open global interoperability, especially when values are added as a condition for adoption. Coalitions with like-minded countries are therefore essential for advocacy, persuasion, alliance-building and market access. Standardisation fora are becoming the places where tensions between technical excellence, values, and efficiency are negotiated.

Conclusion

Standards have moved from the margins to the centre of digital geopolitics. They are no longer merely technical supports for markets; they are mechanisms through which power, values and dependency are organised in the digital order. They are strategic because they shape markets, infrastructures and technological trajectories, but they are also constitutive of the openness on which the global internet and digital trade depend.

20 years ago, the EU Global Europe Strategy (European Commission, 2006) presented the EU as well-placed to become a reference point for global standards. Fast forward twenty years, and strategic rivalry is increasingly infiltrating the global standardisation system, threatening the values and prosperity of democracies worldwide. This calls for an urgent political awakening but, crucially a proactive, concerted engagement in international standard-setting to ensure that existing (such as a free, open, accessible and secure global internet) but also future technologies (such as AI or quantum) remain accessible and interoperable while reflecting our shared values, including rule of law, respect of human rights, dignity, or privacy. EU's digital diplomacy instruments, including Digital Partnerships, DEPAs and TTCs, are actively redefining power and influence in international standardisation. They diffuse EU regulatory approaches, support an open rules-based digital order, and seek to make EU-preferred standards benchmarks for international cooperation. But this should be coupled with sustained domestic investment in strategic technologies (European Commission, 2025a).

Standards can reinforce dependence if they lock ecosystems into proprietary or politically controlled infrastructures. But they can also reduce vulnerability if they preserve portability and interoperability. Owing to its powerful single market, the EU can play a key role in ensuring that the standards it helps shape preserve both agency and interoperability in a world increasingly tempted by technological closure. If that balance can be sustained, standard-setting may become one of the few sites where digital sovereignty and global interdependence are not mutually destructive, but mutually constitutive.

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Standards for Data Interoperability and Trust

Yoshiko Naiki

Introduction: data sharing and Cross-Border trade

Today, demand for sustainability across global supply chains has grown across sectors and product categories. As a result, there is increasing interest in tools, systems, and guidelines that enable accurate, reliable sustainability information to be shared across complex global supply networks. To protect business secrets, however, such information should be shared only with supply-chain participants who have a legitimate interest. Beyond direct benefits to operators, the solutions regarding sustainability information may offer a pathway out of the growing internet fragmentation and the lack of trust resulting from geopolitical tensions.

At the cross-border level, ensuring the sharing and exchange of sustainability information among numerous actors throughout supply chain networks is particularly challenging. On this point, interoperability in cross-border data sharing becomes a key issue. The main challenge for policymakers and operators throughout the supply chain is to ensure that sustainability information is transmitted and shared in an interoperable manner across different jurisdictions. Data interoperability encompasses several layers, including communication infrastructure for data exchange, data formats, and data semantics. Moreover, in the context of cross-border data sharing, legal, organisational, and policy interoperability—such as differences in data-localisation laws—also become relevant.

International standardisation can play an important role in cross-border data sharing, especially through standard-setting initiatives that address data interoperability and trusted supply-chain information. The first involves earlier efforts by the United Nations Economic Commission for Europe (UNECE) to develop recommendations guiding industry stakeholders and policymakers toward greater traceability and transparency in sustainable value chains. The second is the International Standardization Organisation (ISO), specifically Technical Committee (TC) 292, which addresses security and resilience and develops standards for data sharing across supply chains. While the earlier UNECE standardisation process remained relatively neutral in encouraging stakeholders to take action, discussions within ISO/TC 292 have evolved into a more strategic process that involves particular business interests.

These developments are closely linked to the EU's ambition to regulate sustainable battery supply chains by introducing digital product passports (DPPs). The EU battery passport requirement was introduced under the EU Batteries Regulation, which entered into force in August 2023 (Council of the EU (2023). In essence, DPPs function as digital repositories of product-specific information. According to the Ecodesign Regulation, a DPP is defined as "a set of data specific to a product that includes the information specified … and that is accessible via electronic means through a data carrier." The EU Batteries Regulation requires manufacturers exporting batteries and electric vehicles to the EU to establish such passports. Although this represents the first use case of EU DPPs and remains in a complex implementation phase—with compliance becoming mandatory in February 2027—the Ecodesign for Sustainable Products Regulation adopted in July 2024 foresees the introduction of DPPs for additional sectors and product groups. Both regulations form part of the EU's broader circular economy strategy (Council of the EU, 2024).

These developments illustrate how sustainability regulation, data governance, and international standardisation are becoming increasingly intertwined. As governments seek to improve traceability and transparency in global supply chains, technical standards for interoperability and trust become critical mechanisms for enabling cross-border regulatory implementation.

Standard-Setting activities in the UNECE

The UNECE's work on trusted sustainability information across supply chains emerged in response to growing demands for responsible business conduct, particularly in the garment and footwear sectors. The UNECE Recommendation "Enhancing Traceability and Transparency of Sustainable Value Chains in the Garment and Footwear Sector" (UNECE, 2022) introduced tools for manufacturers and governments to enhance traceability and transparency (Rinaldi et al., 2022).

Although the 2022 Recommendation discusses traceability and outlines relevant principles and concepts, it provides only a limited discussion of interoperability. It notes that because value chains include multiple partners, interoperability—the ability to exchange data with minimal transformation—is essential, and that agreeing on a common standard for both IDs (unique identifiers) and data formats is the best way to achieve it (UNECE, 2022: para. 70).

By contrast, the UNECE Recommendation "Transparency at Scale: Fostering Sustainable Value Chains" places greater emphasis on interoperability (UNECE, 2025). This reflects the Recommendation's broader objective of enhancing trust in sustainability information (UNECE, 2025: para. 1). The document explicitly recognises the role of DPPs in achieving this goal, noting that widespread adoption of DPPs can provide reliable and verifiable sustainability information conveyed through clear and standardised terminology (UNECE, 2025: para. 16).

At the same time, the Recommendation acknowledges that the introduction of DPPs raises interoperability challenges across industries and borders (UNECE, 2025: para. 38). To address these challenges, it outlines interoperability design principles such as the use of identifier schemes and the need to build bridges between different traceability platforms and ecosystems (UNECE, 2025: para. 40).

The UNECE process illustrates an early stage of international coordination, where the emphasis is placed on principles and voluntary guidance rather than binding technical standards. In this sense, the UNECE recommendations help establish a conceptual framework for traceability and interoperability, while leaving the detailed operationalisation of these principles to industry actors and other standard-setting bodies.

Standard-Setting activities under the ISO

As discussed above, the trusted exchange of sustainability information among numerous supply-chain participants is a central challenge in cross-border data sharing. Within ISO/TC 292, a new standard titled "Framework for Establishing Trustworthy Supply and Value Chains" was finalised in November 2025 (ISO22373: 2025). The initiative was driven in part by German and Japanese industries, reflecting their strong interest in data-sharing standards linked to collaboration between the EU and Japan in the automotive and battery sectors. The implementation of the EU Batteries Regulation partly drives this cooperation.

Compared with the UNECE Recommendation, the ISO standard provides a more detailed process for establishing trust in data exchange. The concept of trustworthiness is defined as the ability to meet stakeholders' expectations in a verifiable way (ISO22373: 2025, para. 3.3). To operationalise this concept, the standard proposes that actors involved in data sharing exchange lists of "trustworthiness expectations" and "trustworthiness capabilities." Trustworthiness expectations refer to requirements necessary to achieve desired trust attributes (ISO22373: 2025, para. 3.10), while trustworthiness capabilities describe verifiable properties that demonstrate an actor's ability to meet those expectations (ISO22373: 2025, para. 3.11). When these lists are exchanged and mutually agreed upon, actors sharing data can evaluate and confirm each other's level of trustworthiness (Kawabata & Furukawa, 2025). In this way, trustworthiness is conceptualised as a structured process for assessing trust in data exchanges.

These standardisation efforts also reflect a memorandum of understanding between the EU and Japan recognising the importance of advancing the sharing and practical use of automotive data, including interoperability between their respective data-sharing systems (IPA, 2024). For Japanese manufacturers and exporters, sharing sensitive information regarding supply sources and business partners raises significant concerns. Consequently, the Japanese automotive and battery industries have strong incentives to test interoperability between Japanese and EU data spaces and to demonstrate secure and trusted data transfers to the EU. The EU and Japan have been promoting projects to test the interconnectivity between their data spaces (Santana et al., 2025).

The ISO standardisation process, therefore, reflects not only technical considerations but also broader economic and geopolitical dependencies, because battery passports address the geopolitical realities of battery production, namely securing the supply of critical raw materials for batteries (Ducuing, 2026; Huang & Nottage, 2025). Efforts to standardise trusted data-sharing frameworks are closely linked to emerging regulatory requirements and to strategic industrial cooperation among major economic partners.

Conclusions

Additional standard-setting initiatives related to DPPs are emerging. One example is an initiative under the joint working group between the UNECE and ISO/TC 154 (ISO/AWI 25534-1), which reflects growing interest in implementing DPPs beyond the EU's regulatory framework. As these initiatives develop, the arena of DPP standardisation is likely to become an important focal point for cross-border data sharing and interoperability, ultimately promoting more openness in the online environment.

Beyond enabling reliable sustainability information sharing, the adoption of DPPs is increasingly linked to broader geopolitical considerations in supply chains, particularly those involving strategic product groups such as batteries and, potentially in the future, chemicals and electronics. Future research should therefore pay closer attention to the geopolitical interests that shape DPP standardisation processes. These dynamics may influence how standards evolve and may contribute to new forms of power asymmetry in the emerging digital economic order.

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Something Does Not Compute? Trade-offs Between EU Cloud Sovereignty and AI Sovereignty

Julia Carver and Dennis Broeders

Introduction

Cloud technologies have become central to the operation of both public and private services in the European Union. However, governments' move to the cloud has occurred under increasingly fraught conditions of geoeconomic competition and fears of digital dependence in Europe. Pressures to increase cloud uptake across the bloc have coincided with rising fears that Europe lacks control over the infrastructure upon which its digital economy depends. Over 80 % of European data is stored on servers owned by American or Chinese companies (Rone, 2024; Ferreira Gomes and Okano-Heijmans, 2024). Furthermore, legislation such as the United States' CLOUD Act has heightened concerns about Washington's ability to exploit the outsized influence of American Big Tech over the design, provision, and maintenance of global digital infrastructure, including cloud technologies. Ironically, it has been the openness and interoperability of the internet—in combination with the openness of the European market—that facilitated the rise of an economic/technological oligarchy in the international and the European digital economy. The current challenge is how the EU can reconcile the need for European digital sovereignty and strategic autonomy with the openness and interoperability of the global internet and global internet economy.

These concerns have added new impetus to the Union's 'European digital sovereignty' ambitions, a wide-ranging policy agenda covering digital technologies, including artificial intelligence (AI), semiconductor chips, cloud technologies, cyber capacity building, and digital diplomacy (Carver, 2024; Broeders et al., 2023). While these initiatives signal a shift in the EU's approach to economic competitiveness, industrial policy, and global affairs, they also generate new policy tensions.

This contribution argues that the dominant strategic priorities underlying European cloud sovereignty and AI sovereignty are increasingly in conflict. Given the structural realities of the EU's digital market and the AI innovation ecosystem, policymakers face a difficult choice: prioritise global competitiveness in AI development or pursue strategic autonomy in cloud infrastructure.

Geoeconomic entanglements between AI and cloud infrastructures

'Cloud sovereignty' issues were among the earliest policy debates on European digital sovereignty. Before the rise of large language models and generative AI systems, European policy debates focused primarily on cloud infrastructure not used for high-performance compute functions (i.e., not cloud for AI). In this earlier context, 'cloud sovereignty' concerns were closely linked to data localisation in Europe, given the importance of data centres as sites of infrastructural power (Rone, 2024; Blancato and Carr, 2024). However, the US CLOUD Act significantly altered these debates by enabling American authorities to request access to data held by American companies—regardless of where this data is physically stored. The US CLOUD Act introduced the possibility that sensitive government or corporate data stored on American cloud platforms within European data centres—on European soil—would be subject to foreign jurisdiction.

Given these developments, the focus of European cloud sovereignty has expanded to include concerns about reliance upon foreign hyperscalers, particularly Europe's infrastructural dependence on American cloud services (Blancato and Carr, 2024). While shifting crucial government, corporate, and private data into American cloud infrastructure once made budgetary sense for Europeans, it is now politically unviable. Particular attention has been paid to the Union's outsized dependence on large hyperscalers such as Microsoft, Alphabet (Google), and Amazon for data collection, processing, and storage (Baur, 2024).

After the COVID-19 pandemic, the rapid rise of AI technologies—particularly the public release of several popular large language models (LLMs)—has added a new dimension to European discussions about cloud sovereignty. Producing leading commercial innovations in AI, including foundational AI models such as LLMs, rests upon scalable cloud computing infrastructure. Globally, Microsoft, Amazon, Alphabet (Google), and NVIDIA now provide much of the computational capacity used to develop and deploy advanced AI models and systems (Hawkins et al., 2025). Notably, in the European digital ecosystem, major American technology companies dominate both the 'traditional' cloud storage and the 'AI for cloud' computing markets (Hawkins et al., 2025).

Against this background, the Union has recognised the growing entanglement between AI innovation and cloud infrastructure and their significance for European digital sovereignty (European Commission, 2025a). Concerns about European digital dependency have extended to AI innovation and governance, given American private-sector dominance in frontier AI development and in providing the cloud infrastructure necessary for it (Petit and Teece, 2021). However, Brussels' dominant narratives about 'cloud sovereignty' and 'AI sovereignty' contain logical contradictions and practical challenges which hinder the EU's pursuit of a coherent, actionable digital sovereignty agenda.

Tensions between european AI sovereignty and cloud sovereignty

As noted above, achieving European 'cloud sovereignty' is often defined as reducing dependence upon foreign cloud service providers—thereby improving European strategic autonomy by ensuring regionalised solutions. Recent debates on *cloud sovereignty* largely reproduce this rationale by emphasising the risks of European dependence on foreign hyperscalers for AI innovation. This logic is reflected in the EU's '*Cloud Sovereignty Framework*', which prioritises supply chain transparency, technological independence, and control over data assets and AI services (European Commission, 2025b). Under this framework, the highest levels of sovereignty are associated with cloud services that "*remain under EU control* or [not] *exposed to non-EU* *dependencies"* (European Commission, 2025b). Notably, cloud sovereignty also includes a "*focus on the protection, control, and independence of data assets and AI services within the EU,*" including '[...] *how data is secured, where it is processed, and the degree of autonomy customers retain over AI capabilities*" (European Commission, 2025b).

By contrast, recent policy discussions surrounding European AI sovereignty have tended to follow a different narrative. Rather than focusing on autonomy, an emerging dominant narrative for AI sovereignty in Brussels, as advanced by the second von der Leyen Commission, has emphasised the need to ensure European technological leadership in the global AI race (Mügge, 2024). This logic is clearly laid out in Mario Draghi's (2024) report on *European Competitiveness*, which underlined the importance of Europe capitalising on the 'AI revolution' to 'catch up' with the United States and China and regain its global economic competitiveness (Draghi, 2024; European Commission, 2026). Similarly, recent EU strategy documents, including the AI Continent Action Plan, have framed 'AI sovereignty' as pursuing European global leadership over frontier AI technologies and ensuring global AI competitiveness (European Commission, 2025c; Hess and Sieker, 2025).

This logic follows Brussels' approach to AI investments, whereby the Commission and bodies such as the European Investment Bank Group have prioritised 'deeptech' AI innovation—that is, financing the development of foundational AI (such as LLMs) which can be scaled to the European market and beyond (Saari, 2026). Such investments include the EU's 'AI Gigafactories', facilities that produce high-performance semiconductor chips for developing advanced (frontier) and large-scale AI models, supported by a €20 billion project investment (European Commission, 2025d). Under this current approach, Brussels has largely emphasised the development of commercially scalable, globally competitive AI systems that rival leading American and Chinese LLMs, as well as overcoming fragmentation in the European AI ecosystem.

Yet, under current market conditions, prioritising European global competitiveness in frontier AI technologies remains crucially dependent on the scaling capacity and compute power provided by foreign hyperscalers (van der Vlist et al., 2024; Saari, 2026). Global compute capacity is heavily concentrated between companies headquartered in the US and China, with American companies dominating the so-called European 'sovereign cloud' market. (van der Vlist et al., 2024; Saari, 2026) As noted previously, leading innovations in AI foundational models and cloud computing are provided and enabled by Big Tech hyperscalers, which have forged close partnerships with leading AI startups, including OpenAI (with Microsoft), Anthropic (with Amazon AWS), and Google DeepMind (with Google Cloud). Even European cloud and/or digital 'stack' initiatives, including European 'neoclouds', rely upon 'AI Accelerator' computer chips, and they are heavily financed and supplied by the American firm NVIDIA (NVIDIA, 2025; Saari, 2026). Accordingly, NVIDIA predicted that its market share in providing chips for sovereign AI compute initiatives would increase by 100% from 2024 to 2025, amounting to $20 billion USD in chips (Jarvis et al., 2025). Such hardware and infrastructure challenges—together with Europe's localised approaches to cloud and data sovereignty—pose multiple scaling challenges to the EU achieving a globally competitive alternative to American frontier AI models.

A sovereign dilemma?

This structural reality creates a dilemma for European policymakers. Particularly in the area of cloud computing, it is unrealistic for European Member States to pursue both 'cloud sovereignty', which is premised upon a logic of European strategic autonomy, and 'sovereign AI', which is predominantly premised upon a globally competitive race logic. After all, powerful American firms dominate the menu of options available to European states seeking to grow their AI and cloud ecosystems. Altogether, European decoupling from foreign companies, particularly American firms, is stymied by structural challenges in the cloud/AI market, such as attracting the requisite talent/expertise towards AI innovation, sustaining the demand for a European AI compute innovation ecosystem, incentivizing Europe-wide approaches to financing cloud projects (market fragmentation), and allocating sufficient long-term (venture) capital, which can help prevent 'European unicorns' from being bought out by Big Tech monopolies. (Calcara, 2025; Saari, 2026)

European policymakers therefore face a strategic trade-off between achieving 'scalable' European sovereign AI (as envisioned by Brussels) and their goals of European autonomy, defined as European control over cloud computing infrastructure and European data (cloud sovereignty). Efforts to achieve sovereign AI may, *by definition,* undercut Brussels' cloud sovereignty goals and vice versa.

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Balancing Sovereignty and Scale: The Future of EU Cloud Governance

Anushka Kaushik

Introduction

The ongoing debates around cloud governance in the European Union (EU) reflect broader ambitions to preserve sovereignty in key technological domains. Most contemporary definitions of technological sovereignty in the EU emphasise autonomy over technology policies, standards, and structural processes. A central component of the EU perspective concerns citizens' data—where it is stored and whether adequate privacy safeguards are in place (Kaushik, 2024).

This discourse is also being shaped by a central question: can digital sovereignty be strengthened without undermining the openness and interoperability that cloud infrastructure depends on? Balancing sovereignty ambitions with the strategic realities of dependence and the benefits of participation in open, globally interconnected digital ecosystems is at the core of the EU cloud governance challenge.

According to recent estimates, the EU cloud market was worth EUR 87 billion in 2022 and is projected to reach EUR 200 billion by 2028 (Draghi, 2024). Cloud adoption is also a strategic priority for the Union, which has set a target for 75% of European businesses to use cloud-edge technologies by 2030 (European Commission, 2025).

Cloud services offer particular benefits for small and medium-sized enterprises (SMEs) and local businesses. With scalable infrastructure, organisations can respond to surges in data demand and adjust storage capacity to fluctuating needs. This scalability is closely linked to cost optimisation, as businesses can adopt "pay-as-you-go" models rather than investing in expensive physical infrastructure. Beyond operational efficiencies, cloud infrastructure also provides resilience through geographically distributed data storage. A prominent example occurred in 2022, when Ukraine ensured the continuity of government services by migrating national data from domestic data centres—under threat of Russian attacks—to the cloud.

Market asymmetries and the politics of certification

The EU cloud policy debate is shaped by concerns about technological dependence, data control, and foreign access to critical infrastructure, all within the context of clear market asymmetries. Nearly 70% of the EU cloud market is controlled by Amazon, Microsoft, and Google, while European cloud providers account for only around 15% (Synergy Research Group, 2025).

The ongoing discussions surrounding the European Union Cybersecurity Certification Scheme for Cloud Services (EUCS) illustrate these concerns. In December 2019, the European Commission tasked ENISA, the EU cybersecurity agency, with developing a certification framework to establish a common technical standard across the 27 Member States (European Commission, 2019). The aim was to enable governments and consumers to make informed decisions based on third-party certification while strengthening cybersecurity standards, encouraging customer adoption, and promoting best practices.

The initial draft introduced three assurance levels based on different security criteria. In later iterations, proposals emerged to add a condition at the highest assurance level: requiring not only EU-based data storage and processing, but also European headquarters and majority ownership. This proposal proved contentious. Denmark, Estonia, Greece, Ireland, the Netherlands, Poland, and Sweden opposed the provision in a 2022 non-paper, while France, Italy, Spain, and Germany supported it (Bômont, 2025).

While some governments viewed the measure as necessary to reduce structural vulnerabilities, others argued that it politicised a technical certification scheme and risked excluding major cloud providers. The March 2024 draft ultimately removed the clause and left such decisions to national regulators, though the adoption of the final scheme remains pending.

Another important aspect of this debate concerns the scale of US cloud providers. Cloud infrastructure requires substantial capital investment, which smaller European providers may struggle to match. The Draghi Report argues that attempting to develop fully competitive European challengers to US hyperscalers would require enormous investment and could divert resources from sectors where Europe has stronger innovation potential (Draghi, 2024).

European competition authorities have also identified structural barriers that may limit competition, including high switching costs and technical obstacles to interoperability. At the same time, scale provides certain security advantages. Large providers can segment and distribute data across multiple global data centres to reduce exposure risks and can use global traffic patterns to detect emerging cybersecurity threats more rapidly (Meyers, 2023). Moreover, Europe's growing demand for cloud services—particularly from firms higher up the technological stack requiring secure and reliable Infrastructure-as-a-Service (IaaS)—may be difficult to meet solely through domestic providers.

Factors influencing the future of EU cloud governance

In this dynamic context, three factors are likely to shape the future trajectory of EU cloud governance: continued divergence among Member States on digital sovereignty, the development of sovereign cloud solutions, and the emergence of sectoral data hubs.

Technological sovereignty remains a contested concept without a shared operational definition. At the EU level, it is frequently embedded within the broader framework of strategic autonomy, understood as the Union's capacity to make independent policy decisions and reduce structural dependencies in critical sectors. In the context of cloud governance, this has translated into debates about data localisation, foreign jurisdiction risks, and the role of non-EU providers.

Member States differ in their approaches, shaped by historical experiences, economic size, geopolitical considerations, and strategic priorities (Kaushik, 2024). For instance, several Central and Eastern European (CEE) countries express reservations about expansive interpretations of strategic autonomy and technological sovereignty. These concerns are closely tied to security considerations and the importance of maintaining strong transatlantic relations, particularly in light of the Russian threat (Marusic & Brudzińska, 2021). In addition, SMEs constitute the majority of enterprises in the CEE region, many of which rely heavily on US-based technology providers.

A second factor concerns the extent to which genuinely sovereign cloud solutions can be developed within Europe. The Draghi Report recommends a middle path between supporting domestic cloud industries and maintaining access to critical technologies provided by US hyperscalers. This approach would involve adopting EU-wide data security policies governing collaboration between EU and non-EU providers, with a strong emphasis on security and encryption. At the same time, mandatory standards for public sector procurement could help level the playing field for European firms while maintaining broader participation in the transatlantic digital marketplace outside strictly "sovereign" domains.

Between 2022 and 2023, major US cloud providers introduced sovereign cloud solutions in response to the EU's digital sovereignty agenda. These offerings typically involve infrastructure located entirely within the EU, operational control exercised by EU-resident personnel, and additional safeguards for data access and encryption. In principle, such arrangements allow European governments and companies to benefit from the technological capabilities and economies of scale of hyperscale providers while addressing regulatory concerns about jurisdictional control over data.

Nevertheless, scepticism remains regarding the extent to which these solutions are truly sovereign. US legislation—including the CLOUD Act (2018) and Section 702 of the Foreign Intelligence Surveillance Act (FISA)—may allow US authorities to request access to data held by American companies, even when that data is stored abroad. As a result, locating infrastructure entirely within the EU may not fully eliminate foreign jurisdictional risks.

Member States have responded differently to these developments. Some governments view sovereign cloud offerings as a pragmatic compromise, while others—notably France—remain more sceptical and favour stricter interpretations of sovereignty that prioritise European ownership and operational control.

A third factor shaping EU cloud governance is the emergence of sectoral data hubs such as Catena-X, developed under the broader Gaia-X initiative. Originally launched as a Franco-German project, Gaia-X aimed to establish a decentralised cloud ecosystem that would enable users to retain greater control over their data.

Within this framework, Catena-X has emerged as a collaborative data ecosystem for the automotive industry, connecting solution providers and service partners across the entire value chain. These initiatives do not directly compete with US hyperscalers. Many participating companies continue to rely on partnerships with them.

Instead, as Calcara (2025) argues, these hubs aim to create critical mass among European firms to strengthen their negotiating position vis-à-vis global cloud providers. While companies remain competitors in the marketplace, they cooperate in building shared data infrastructures to improve bargaining power and establish more balanced digital ecosystems. Similar initiatives are emerging in sectors such as agriculture and finance, suggesting a model of structured interdependence rather than direct technological competition.

Conclusion

Cloud infrastructure is central to Europe's digital transformation, and the EU is actively promoting its adoption through ambitious targets under the Digital Decade strategy. However, this expansion takes place in a context of significant market asymmetry and growing concerns about dependence on a small number of largely non-European providers.

The scale of US hyperscalers remains a key structural factor. Their global infrastructure enables cybersecurity and resilience capabilities that rely on extensive data networks and operational reach. At the same time, divergent views among Member States on technological sovereignty continue to shape debates around EU cloud governance.

Efforts to strengthen Europe's domestic cloud ecosystem are ongoing, supported by funding under programs such as Horizon Europe and the Digital Europe Programme. However, whether European providers can compete directly with US hyperscalers remains uncertain. The Draghi Report therefore proposes a middle path: supporting domestic capabilities while maintaining access to global cloud technologies.

In response to Europe's sovereignty ambitions, major cloud providers have introduced sovereign cloud offerings. While some Member States view these as workable compromises, concerns remain about potential foreign access to European data—even when infrastructure is located within the EU.

Against this backdrop, sectoral data hubs are emerging as an alternative governance model. Rather than competing directly with hyperscalers, these initiatives seek to establish European data ecosystems that improve bargaining power and create more balanced digital markets.

Ultimately, the success of EU cloud governance will depend on whether efforts to strengthen technological sovereignty can avoid regulatory fragmentation and policy deadlock. Resolving debates such as the EUCS certification framework will be crucial in ensuring that Europe can harness cloud infrastructure to support cybersecurity, competitiveness, and innovation.

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Interoperability and Sovereignty as Economic Strategies

Sara Çela

Introduction

In Europe's internet economy, sovereignty and interoperability in digital infrastructures are not merely technical endpoints but important strategic levers. When embedded into infrastructure design, they influence SME competitiveness, cross-border trade dynamics, and Europe's capacity to project regulatory standards globally. These developments should also be understood within broader debates about the future of the open internet. While early visions of the internet emphasised borderless connectivity with minimal governance, contemporary digital infrastructures increasingly operate within competing regulatory models. Europe's approach seeks to reconcile openness with accountability by embedding governance principles directly into infrastructure design.

Three initiatives operating at distinct infrastructure layers—GAIA-X, Wire, and Sover—highlight both progress and structural limits in adoption, scale, and systemic integration. GAIA-X reflects European efforts to construct a federated cloud and data infrastructure designed to reduce dependency on non-European hyperscale providers. Wire demonstrates how secure communication systems embed privacy and sovereignty at the protocol level. Sover provides an applied case of how modular compliance and identity architectures extend sovereignty into citizen-facing services.

Regulatory foundations of europe's digital infrastructure strategy

Europe's approach to digital regulation is often described as statutory and rules-based. Increasingly, however, these regulatory frameworks serve as the architectural foundations of digital governance. European companies tend toward incremental innovation within institutionalised frameworks, and recent regulatory initiatives appear calibrated to induce structured adaptation rather than radical innovation (Blind, Niebel & Rammer, 2023).

The General Data Protection Regulation (GDPR), the Data Governance Act (DGA), the Data Act, and the AI Act define how infrastructures are built, managed, and how data flows across borders (European Commission, 2020; Council of the EU, 2022; Council of the EU, 2023; Council of the EU, 2024). In effect, Europe is regulating at the level of architecture, not merely behaviour. The GDPR establishes sovereignty through enforceable data protection and accountability. The Data Governance Act introduces institutional mechanisms for trusted data sharing. The Data Act transforms interoperability into a legal obligation across industrial ecosystems. The AI Act embeds transparency, traceability, and risk classification directly into algorithmic systems.

Together, these frameworks reconfigure Europe's internet economy by translating regulatory values—trust, transparency, and accountability—into technical specifications and infrastructure design principles. This reflects what may be termed "sovereignty through design," as Europe seeks to express political autonomy and economic strategy by codifying governance principles directly into the infrastructures that underpin its digital economy (Bendiek and Schallbruch, 2024; Bradford, 2023).

Operationalising sovereignty through digital infrastructure

Building technological systems represents the operational dimension of sovereignty: the point at which governance principles translate into functioning infrastructures. Federated clouds, sectoral data spaces, digital identity frameworks, and secure communication systems form interdependent layers through which sovereignty and interoperability are enacted in practice (European Commission, 2023b; OECD, 2023).

These infrastructures do not operate as isolated platforms but as coordinated environments structured by shared standards. Their strategic function lies in preserving jurisdictional control while enabling cross-border data circulation and market integration. Federated cloud architectures reduce dependency on external hyperscalers; data spaces institutionalise trusted sharing; identity systems secure participation; and encrypted communication embeds compliance directly at the protocol level (GAIA-X Association, 2023; ENISA, 2023).

Together, these layers demonstrate that Europe's approach to innovation does not consist in digital enclosure but in cross-layer institutional design. Legal norms are embedded directly into technical architectures, turning sovereignty into an operational grammar of coordination and trust. The interaction between regulatory frameworks and technological infrastructures becomes visible when examining concrete implementations.

Sovereignty through federation: GAIA-X

GAIA-X is Europe's federated cloud and data infrastructure initiative connecting national and private cloud providers through shared technical and governance standards (GAIA-X Association, 2023). Rather than replacing global hyperscalers, it introduces a European trust layer emphasising transparency, portability and interoperability (European Commission, 2023b). GAIA-X illustrates sovereignty as a federation, preserving national control while enabling cross-border data flows and competitive participation.

However, implementation of such systems has revealed governance complexity and uneven adoption across Member States (European Court of Auditors, 2023). This demonstrates a structural tension where regulatory ambition may outpace coordinated execution if technical baselines and incentives remain fragmented (Bendiek and Schallbruch, 2024).

Sovereignty through design: Wire

Wire operates at the protocol layer as a secure communication and collaboration platform, embedding end-to-end encryption and open-source transparency into its architecture. Wire’s infrastructure is designed so that only communicating parties can decrypt content, and no operator or intermediary holds users’ encryption keys. This is a design choice that aligns the platform structurally with principles of privacy by design and technical accountability (Wire, Security & Privacy, 2025).

Wire represents sovereignty through cryptographic design. Privacy and jurisdictional control are not contractual assurances but embedded technical features. Economically, this positions compliance as competitive differentiation particularly in public procurement and regulated industries. Yet Wire also reflects the persistent challenge of network effects. Competing with dominant global platforms, whose scale is driven by massive user networks, requires balancing regulatory alignment with scalability and interoperability across ecosystems (IMF, 2024).

Sovereignty through modular architecture: Sover

Sover operates at the application and service layer, developing modular identity and compliance infrastructures aligned with EU regulatory requirements. According to its publicly available technical documentation, the platform is structured around interoperable APIs, jurisdiction-aware data governance modules and verifiable data control mechanisms designed to integrate with existing public and private systems (Sover, 2025).

Through this modular architecture, Sover extends sovereignty into citizen-facing and enterprise services. Rather than positioning compliance as an external legal obligation, the platform embeds regulatory logic into service design by aligning identity verification, data traceability and access governance with European data protection and interoperability requirements. However, the scalability of such models remains contingent upon consistent technical and regulatory implementation across Member States. Divergent interpretations of interoperability requirements or digital identity frameworks risk fragmentation if coordination mechanisms remain insufficient (European Court of Auditors, 2023).

Sovereignty, interoperability, and the architecture of an open internet

Within Europe’s digital architecture, sovereignty and interoperability are embedded design functions shaping how value is created and protected. Sovereignty lies in defining the rules of participation, such as data location, governance standards, and accountability structures.

Each system, whether GAIA-X, Wire or Sover, operates as a point where control and openness meet and ensure that innovation can scale without surrendering regulatory integrity. Interoperability, in turn, lies in the operational logic that connects those rules.

It allows different system providers to communicate through shared protocols, ensuring continuity across jurisdictions and technologies. It transforms fragmentation into a coordinated marketplace where compliance is not a constraint but an enabler of market access. A functioning global system where sovereignty and interoperability coexist is the international airport network. Each airport operates under national jurisdiction, yet interoperability in logistics, security and data exchange is governed by shared global standards. Sovereign control coexists with operational coordination. Digital regulatory ecosystems should aspire to a similar model with sovereign control at the institutional level, interoperability at the operational level and trust as the currency connecting both.

Economically, this architecture transforms regulatory certainty into competitive advantage (Bradford, 2023). Opportunities include reduced dependency on monopolistic providers, enhanced SME scalability and strengthened bargaining power in global digital negotiations (OECD, 2023). Europe's value proposition becomes innovation within trusted frameworks. At the macro level, interoperability enables Europe to project market norms globally and extend its regulatory influence beyond territorial borders (European Commission, 2023a).

However, systemic challenges persist. Fragmented national implementation risks undermining interoperability baselines (European Court of Auditors, 2023). Regulatory density may slow experimentation if compliance is perceived as a cost rather than an enabler. Globally, Europe must balance normative leadership with competitive pressures from less regulated ecosystems (IMF, 2024). Europe's distinct strength lies not in digital scale but in the credibility of its standards (Bradford, 2023). If coherently implemented, sovereignty and interoperability function as mutually reinforcing engines of competitiveness.

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Aligning Digital Sovereignty With Physical Territory: The EU's Evolving Perspectives

Arno R. Lodder

Introduction

The tension between sovereign control and global interdependence defines the internet from a legal perspective. The classic way for a government to regulate is to govern and control what happens within its borders. The internet's architecture is inherently global and relies on borderless functionality. This creates a fundamental jurisdictional mismatch: the internet lacks a 'natural fit' for traditional legal frameworks (Johnson and Post, 1996). It is not intuitively clear how to regulate the internet or how to apply the law online. The European Union has been developing an optimal approach to law and the internet and to ensuring its law adequately covers internet activities (Savin, 2025). The EU legal framework has undergone a significant transformation.

The following discussion traces the evolution of jurisdictional criteria in EU digital law. It begins with the predecessor to the GDPR, the Data Protection Directive 95/46/EC, which used the physical location of hardware and processing equipment as a criterion for determining the applicability of the law. This territorial focus shifted with the E-Commerce Directive 2000/31/EC, which established the 'country of origin' principle, centring applicability on the place of establishment for legal entities or the residence of natural persons. Over the last ten years, the focus has shifted toward the destination-based principle, first introduced by the GDPR (2016) as a hybrid model alongside its establishment. In 2022, the DSA, a regulation amending the E-Commerce Directive, prioritises only the location where the service is received, moving beyond physical or corporate presence to address the borderless nature of the digital economy.

1995: location of server matters (Data Protection Directive)

At the infancy of the internet, in 1995, the Data Protection Directive 95/46/EC was enacted (Data Protection Directive 95/46/EC). Unlike the modern "targeting" approach of the GDPR, back then, the criteria for determining the applicability of national law were grounded in physical infrastructure. Under Article 4(1)(a), national law was applied based on the controller's establishment. The reason to discuss this Directive here is that Article 4(1)(c) used the location of the servers as the criterion to apply the law:

> "Each Member State shall apply the national provisions (...) to the processing of personal data where: c) the controller is not established on Community territory and (...) *makes use of equipment (...) situated on the territory of the said Member State*".

Based on this criterion, national law applies to the processing of personal data by controllers and processors established even outside the EU, provided they use servers physically located within the EU. This was a time when cloud computing was not yet common, so accessing data remotely was rare. Still, if servers abroad were used to process data, national law also applied to those controllers operating from outside the EU. To ensure the enforceability of EU law against these foreign-based controllers, Article 4(2) mandated the appointment of a local representative within the Member State where the processing equipment was situated.

2000: only establishment matters (E-commerce Directive)

Some years later, while drafting the Directive 2000/31/EC on e-commerce (E-commerce Directive, 2000), there was uncertainty about how to apply the law to internet activities. This uncertainty was qualified in recital 5 of the Directive as a legal obstacle:

> "a number of legal obstacles to the proper functioning of the internal market (...) these obstacles arise from divergences in legislation and *from the legal uncertainty as to which national rules apply to such services*"

Options considered to end this uncertainty were to apply the law based on the server's location, similar to the just-discussed Directive 95/46/EC, or on the domain name. The reason not to choose a domain name or the equipment's location was that it could lead to circumventing the law by locating servers outside the EU, or by choosing a domain name from a non-EU country (including the most popular generic domain name, .com). Instead, the establishment of the provider (or the place of residence of the natural person) was considered the appropriate criterion to determine the applicability of the law. The physical location of equipment was no longer used as a criterion to apply the law.

An established service provider is defined in Article 2(c) as:

> "a service provider who effectively pursues an economic activity using a fixed establishment for an indefinite period. *The presence and use of the technical means and technologies required to provide the service do not, in themselves, constitute an establishment of the provider*."

The so-called home state control criterion makes sense, particularly from an enforcement perspective. When companies and individuals are physically present within a state's territory, that state can, in line with the traditional concept of sovereignty, exercise control over who is within its borders. There are, however, at least two peculiar consequences (Lodder, 2022).

First, a company established within the EU but operating solely on a market outside the EU must still comply. Once, a Dutch attorney represented a provider of online pornographic content solely targeting the Japanese market. Service providers should adhere to the regulatory standards of the jurisdiction where they are established. However, from an enforcement perspective, such an expectation appears largely illusory. It is unlikely that regulatory authorities would monitor a foreign-language website aimed at another jurisdiction, particularly where linguistic barriers limit their ability to assess its content.

Second, a company with a .nl (Netherlands) domain name and services in Dutch does not have to comply with the E-commerce Directive if it is not established within the EU. This is counterintuitive. One would expect that, on a Dutch-language website with a Dutch domain name, the provider complies with EU law. Note that the e-commerce directive does not touch upon private international law (cf. Article 1(4)), so a consumer may still seek protection on the ground that a website, like this one, is directed at a specific jurisdiction.

2015: destination-based principle (GDPR)

Some 15 years later, the above-mentioned second peculiarity was solved by the GDPR, the successor of the earlier addressed Directive 95/46. In Article 3(1), the establishment of the controller, as in the e-commerce directive, remains. New is the criterion introduced in Article 3(2), which at first glance feels somewhat pedantic. Regardless of the location or establishment from where the service is offered, compliance with EU law is required the moment these services are provided to recipients inside the EU. However, on second sight, it does make sense. The internet is open, global, and accessible from everywhere.

As a consequence, what is communicated can have an impact everywhere. It makes sense not to restrict legal consequences to those providers located within the EU (De Hert and Czerniawski, 2016). Equally, if an EU citizen defames a US citizen, the US citizen can remedy the defamatory communication (or the other way around). With the introduction of the GDPR, a US company established in the US but offering services in the EU has to comply with EU law.

The destination-based principle of the GDPR reflects a shift toward a "targeting criterion" that resembles established approaches in tax and consumer protection law. VAT is applied based on where a product is consumed, e.g. requiring an Italian citizen buying from a Brazilian website to pay Italian VAT. Similarly, a consumer can invoke consumer protection laws of their home country against foreign entities. What distinguishes the GDPR is that data protection authorities can directly impose the norms upon companies operating from abroad. While international enforcement remains a challenge, the legislative intent is to bind non-EU companies to EU standards. This regulatory approach has since been adopted by subsequent frameworks, including the Digital Services Act (DSA) and the AI Act.

2022: just directed to EU citizens, establishment irrelevant (DSA)

The Digital Service Act (DSA) amended the already discussed e-commerce directive (Digital Services Act 2022/2065). Article 2(1) no longer uses establishment as the way to determine the applicability of the law. Where the GDPR used establishment and added the destination-based principle, the DSA uses just the latter:

> “intermediary services offered to recipients of the service that have their place of establishment or are located in the Union, *irrespective of where the providers of those intermediary services have their place of establishment*.”

The approach followed in the DSA addresses both the peculiarities of the E-commerce directive's home-state principle. The only criterion is the destination-based principle. First, this means that it does not matter where a provider is established; only the place where services are received determines the applicable law. Second, since establishment is no longer a criterion, providers operating from within the EU but offering services only outside the EU do not have to comply with the DSA's norms. Whereas the AI act still uses the establishment criterion for some activities, it also applies a pure destination-based principle. Article 2(1)(a) AI Act indicates that national law applies to providers who put AI systems into service, *irrespective* of where these providers are established.

Directive 2000/31/EC on e-commerce was considered the framework directive for online commerce, often referred to as the mother directive. The DSA amends the e-commerce directive and, in particular, adds norms on platforms and online content. As such, the DSA has a huge impact on the digital economy.

Goldsmith and Wu (2006) mentioned Directive 95/46/EC as an example of global norms, not because it was applicable outside the EU, but because of the EU's normative impact in the area of data protection abroad. Later, this phenomenon was coined the Brussels effect (Bradford, 2012). The GDPR extended this effect to direct norm enforcement of data protection norms abroad. Whereas the Brussels effect likely applies to the DSA, what is more important is its direct norm application based on EU market access. It does so at the core of the present-day online economy: online content and platforms.

There is no place here to further elaborate on two less-ideal aspects of the online application of law as it is now per DSA, but for the sake of completeness, they are briefly mentioned. There may be online activities by providers established within the EU that provide services outside the EU that we do not feel comfortable with, and they cannot be sanctioned for it. First, certain online activities conducted by providers established within the EU but serving only non-EU markets may raise ethical or legal concerns that remain unactionable under the DSA enforcement framework. Second, the application of law abroad depends on how the internet is accessed. If a cell phone is used outside the EU via an EU telecommunication subscription, EU law applies; if local WIFI is used, it does not (Lodder, 2017).

Conclusion

A foundational principle of internet law is that online activities must be attributed to a relevant jurisdiction by linking them to a specific actor (such as a person, company, or state) and/or to the technical infrastructure (Lodder, 2013). As the discussion of the EU approaches over the years shows, correctly linking internet activities to a particular jurisdiction is not easy.

After an initial approach in which the location of the technology mattered, it is understandable that the EU chose establishment as the deciding criterion in the e-commerce directive. Hence, enforcement is straightforward for the enforcing state if companies or persons are located within its borders.

With the GDPR, the focus also shifted to include who is targeted by the internet activities.

Finally, the DSA abandoned the establishment and focused solely on those targeted by internet activities. In terms of enforcement, it can be difficult when providers are in faraway countries. However, these providers often want to provide services in the EU and, as such, have an interest in complying with EU law.

With the approach taken in the DSA, the EU has perfected how to apply the law to internet activities. The objective of the law is to regulate what is happening within the EU, and that is what the DSA does.

The sketched development impacts the concept of the open internet. Initially, the lack of borders facilitated worldwide activities largely unhindered by foreign laws. The home-state control principle of the E-commerce Directive emphasised that national norms are central to internet-based commercial activities. Even before the GDPR, however, companies increasingly showed a willingness to comply with foreign legal norms, e.g., on defamation or national rules governing social media activity before elections. With the GDPR, the openness of the internet became more clearly constrained, as the EU imposed data protection law directly on companies targeting the EU market. Access to the market now determines which law applies. The DSA extended this principle to platform activity and online content. As a result, while the open internet technically facilitates easy reach of a global audience, the legal environment has fundamentally changed: access to the EU market is conditional on compliance with EU norms.

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Digital Sovereignty in a Multipolar World: The BRICS Approach

Rebeca Rabêlo

Introduction

Artificial intelligence has become a structural determinant of geopolitical power. Control over semiconductor production, cloud infrastructure, and foundational models increasingly defines economic competitiveness and national security. Yet the global AI ecosystem remains heavily concentrated in two dominant technopoles, the United States and China (Mayer & Lu, 2025). This concentration generates patterns of asymmetrical interdependence that constrain the policy autonomy of emerging economies (Lukings & Lashkari, 2023). In this setting, the concept of an open internet, characterised by decentralised infrastructure, equitable access to digital resources, and the absence of monopolistic control over technological systems, provides a critical normative and institutional counterpoint to such concentration.

Within this context of asymmetric interdependence and contested digital governance, BRICS has increasingly positioned itself as a relevant actor in the digital domain. Originally formed as a coalition of major emerging economies seeking greater influence in global governance, the group has expanded its agenda to include digital infrastructure, technological development, and cooperation on artificial intelligence. However, there remains a clear research gap in understanding how BRICS countries pursue digital sovereignty within a structurally concentrated AI ecosystem, and what this reveals about the prospects for multipolarity in the evolving digital order.

As a heterogeneous bloc composed of both major technological powers and developing economies, BRICS provides a revealing site for examining how digital sovereignty is articulated under conditions of structural asymmetry. Rather than pursuing technological autarky, BRICS countries appear to adopt a hybrid strategy of negotiated interdependence, combining infrastructural investment, regulatory assertion, and open-source AI development. Importantly, this strategy is internally differentiated, with member states exhibiting distinct levels of ambition and strategic focus.

Digital sovereignty and structural concentration

The modern state has traditionally been grounded in territorial authority and the capacity to formulate and enforce law within defined borders. However, the architecture of the internet disrupts this logic by enabling transnational data flows largely governed by private technology corporations that regulate platforms, infrastructures, and digital markets (Chander & Sun, 2021; Gu, 2023). Digital sovereignty emerges as an attempt to reconcile territorial authority with deep digital interdependence. However, the concept remains contested: it has been defined as the state's ability to regulate cross-border data and strategic technologies, as broader collective control over data governance, or as varying degrees of technological autonomy (Robles-Carrillo, 2023; Crespi *et al.*, 2021).

Fratini *et al*. (2024) capture this diversity through four ideal-typical models—rights-based, market-oriented, centralising, and state-led—emphasising that, in practice, states adopt hybrid configurations shaped by domestic capacity and geopolitical positioning.

However, these models cannot be understood solely in terms of regulatory design or normative orientation. Digital sovereignty is materially conditioned by control over the infrastructures that sustain digital systems. As highlighted in the *BRICS Strategy and Technology Review* (2025), the rise of "functional sovereignty" exercised by corporations dominating cloud services, software ecosystems, and global connectivity networks embeds public authority within privately governed technological architectures. This structural constraint is particularly evident in advanced semiconductor production, where high-performance chips essential to AI development remain geographically concentrated and capital-intensive. Consequently, the feasibility of any sovereignty model, whether rights-based, market-oriented, or state-led, depends not only on legal frameworks but on access to hardware supply chains, computational capacity, and energy infrastructure, underscoring its embeddedness in the global political economy of technology.

AI, infrastructure and digital colonialism

Artificial intelligence amplifies existing structural asymmetries in the global political economy. The development and deployment of advanced AI systems depend on high-performance GPUs, large-scale datasets, reliable energy infrastructures, and capital-intensive data centres. Training frontier models requires thousands of specialised processors operating continuously for extended periods, resulting in substantial energy consumption and financial costs (Aguiar, Berwaldt, & Nunes, 2025). AI is therefore not a frictionless or purely digital innovation; it is deeply embedded in material infrastructures whose ownership and geographic distribution shape access to technological capabilities.

This infrastructural concentration has been conceptualised as a form of digital colonialism, in which data extraction and control over computational architectures reproduce historical hierarchies through technological means (Couldry & Mejias, 2019; Faustino & Lippold, 2023). The expansion of data centres in the Global South, frequently without substantive technology transfer or local value capture, risks entrenching dependency rather than fostering autonomous development (Diniz, 2025). Infrastructure, in this sense, becomes both a site of accumulation and a mechanism of structural constraint.

Under these conditions, reliance on proprietary AI systems developed within dominant technopoles risks reinforcing asymmetrical interdependence. For BRICS countries, securing meaningful participation in AI's infrastructural and epistemic foundations becomes the central question. The challenge is to expand domestic capacity without reproducing extractive digital hierarchies, by transforming interdependence from a source of vulnerability into a basis for negotiated sovereignty.

Differentiated ambitions within BRICS AI strategy

The BRICS coalition presents a constellation of differentiated national strategies. China emphasises strong state control and data localisation; India combines regulatory assertiveness with digital public infrastructures; Brazil advances data protection and state-backed payment systems; Russia invests in autonomous internet architectures (Fratini *et al.*, 2024; Queiroz & Direito, 2025).

These strategies reflect ideal-type models of digital sovereignty but rarely function in isolation. In practice, BRICS countries adopt hybrid configurations influenced by domestic capacity, geopolitical stance, and developmental priorities. The investment in alternative infrastructures, such as submarine cables, national payment systems, and digital public platforms, demonstrates efforts to increase strategic autonomy (Diniz, 2025; Chander & Sun, 2022).

Within this hybrid framework, open-source AI plays a distinctive role. Unlike strict data localisation or market closure, open-source initiatives enable participation in global knowledge networks while preserving adaptability. By supporting collaborative innovation and reducing barriers to entry, open-source ecosystems can partially offset the "functional sovereignty" of Big Tech firms (Gu, 2023).

While the BRICS is often treated as a cohesive geopolitical bloc, its AI strategies reveal significant internal differentiation in ambition, strategic focus, and positioning within the global technological hierarchy. Document analysis of national contributions and policy statements indicates that member states adopt differentiated strategic orientations.

Table 1. Differentiated AI Strategic Positioning within BRICS

Country	Strategic Ambition	Technological Orientation	Sovereignty Approach	Institutional Maturity
China	Comprehensive Global Leadership	Advanced R&D, neuromorphic computing, military AI, full-stack development	End-to-end technological sovereignty	Consolidated and expansionary
Russia	Strategic Technological Leadership	Advanced research, autonomous infrastructure, national hardware development	Security-driven technological sovereignty	Consolidated but state-centric
Brazil	Normative and Developmental Influence	Ethical AI, data protection, digital public infrastructure	Sovereignty of interdependence	Developing, framework-oriented
India	Developmental and Inclusive Scaling	Data ethics, digital public infrastructure, large-scale talent formation	Hybrid sovereignty model	Expanding institutional capacity
Indonesia	Long-term structural development	National AI roadmap, infrastructure alignment	Capacity-building sovereignty	Early-stage institutionalization

Country	Strategic Ambition	Technological Orientation	Sovereignty Approach	Institutional Maturity
South Africa	Institutional Consolidation	Governance frameworks, digital inclusion	Development-focused sovereignty	Framework-based, emerging
Egypt	Capacity and Infrastructure Expansion	National strategy, infrastructure development	Developmental sovereignty	Early-stage consolidation
Iran	Selective Autonomy	National AI plan, partial self-sufficiency	Semi-autonomous sovereignty	Developing with geopolitical constraints
Ethiopia	Infrastructure-led Catch-Up	Energy-linked infrastructure strategy	Structural capacity-building	Early-stage
United Arab Emirates	Strategic International Positioning	Best-practice adoption, data-sharing platforms	Market-integrated sovereignty	Advanced regulatory orientation

Source: Author's elaboration based on national AI strategies

The structural asymmetries discussed above do not produce uniform responses within BRICS. Instead, they generate differentiated strategic adaptations. To systematise these variations, Table 1 maps member states along four interrelated dimensions: strategic ambition, technological orientation, sovereignty approach, and institutional maturity.

Rather than treating BRICS as a cohesive challenger group, the matrix reveals a layered configuration of sovereignty strategies. China and Russia occupy the upper tier of strategic ambition, pursuing end-to-end technological sovereignty and investing heavily in advanced research domains such as neuromorphic computing, autonomous infrastructures, and national hardware development. Their positioning reflects both accumulated institutional capacity and long-term geopolitical objectives centred on technological leadership.

In contrast, countries such as Brazil and India adopt intermediate positions characterised by normative engagement and developmental scaling. Their strategies emphasise digital public infrastructure, data governance, and ethical frameworks, reflecting what may be described as a sovereignty-of-interdependence model: autonomy pursued within, rather than against, global integration. Meanwhile, Indonesia, South Africa, Egypt, Ethiopia, and Iran prioritise capacity building and infrastructure consolidation, focusing on talent development, regulatory frameworks, and access to AI-compatible infrastructure.

This stratification demonstrates that digital sovereignty within BRICS operates as a spectrum conditioned by material capabilities and institutional maturity. The bloc's cohesion lies in a shared structural objective of reducing vulnerability within a concentrated AI ecosystem.

Conclusion

In a multipolar digital order characterised by infrastructural concentration and technological rivalry, BRICS countries pursue digital sovereignty not through technological isolation, but through differentiated strategies of negotiated interdependence. Their approaches combine infrastructure investment, regulatory assertiveness, and open-source AI development, calibrated to domestic capacity and geopolitical positioning. The internal variation revealed in this analysis demonstrates that sovereignty within BRICS operates along a spectrum of ambition and institutional maturity rather than as a unified project of strategic autonomy.

This differentiation reveals an important feature of contemporary multipolarity. Rather than fragmenting the digital order into insulated blocs, emerging powers seek to recalibrate their position within concentrated technological systems. Sovereignty in the AI era thus emerges as a relational and infrastructurally conditioned practice, shaped by access to hardware, computational capacity, and governance frameworks. Multipolarity, in this sense, does not imply symmetry of power, but the strategic management of asymmetrical interdependence within a hierarchically structured technological order.

From this perspective, the open internet emerges less as an empirical reality than as a normative horizon that informs and constrains these strategies. While BRICS initiatives partially align with its principles, particularly in their emphasis on decentralisation, access, and open technological ecosystems, they also reflect the enduring limits imposed by structural concentration. The pursuit of digital sovereignty unfolds as a situated and contested process within the emerging digital order.

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III. Data, Trade & Cross-Border Flows

Cross-border data flows are the connective tissue of the digital economy, enabling trade in goods, services, and knowledge. Yet, the growing divergence in regulatory approaches—ranging from the EU's rights-based model to Asia-Pacific's digital trade frameworks such as DEPA and CPTPP—illustrates competing visions of openness and control. This section looks at how digital trade agreements and regional governance mechanisms mediate the tension between economic integration, data protection, and strategic autonomy.

Martina F. Ferracane opens the chapter by mapping the current landscape of cross-border data regulation, documenting the accumulation of both restricting and enabling measures across different legal orders—a layering she describes as producing a "spaghetti bowl" of partially compatible disciplines that generates uncertainty for businesses despite the central role of data in trade architecture. María Vásquez Callo-Müller traces how digital economy agreements have moved from early commitments to free data flows toward more qualified architectures that embed rights-based safeguards, reflecting broader political contestation over the appropriate balance between liberalisation and regulatory autonomy. Georgios Dimitropoulos examines a region often absent from the dominant policy literature—the Middle East and North Africa—where data localisation, industrial strategy, and evolving bilateral relationships interact in ways that sit uneasily within existing governance templates.

Franziska Sucker's analysis of the AfCFTA Digital Trade Protocol shows how African states have developed a governance model that pairs transfer obligations with shared continental data protection standards and explicit development commitments, an approach that does not fit neatly within the conventional binary of "free flow" versus "adequacy." The chapter as a whole illustrates that regulatory divergence need not produce fragmentation in the technical sense, but that the political costs of legal incompatibility fall unevenly, and most heavily on smaller economies with less capacity to negotiate exceptions or enforce reciprocity.

The State of Cross-border Data Regulation: Where Do We Stand?

Martina F. Ferracane

Data flows are the backbone of digital trade

Data serves as the means through which digital trade occurs, as a means of production, and as an asset that can be traded itself (Lopez González *et al.*, 2023). It is also the core of new 'information industries' such as cloud computing, big data analytics, and artificial intelligence, while contributing to nearly all sectors as an input into R&D, product design, production processes, logistics, marketing, sales, and customer engagement (OECD, 2022).

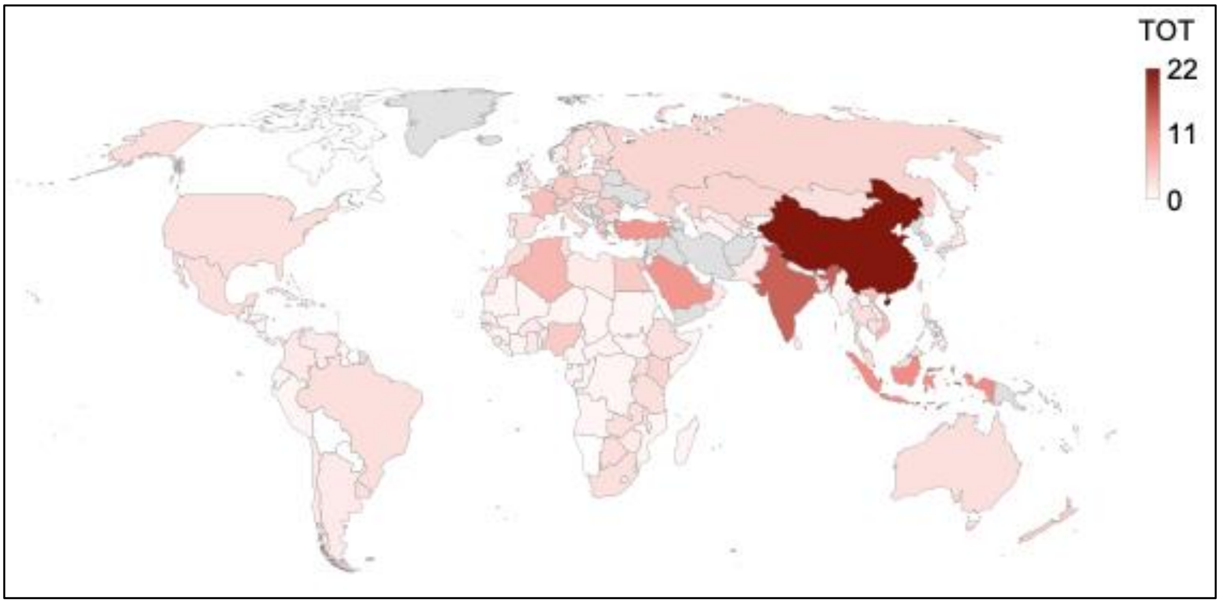
Data flows have become the backbone of digital trade, and empirical evidence confirms that they support trade, productivity, growth, and investment (van der Marel & Ferracane, 2021; Ferracane et al., 2020; Jia et al., 2018). However, the concentration of data in the hands of a few powerful players, coupled with evidence of government surveillance that exploited the global nature of the internet, heightened political concerns and claims to exert governments' sovereignty over the digital space (Zuboff, 2019; Fratini et al., 2024).

These dynamics are closely linked to ongoing debates on the governance of the open internet, which relies on cross-border data flows. The resulting policy landscape reflects a trade-off between maintaining an open and integrated digital environment and addressing domestic regulatory objectives.

Restrictions on data flows are on the rise

The rise in the importance of data flows in the economy has gone hand in hand with governments' growing interest in regulating and controlling them. These restrictions can take different forms, ranging from outright bans on data transfers to imposing certain conditions for transferring data abroad (Ferracane, 2024). As such, these policies share the trait of incentivising private entities to process their data within the jurisdiction imposing the requirement, thereby increasing the costs of conducting business across borders. Using data from the DTI database, Figure 1 shows the geographic distribution of data flow restrictions worldwide for the latest year available, that is, 2024.

Figure 1: Number of policies restricting data flows per country, 2024.



Source: Author based on data from indicators 6.1, 6.2, 6.3 and 6.4 of the DTI database retrieved on 20 February 2026 from <https://dti.eui.eu/database>. Note: the countries in grey are not included in the database.

The DTI database contains regulatory information on digital trade policies across 160 jurisdictions, including data governance regulations (Ferracane *et al.*, 2026). The database lists 331 policies restricting data flows across borders. About half (53%) of the policies are local processing requirements and infrastructure requirements, thereby requiring companies to process data locally and, in some cases, to use a specific data centre for processing. One-third of the policies impose conditional regimes, most of which are inspired by the European General Data Protection Regulation (GDPR). Under this regime, such transfers abroad can be made only if certain conditions are met, such as the adequacy of the recipient country or the data subject's consent. The remaining policies are local storage requirements, which require keeping a copy of certain data in the country, while processing can be undertaken in any country.

The majority of policies apply horizontally across all sectors (48%), followed by sector-specific policies for the financial sector (20%) and the public sector (10%). Moreover, most of the restrictions apply to personal data (39%), followed by financial data (18%) and accounting records (11%). The countries with the highest number of data flow restrictions are China (22 policies), India (15 policies), Indonesia (11 policies), Saudi Arabia (10 policies), and Türkiye (10 policies). In contrast, several countries do not impose any regulations that affect data flows across borders, largely due to a general lack of regulation of the digital economy. These are Belize, Bolivia, Burundi, Canada, Comoros, Djibouti, Eritrea, Fiji, Guinea-Bissau, Haiti, Namibia, Paraguay, Suriname, and Trinidad and Tobago.

Overall, the evidence points to a clear global trend toward the increasing regulation of cross-border data flows. The uneven distribution of such measures across countries highlights a fragmented regulatory landscape, with far-reaching implications for digital trade, market integration, and the future openness of the global digital economy.

Enabling measures to facilitate data flows are also increasing

While restrictions on data transfers could be interpreted as restrictions on the provision of services under the World Trade Organization (WTO), there is no explicit commitment on data flows undertaken at the multilateral level, making the application of WTO rules on data flows challenging at the very least (Ferracane, 2019; Mitchell & Mishra, 2021). Commitments on data flows have also been removed from the final text of the Joint Statement Initiative on e-Commerce, a plurilateral agreement negotiated at the WTO, whose future remains uncertain (Whittle, 2024).

Instead, international commitments on data flows have primarily taken place in the context of preferential trade agreements (PTAs) and, more recently, in digital economy agreements (DEAs). Figure 2 provides an overview of countries that have joined at least one agreement with a binding commitment to free flows of data for business purposes, highlighted in dark green. The remaining countries have not yet joined any agreement with binding commitments on data flows or, in the case of African countries, the commitments are part of an agreement that has not yet entered into force.[4]

In addition to commitments under trade agreements, governments have negotiated other bilateral and regional mechanisms to facilitate data transfers, especially for personal data flows, with commitments dating back to the early 1980s.[5] These agreements often link the free flow of personal data to safeguards embedded in the recipient country's regulatory system, thereby reconciling data flows and data protection. However, despite the substantial trade benefits linked with these agreements (Spiezia & Tscheke, 2020), and in particular with adequacy determinations (Ferracane et al., 2026; Ferracane et al., 2025), these mechanisms remain opaque and subject mostly to political (rather than legal) considerations (Yakovleva, 2024; Ferracane, 2025), as also testified by the recent commitment by Argentina to grant adequacy to the United States through a trade agreement.[6]

As a result, cross-border data flows are subject to selective liberalisation contingent on participation in specific agreements or regulatory alignments, rather than governed by universal rules.

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[4] The 48 State Parties of the African Continental Free Trade Area (AfCFTA) adopted the Protocol on Digital Trade (DTP) in 2024, which contains binding commitments on data flows. The Protocol will enter into force 30 days after the ratification process is completed, and State Parties will have a further five years to align their national laws with the new commitments. See Sucker & Beyleveld (2025).

[5] See Convention for the protection of individuals with regard to automatic processing of personal data, Council of Europe, 28 January 1981, European Treaty Series, No. 108.

[6] The commitment can be found in Section 2.1 of the Agreement between the United States of America and Argentina on Reciprocal Trade and Investment, which was signed in February 2026 and will enter into force 60 days after both countries notify each other of the completion of their respective domestic legal requirements.

Figure 2: Participation in at least one agreement with binding commitments on free data flows, 2024.

Source: Author based on data from the TAPED database, retrieved on 30 November 2025, and indicator 6.5 of the DTI database retrieved on 20 February 2026. Note: for the calculation, I consider only agreements which have entered into force. Countries in grey are not included in the database.

Conclusions

The global landscape of cross-border data flows is characterised by rising fragmentation, both in restricting and enabling policies, resulting in a spaghetti bowl of commitments that increases uncertainty for businesses despite the crucial role of data in today's trade architecture. This fragmentation can hardly be overcome amid current geopolitical instability, though there are signs of potential coordination at the regional level. However, for these efforts to bear fruit and provide the basis for a multilateral agreement, governments must be willing to agree on rules that prioritise privacy-by-design, security-by-design and shared governance over sovereign claims of regulatory control, while ensuring that data is not concentrated in the hands of a few powerful players.

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Cross-border Data Flows Regulation in Digital Economy Agreements

María Vásquez Callo-Müller

Introduction

Data has become an essential factor of production and a prerequisite for growth and innovation. As a result, data policies are at the heart of the current economy and, in particular, of international trade policymaking, including digital trade. Within the latter, the role of cross-border data flows (CBDF) takes centre stage for a simple reason: digital trade is highly dependent on the seamless movement of data across borders. Domestic policies affecting CBDF can affect and reduce trade flows (Ferracane, 2021). Restrictions on CBDF can be applied for a variety of reasons, including privacy and national security justifications. When such restrictions go beyond legitimate public policy reasons and are discriminatory, they can undermine market openness and impose high costs on enterprises, for instance, due to limitations imposed by the use of cloud storage or other third-party services (e.g., software) that would significantly affect their operations. The costs of such measures are borne not only by foreign businesses but also by local ones, impacting the economy as a whole.

One way to address challenges to CBDFs is through Digital Economy Agreements (DEAs), a new type of trade agreement that specifically regulates digital trade. All ten DEAs[7] concluded to date include CBDF provisions, including a ban on data localisation requirements. DEAs are a novel type of international treaty being advanced primarily by middle-sized economies with a high interest in facilitating digital trade. More recently, the European Union (EU) and the African Free Continental Trade Area (AfCFTA) have embraced the DEA trend, underscoring the centrality of digital trade frameworks as tools for deeper economic integration and growth, even in regions pursuing sovereignty-related policies, such as the EU's concept of strategic autonomy.

Cross-Border data flows regulation in digital trade law

The regulation of CBDF internationally, insofar as it relates to international trade, primarily occurs through Preferential Trade Agreements (PTAs). Although CBDF and related provisions on data localisation were first included in PTAs at the initiative of the United States (US) in the early 2000s, the first agreement with binding commitments on CBDF was the US-led Trans-Pacific Partnership (TPP) Agreement. The TPP established that "[e]ach Party shall permit the cross-border transfer of information by electronic means, including the personal information, when this activity is for the conduct of the

[7] US–Japan DTA; Digital Economy Partnership Agreement (hereinafter 'DEPA'); Australia – Singapore DEA; UK–Singapore DEA; Korea–Singapore DEA; UK–Ukraine DTA; AfCFTA Digital Trade Protocol and the Annex on Cross-Border Data Transfer; EU–Singapore DTA; EU–Korea DTA; EFTA–Singapore DEA.

business of a covered person”.[8] This was complemented by another mandatory obligation banning the localisation of computing facilities: “[n]o Party shall require a covered person to use or locate computing facilities in that Party’s territory as a condition for conducting business in that territory”.[9] Both provisions are subject to exceptions. This liberal approach to CBDF (in the sense of mandating the free flow of data) was also taken up by the Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP), which retained the TPP’s e-commerce chapter, making this US-led/CPTPP template a milestone in the governance of CBDF.

In other regions, such as the EU, treaty provisions on CBDF and data localisation in PTAs are relatively recent developments. The EU–UK Trade and Cooperation Agreement was the first agreement in which the EU committed to binding data provisions. This approach to regulating CBDF (by also embedding a ban on data localisation restrictions within them) has become a mainstream approach in the EU’s trade agreements. For instance, the EU agreements with New Zealand and Japan, as well as the recently concluded DEAs with Singapore and South Korea, include CBDF obligations,[10] notwithstanding an evolving approach to the incorporation of exceptions to these commitments.

Several recent developments are noteworthy. To begin with, addressing CBDF provisions in international agreements, coupled with data localisation commitments, is no longer exclusively a US domain. In fact, the US retreat from global digital trade rulemaking has created an opportunity for other actors to fill the vacuum left by Washington. Singapore, for example, is an avid promoter of CBDF obligations, having negotiated the most CBDF and data localisation obligations to date. These are found in 16 PTAs that Singapore concluded bilaterally and as part of the Association of Southeast Asian Nations (ASEAN).

Furthermore, new forms of treaty-making have emerged since 2019, namely the DEAs, which exclusively regulate the digital economy, and mainstream provision on CBDF. Indeed, although CBDF remains a contested issue in international negotiations, there is a growing tendency to include more commitments in this regard (see Figure 1).

Cross-Border data flow regulation in digital economy agreements

All ten existing DEAs to date include provisions on CBDF.[11] However, these provisions include exceptions for legitimate public policy objectives or adopt the tests under the WTO general exceptions of Article. XX GATT 1994 and Art. XIV GATS. In addition, all DEAs

[8] Trans-Pacific Partnership Agreement, art. 14.11.2 (hereinafter ‘TPP’).

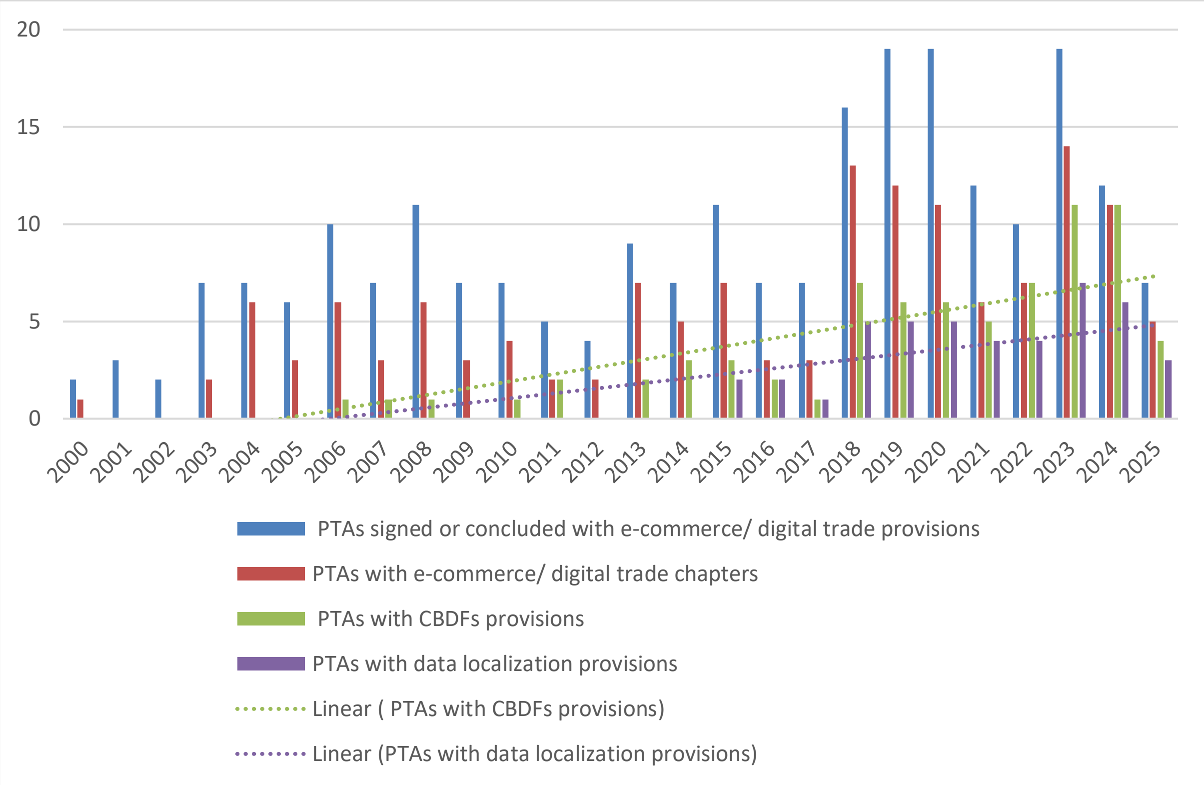
[9] TPP, art. 14.13.2.

[10] EU–New Zealand FTA, art. 12.4; EU–Japan FTA as amended by Article 3 EU–Japan Protocol, art.8.81; EU–Singapore DTA, art. 5; EU–Korea DTA, art. 5.

[11] US–Japan DTA, art. 11; DEPA, art. 4.3; Australia – Singapore DEA, art. 23; UK–Singapore DEA, art. 8.61-F; Korea–Singapore DEA, art. 14.14; UK–Ukraine DTA, art. 132-K; AfCFTA Digital Trade Protocol, art. 20 and the Annex on Cross-Border Data Transfer; EU–Singapore DTA, art. 5; EU–Korea DTA, art. 5; EFTA–Singapore DEA, art. 36-M.

prohibit data localisation requirements.[12] Despite these common denominators, DEAs show variation, with two distinct patterns, depending on whether the EU participates.

Figure 1: Evolution of Cross-border Data Flows and Data Localisation Commitments in PTAs and DEAs (2000-2025)



Source: Authors' own elaboration based on the TAPED dataset (Burri, M., & Vásquez Callo-Müller, M., 2026)

As per the DEAs where the EU does not participate, the Digital Economy Partnership Agreement (DEPA) is an influential example of CBDF regulation. DEPA is considered state-of-the-art in digital trade regulation, with many countries seeking to accede to it. Art. 4.3(2) of DEPA expressly requires parties to allow CBDF, provided that the activity pertains to the business operations of the concerned person. While this provision establishes a binding obligation, it also acknowledges that each party may maintain its own regulatory requirements concerning such transfers.[13] This approach to allowing CBDF is similar to the provisions in the CPTPP, which is unsurprising as all DEPA signatories are also CPTPP parties. The provision on data flows in DEPA was made subject to dispute settlement only through the DEPA amending protocol, which entered into force on 19 May 2024, showcasing a cautious approach to regulating this issue at the

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[12] US–Japan DTA, art. 12; DEPA, art. 4.4; Australia – Singapore DEA, art. 24; UK–Singapore DEA, art. 8.61-G; Korea–Singapore DEA, art. 14.15; UK–Ukraine DTA, art. 132-L; AfCFTA Digital Trade Protocol, art. 22; EU–Singapore DTA, art. 5; EU–Korea DTA, art. 5; EFTA–Singapore DEA, art. 36-M.

[13] DEPA, art. 4.3(1) read together with arts. 5 and 8(b) of the DEPA Protocol.

international level, first from an unenforceable commitment to a binding commitment enforceable under dispute settlement.

Concerning the DEAs to which the EU is a party, the EU Digital Trade Agreements (DTAs) with Singapore and Korea differ from the rest of the DEAs by including a fully-fledged language on exceptions, which, to some extent, reflects concerns about EU strategic autonomy. In contrast to other DEAs, which provide that measures to protect legitimate public policy objectives should not amount to arbitrary or unjustifiable discrimination or a disguised restriction on trade, and should not impose restrictions greater than are required to achieve the legitimate objective,[14] the EU DEAs go a step further. Namely, they clarify, through a non-exhaustive list, what is understood by legitimate public policy objectives and mention a great variety of measures that qualify. Moreover, EU DEAs also clarify that the interpretation of those objectives should take into account the 'evolving nature of digital technologies'.[15] Article 5 of the EU–Singapore DTA illustrates the type of exceptions included in EU DTAs that subject a binding CBDF obligation to a legitimate public policy exception, and which shall be interpreted:

> 'in an objective manner and [which] shall enable the pursuit of objectives such as the protection of public security, public morals, or human, animal or plant life or health, the maintenance of public order, the protection of other fundamental interests of society such as online safety, cybersecurity, safe and trustworthy artificial intelligence, or protecting against the dissemination of disinformation, or other similar objectives of public interest, taking into account the evolving nature of digital technologies'.

This detailed exception, reaffirmed in the textual proposal for the EU-Korea DTA, can only be described as an 'omnibus' exception—an overarching, deliberate approach to preserve regulatory space for a variety of open-ended policy reasons.

This approach is an important aspect to note, as it clearly exemplifies a nuanced way to promote trade integration while safeguarding regulatory space, and to some extent, associated concepts such as digital sovereignty. It also shows how countries and regions have sought to imprint their own way into the regulation of CBDF, differing prominently from the CPTPP model. This 'evolving' approach to the governance of CBDF in DEAs (Vásquez Callo-Müller and Sucker, 2025) is also evident in the AfCFTA Digital Trade Protocol (discussed in another contribution to this edited collection), which has binding provisions on both CBDF[16] and data localisation,[17] but with a clear orientation to primarily facilitate CBDF within the African continent.

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[14] DEPA, art. 4.3 (3) read together with arts.5 and 8(b) of the DEPA Protocol.
[15] See EU–Korea DTA and EU–Singapore DTA, both respectively arts. 5.
[16] AfCFTA Digital Trade Protocol, art. 20.
[17] AfCFTA Digital Trade Protocol, art 22.

Conclusions

PTAs have paved the way for the inclusion of CBDF and related bans on data localisation in digital trade rulemaking. This is clearly exemplified by the fact that all DEAs to date include binding provisions on both aspects. The emerging pattern among the DEAs' parties, which include a mix of developed, developing, and least developed countries, is to agree to these types of commitments to facilitate digital trade and economic integration. Concurrently, there is still a contested evolution in DEAs (as well as in PTAs) on how to preserve policy space in this regard, particularly considering the rapid advancement of digital markets and technologies. The EU approach provides one approach to the design of escape clauses to otherwise binding CBDF and data localisation commitments, reflecting, to some extent, digital sovereignty considerations. Other regions, such as Africa, show a similar effort to carve out one's own path in addressing these issues. Such tailored approaches will continue to expand in digital trade rulemaking, showcasing the importance of how countries and regions seek to retain oversight over the free movement of data.

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Data, Trade & Cross-Border Flows: Trends in the Middle East and North Africa Region

Georgios Dimitropoulos

Introduction

It is now widely acknowledged that the regulation of personal data is structured around three dominant regulatory models—often described as competing "data realms" (Aaronson and Leblond, 2018), "digital empires" (Bradford, 2020) or "digital kingdoms" (Gao, 2023). These are commonly associated with the European Union's comprehensive and rights-based regulatory approach, China's approach emphasising data sovereignty and national security, and the United States' market-oriented model. Each model reflects distinct political, industrial, and trade policy objectives (Dimitropoulos, 2025).

These dynamics are closely linked to broader debates about the future of the open internet, particularly the extent to which cross-border data flows (CBDFs) can be sustained in an increasingly fragmented regulatory environment. Against this backdrop, the interplay between trade and data governance in the Arab countries of the Middle East and North Africa (MENA) region has received comparatively limited scholarly attention. Patterns in domestic data laws governing CBDFs across the region, alongside evolving approaches within Preferential Trade Agreements (PTAs), illustrate how states navigate the tension between openness and control. In this context, CBDFs are not only a matter of trade but also a determinant of whether digital ecosystems remain interconnected or become more territorially segmented. These developments can therefore be understood in relation to the wider global landscape shaped by the three dominant models of data governance and digital trade.

Cross-Border data flows in domestic laws

The MENA region demonstrates a degree of internal diversity in the regulation of CBDFs—reflecting differences in country size, trade and industrial policies, and economic and digital policy priorities. Some countries, such as Iraq and Yemen, do not currently have comprehensive national laws governing personal data processing and cross-border data flows. Most countries in the MENA region pursue a European-style approach, while the other two approaches are also represented. Four major patterns have emerged across the region.

Adequacy and/or safeguards as the default rule for CBDF

Across the MENA region, international transfer-related provisions in new data laws require adequacy of level of protection decisions and/or appropriate safeguards as the default rule for permitting CBDFs—along with some narrow exceptions. Egypt's Personal Data Protection Law, Law Number 151 of 2020, requires that any data transfers outside the country meet the security and protection standards set out in the law and be licensed by the Personal Data Protection Center. The law lists exceptions such as life and health, judicial cooperation, contracts, public interest, and money transfers—see Articles 14-16. Morocco's Law No. 09-08 of 2009 on the Protection of Individuals with Regard to the Processing of Personal Data requires that the destination country ensure an adequate level of protection for privacy and fundamental rights—see Articles 43-44. Jordan's Personal Data Protection Law No. 24 of 2023 provides that personal data shall not be transferred to any person outside the Kingdom, if the level of protection afforded to such data is lower than that stipulated in the law—see Article 15. Federal Decree-Law No. 45 of 2021 on the Protection of Personal Data of the United Arab Emirates (UAE) allows transfers to jurisdictions deemed adequate and approved by the Data Office. Transfers are also permitted based on standard contractual clauses, binding corporate rules, consent, contractual necessity, judicial cooperation, or public interest—see Articles 22-23.

Primacy for national security

Saudi Arabia is widely regarded as one of the most politically and economically influential countries in the region. The Saudi Personal Data Protection Law of 2021, which was amended in 2023, shows similarities to the GDPR's adequacy-based approach to cross-border data transfers, while giving explicit priority to national security and vital public interest considerations. Article 29(1) permits cross-border data transfers where they are necessary to comply with international treaty obligations, serve a public interest, or are required for the performance of a contract involving the data subject. However, Article 29(2)(A) explains that *"[t]he Transfer or Disclosure shall not cause any prejudice to national security or the vital interests of the Kingdom."* The national security clause is followed by an equivalent adequacy-of-protection clause in Article 29(2)(B). Article 29(3) explains that Article 29(2) *"shall not apply to cases of extreme necessity to preserve the life or vital interests of the Data Subject or to prevent, examine or treat disease."*

Primacy by default for CBDF

Qatar's data privacy law, on the other hand, allows, by default, the free flow of data in and out of the country. Article 15 of Law No.13 of 2016 on Protecting Personal Data Privacy in Qatar forbids any controller *"from taking any decision or measure that may limit the Cross-Border Data Flow."* The same article includes a restriction to this prohibition when the *"processing of such data is in breach of this Law, or where such processing may cause serious damage to the Personal Data or to the Individual's privacy."*

Sector-specific localisation common in the health and finance sectors

Even where data protection laws allow CBDFs, sectoral rules may require data localisation or impose tighter controls on data transfers or data to be stored and retained in the country (Ferracane, 2024). In the UAE, for example, health-related data and information may not be stored, processed, generated, or transformed outside the UAE unless an approved exception applies, as determined by the competent federal health authorities—see Article 13 of Federal Law No. (2) of 2019 Concerning the Use of the Information and Communications Technology in Health Fields. Customer financial data, in the UAE, for example, is required to be stored and maintained in the State—see Articles 10(5) and (6) of the Stored Value Facilities (SVF) Regulation, C 6/2020.

Taken together, these developments point to a region that is not converging on a single model of cross-border data governance but rather producing a hybrid, layered regulatory landscape. While adequacy-based frameworks inspired by the European model are emerging as the baseline, they are frequently recalibrated through national security exceptions, sector-specific localisation requirements, and, in some cases, more permissive regimes favouring data mobility. This results in a fragmented yet patterned approach, in which states seek to balance participation in the global digital economy with concerns about sovereignty and industrial policy. From an open internet perspective, this hybridisation suggests that the future of CBDF in the MENA region will be shaped less by wholesale adoption of any single external model, and more by selective adaptation—producing a form of conditional openness in which data flows remain possible, but increasingly mediated by political, regulatory, and sectoral constraints.

Cross-Border data flows in preferential trade agreements

While there has been comparatively less activity in concluding international trade agreements in the region, some PTAs in the MENA region include dedicated e-commerce or digital trade chapters. Many of them, as well as those under negotiation, address CBDFs in these chapters, including through provisions on data localisation requirements and restrictions on data transfers.

GCC level

Some older GCC agreements, including the GCC-Singapore Free Trade Agreement (entered into force 2013, Chapter 7) and the GCC-European Free Trade Association Free Trade Agreement (entered into force 2014, Annex XVI on Electronic Commerce), include dedicated electronic commerce provisions, but do not address CBDFs or data localisation requirements. Instead, they focus primarily on the non-discriminatory treatment of digital products, consumer confidence in electronic transactions, and related cooperative activities. Stronger language and more obligation-imposing provisions will likely appear in the GCC-Republic of Korea Free Trade Agreement (signed in 2023 and not yet in force), as well as in the UK-GCC FTA, negotiations for which are still ongoing.

Other PTAs

The United States-Bahrain Free Trade Agreement (entered into force in 2006) includes a dedicated Chapter on Electronic Commerce (Chapter 13). An identical chapter appears in the United States-Oman Free Trade Agreement (entered into force 2009, Chapter 14). These are brief chapters that focus on promoting electronic commerce through measures such as prohibiting customs duties on electronically transmitted digital products and ensuring non-discriminatory treatment of digital products; they contain no provisions addressing cross-border data flows, data localisation, or related data governance obligations. A similar approach is largely followed in the more recent Trade and Economic Partnership Agreement between the Republic of Türkiye and the State of Qatar (entered into force in 2025 – but originally signed in 2018, Chapter 12).

The UAE stands out as the most active regional signatory to Comprehensive Economic Partnership Agreements (CEPAs), having concluded approximately 15 of them. UAE's CEPAs invariably feature dedicated digital trade chapters. The UAE has also formally applied for accession to the Digital Economy Partnership Agreement (DEPA) and has commenced the accession process. At the same time, most of the CEPAs concluded by the UAE use soft, non-binding language regarding restrictions on cross-border flows of information. They generally employ best endeavour-based formulations—rather than hard, enforceable commitments prohibiting data localisation requirements or mandating unrestricted free cross-border transfer of data as in other PTAs. India-UAE CEPA, Article 9.11, for example, provides that *"[t]he Parties recognise the importance of the flow of information in facilitating trade, and acknowledge the importance of protecting personal data. As such, the Parties shall endeavour to promote electronic information flows across borders subject to their laws and regulatory frameworks."* Indonesia-UAE CEPA, Article 9.12 provides that the Parties *"shall endeavor to consider the possibility of easing barriers to electronic information flows across borders and of promoting cross-border flow of information, to the extent consistent with its laws and regulations."* Digital trade is also generally exempted from dispute settlement.

A notable exception, however, is the Australia-UAE CEPA (entered into force 2025), which represents the UAE's first agreement establishing binding obligations on cross-border data transfers: under Article 12.16, neither party may prohibit or restrict transfer of information by electronic means—including those involving personal data—where necessary for the conduct of business, subject to a public policy exception requiring that any derogating measure be non-discriminatory and not impose restrictions beyond what is necessary to achieve its stated objective.

Sector-specific provisions

Some trade agreements include dedicated financial services chapters or annexes that safeguard the ability of financial service suppliers to transfer and process financial information when necessary for the ordinary course of their business. These provisions simultaneously preserve the right of the Parties to maintain prudential regulation and confidentiality safeguards. For instance, the United States-Oman FTA lists the "provision and transfer of financial information" and related data-processing services as covered cross-border financial services under its financial services chapter (Chapter 12), thereby incorporating financial information flows into its market access commitments.

More recent agreements, particularly certain UAE CEPAs, adopt a similar yet somewhat more explicit approach. They recognise the provision and transfer of financial information, as well as financial data processing (and related software), as covered financial services—for example, under Annex 8-A of the UAE–Korea CEPA.

Taken together, these developments suggest that cross-border data governance in MENA trade agreements remains incremental and rather cautious. While digital trade chapters are becoming more common, they largely avoid binding commitments on CBDF or prohibitions on data localisation. Instead, they rely on soft-law formulations and best-endeavour clauses, preserving regulatory flexibility while signalling openness to digital trade. This reflects a broader regional approach in which states seek to engage in the global digital economy without fully constraining domestic policy space. From the perspective of the open internet, this results in a form of "managed openness," where data flows are enabled in principle but remain contingent on national legal frameworks and sector-specific safeguards.

Conclusion

Overall, these developments suggest that the MENA region is not moving toward a unified model of data governance or digital trade but rather toward a differentiated, strategically adaptive framework shaped by varying regulatory capacities, economic and industrial development priorities, and geopolitical alignments.

While GCC countries are emerging as more active digital trade rule-makers, other states in the region remain more cautious in their engagement. All six members of the GCC are also active participants in the Joint Statement Initiative (JSI) on E-Commerce and are among those who supported the July 2024 stabilised text of the Agreement on Electronic Commerce (World Trade Organization, 2024). In the wider MENA region, several other Arab countries, including Egypt, Morocco, Jordan, and Tunisia, are not participants in the JSI (European Parliament, 2024).

Overall, the MENA region illustrates that, notwithstanding linguistic and cultural affinities, trade and data-flow rules remain context-sensitive. Across both domestic regulation and international trade policy, a common pattern emerges: openness to cross-border data flows is largely conditional, mediated by national security considerations, sector-specific restrictions, and/or flexible legal formulations in trade agreements. From the perspective of the open internet, this points to a trajectory of managed interdependence rather than full fragmentation or full liberalisation, in which connectivity is preserved but remains subject to layered governance constraints. Ultimately, the MENA experience underscores that the future of cross-border data flows—and the openness of digital ecosystems more broadly—will depend less on convergence around a single global model and more on how regions navigate the balance between regional and global integration, sovereignty and industrial policy.

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Cross-Border Data Flows in Africa: Beyond Free Flow and Adequacy

Franziska Sucker

Introduction

The global governance of cross-border data flows (CBDF) is characterised by rising fragmentation, both in restricting and enabling policies (Ferracane, this volume). The cumulative effect is a “spaghetti bowl” of restricting and enabling regimes across different legal orders, contributing to uncertainty for businesses despite the centrality of data to digital trade (Ferracane, this volume). Within this fragmented environment, digital trade agreements (DTAs) have developed distinct approaches to governing CBDF. Early digital trade chapters, most prominently in the Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP), established binding obligations to permit cross-border data transfers and prohibit data localisation requirements, subject to general exceptions (Vásquez Callo-Müller, this volume). More recent agreements, including those concluded by the European Union (EU), integrate transfer commitments into broader rights-based regulatory frameworks and detailed public policy safeguards. Other agreements rely on softer or best-endeavour formulations, while multilateral negotiations at the WTO have yet to consolidate these approaches into comprehensive binding disciplines (Vásquez Callo-Müller, this volume).

Fragmentation is equally evident on the African continent. Domestic data protection frameworks differ significantly in scope, institutional design, and conditions governing cross-border data transfers, with some countries adopting adequacy-style or safeguard mechanisms, others imposing sector-specific localisation requirements, (Ferracane and van der Marel, 2024; Ferracane et al., 2025) and several lacking comprehensive enforcement capacity. At the same time, African countries’ participation in trade agreements that contain CBDF commitments has been very limited (Vásquez Callo-Müller and Sucker, 2025), and regional economic communities have not established continent-wide disciplines in this area (Sucker and Beyleveld, 2023).

Against this backdrop, the adoption of the Protocol to the Agreement Establishing the African Continental Free Trade Area (AfCFTA Agreement) on Digital Trade (AfCFTA DTP) in February 2024 (AfCFTA DTP, 2024) represents a structural shift. The AfCFTA State Parties, currently 49 (TRALAC, 2026), agreed to the first continent-wide legal framework governing digital trade under the AfCFTA, which will form part of the AfCFTA Agreement (AfCFTA Agreement, 2019) upon entry into force (AfCFTA Agreement, Art. 8(3); Techpoint, 2025). In February 2025, the Protocol was complemented by the adoption of eight detailed Annexes, including the Annex on Cross-Border Data Transfers (CBDT-Annex) (African Union, 2025).

CBDF governance under the AfCFTA Digital Trade Protocol reflects a distinct and hybrid legal model that both aligns with and departs from dominant treaty approaches. The

CBDT Annex establishes a layered architecture built around transfer obligations subject to structured public policy exceptions, embedded continental data protection standards that condition and discipline those obligations, and operational mechanisms for trust, reciprocity and interoperability linked to digital development. Together, these elements recalibrate the balance between data flow, regulatory autonomy and developmental objectives. Situated within the Protocol's broader framework on digital trade facilitation and online trust, this CBDF regime positions the AfCFTA as an ambitious but still evolving model of development-oriented digital trade governance.

Cross-border data transfer obligations and structured exceptions

The AfCFTA DTP sets out the core rules governing cross-border data transfers. Article 20 provides that State Parties shall permit the cross-border transfer of data, including personal data, where the activity constitutes digital trade. Article 22 complements this rule by prohibiting the imposition of localisation requirements for computing facilities as a condition for conducting digital trade. Both provisions are subject to public policy exceptions structured along GATT/GATS-style Chapeau and necessity requirements: restrictions must not constitute arbitrary or unjustifiable discrimination, must not amount to disguised restrictions on trade, and must not be more restrictive than necessary to achieve a specified public policy objective. (AfCFTA DTP, Arts. 20(2) and 22(2); GATT 1994, Art. XX; GATS, Art. XIV) This reflects the standard formulation found in many DTAs, combining transfer obligations with public policy exceptions. (DEPA, Arts. 4.3 and 4.2; CPTPP, Arts. 14.11 and 14.13) Three features, however, merit closer attention.

First, while many DTAs apply data transfer obligations to business activity more broadly (DEPA, Arts. 4.3 and 4.2; CPTPP, Arts. 14.11 and 14.13), under the AfCFTA DTP, they are limited to activities constituting digital trade. This is consistent with the Protocol's overall scope (AfCFTA DTP, Art. 3), but its practical reach is uncertain and will depend on future interpretation and practice. One interpretative question concerns the definition of digital trade itself. The joint OECD/IMF/WTO/UNCTAD definition broadly encompasses all trade that is digitally ordered and/or delivered. (OECD et al., 2023) By contrast, the AfCFTA DTP focuses only on digitally enabled trade in goods and services. (AfCFTA DTP, Art. 1(g)) This narrower formulation may exclude digitally enabled economic activities that would fall within the broader understanding of digital trade but do not clearly qualify as trade in goods or services. However, even if digital trade were interpreted broadly to capture all digitally enabled trade, CBDF that arise merely as an incidental element of ordinary business activity would still fall outside the scope of the transfer obligation.[18] In such cases, Articles 20 and 22 would not apply.

[18] Examples: a manufacturing company transferring HR data to a regional headquarters; a multinational firm sending internal compliance or accounting data across borders; a logistics firm transmitting shipment data for physical goods ordered offline; a bank transferring customer data for internal risk management unrelated to a digitally delivered.

Second, the CBDT-Annex establishes a detailed exception framework. Its non-exhaustive list of legitimate public policy objectives expressly includes essential security interests,[19] which, unlike under Article XXI of the GATT, are not framed in self-judging terms but remain subject to the GATT/GATS-style chapeau and necessity discipline of Article 23 of the Annex.[20] This design limits the scope for unilateral invocation of security-based restrictions. (RCEP, Arts. 10.15, 12.14(3), 12.15(3) and 17.13)

The exception framework is further structured by its interaction with the data protection obligations set out in Part II of the Annex and in Article 21 of the Protocol. This interaction operates in a bidirectional manner. Where the shared regional standard is satisfied, the imposition of restrictions on CBDF is precluded (CBDT-Annex, Art. 16(4)), including restrictions that might otherwise be justified under the public policy exceptions. Only where those standards are not met may restrictions be adopted. (CBDT-Annex, Art. 25) The explicit permission to restrict CBDF in case of non-compliance with the shared regional standard, without separate qualification, implies that such non-compliance in itself constitutes a legitimate public policy objective under Article 24. Even then, however, any restriction remains subject to the chapeau and necessity requirements. (CBDT-Annex, Art. 23(2)) The invocation of exceptions is therefore contingent upon non-compliance with shared regional obligations and does not operate as an independent discretionary power.

In addition, Article 26 of the CBDT-Annex introduces detailed rules governing sensitive data. It defines categories of sensitive data and specifies circumstances in which transfers must not be prohibited. Notably, one of the two ‘must-allow’ lists incorporates an additional necessity requirement linked to the facilitation of digital trade (CBDT-Annex, Art. 26(2)), thereby adding to the proportionality analysis. This attention to sensitive data reflects an effort to enhance coherence in an area where domestic regulation across African States differs significantly, while simultaneously constraining the scope for restrictive measures.

Taken together, these elements render the AfCFTA DTP exception regime more structured and coordinated than models relying solely on broadly framed general exceptions. Regulatory autonomy is preserved, but its exercise is embedded within a detailed continental framework that narrows discretion while enhancing predictability and intra-regional consistency.

Third, Article 22(3) introduces a developmental dimension by obligating State Parties to encourage and support the establishment and use of regional computing facilities.[21] By linking data flows to the development of regional digital infrastructure, the Protocol pairs

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[19] CBDT-Annex, Art. 24. This contrasts with Articles 20 and 22 itself, where public policy aims and essential security interests are mentioned separately. The same applies to AfCFTA DTP, Art 4. See Sucker, F., ‘The AfCFTA Digital Trade Protocol’, in Birhanu, F. M., Hailu, M. B., Klimke, R. & Tietje, C. (Eds.), Handbook on the African Free Trade Area. Routledge, parts 4.2 and 7.1.2.

[20] CBDT-Annex, Art. 23 (2) repeats the Chapeau and necessity requirements prescribed in AfCFTA DTP, Arts. 20(2) and 22(2).

[21] See Sucker, (n.d.) in footnote 19 above, part 7.1.1.

liberalisation with continental capacity-building and aligns openness with broader industrial and integration development.

Overall, the AfCFTA DTP does not simply replicate liberalisation disciplines of CBDFs but embeds them within a structured development-oriented continental framework.

Embedding continental data protection standards in a digital trade agreement

A particularly distinctive feature of the AfCFTA DTP is its incorporation of shared substantive principles and standards for personal data protection directly into the legal architecture governing CBDF.

Article 21(1) of the Protocol requires State Parties to adopt or align their domestic legal frameworks with the principles set out in Part II of the CBDT-Annex. These include lawful and transparent processing, data minimisation, purpose and storage limitation, accuracy, security, confidentiality and integrity, and accountability.[22] While these principles resemble those found in Article 5 of the EU's General Data Protection Regulation (GDPR 2016/679) (GDPR), they are not defined in detail in the Annex, thereby leaving discretion for implementation within national legal systems.

Part II further outlines core data-subject rights, including access, correction, erasure and objection to processing, and establishes safeguards against misuse, data protection by design, and remedies in the event of breaches. Article 15 of the CBDT-Annex complements this framework by requiring Parties to promote interoperability and to harmonise national data-protection laws in line with the Annex principles. While harmonisation is framed as a longer-term objective, the provision reinforces regulatory convergence and legal predictability across the continent.

Crucially, the CBDT-Annex links these standards directly to the permissibility of cross-border transfers. Where the agreed standards are met, State Parties must refrain from restricting transfers on data-protection grounds (CBDT-Annex, Art. 16(4)). Where they are not met, restrictions may be imposed, subject to the necessity and chapeau disciplines (CBDT-Annex, Art. 25). Transfers are therefore not contingent upon unilateral adequacy determinations or case-by-case approval; instead, they operate against a collectively agreed baseline of protection embedded within a framework of necessity, proportionality and non-discrimination. This design shifts the regulatory logic from conditional market access to coordinated regulatory alignment.

What is significant is not merely the existence of shared continental standards, but their structural integration into the CBDF regime within a trade agreement. This is highly unusual. Most digital trade chapters, like the CPTPP and DEAs, limit themselves to general commitments to 'adopt or maintain' a legal framework for personal data protection, without specifying its substantive principles (CPTPP, Art. 14.8(2); DEPA, Art. 4.2(2)). EU trade agreements, while closely linked to the GDPR, do not codify GDPR-style

[22] See Vásquez Callo-Müller and Sucker (2025) , p. 12, part D.II.2.

standards within the treaty text itself; instead, cross-border transfers are governed externally through adequacy determinations and related mechanisms.

Operational trust: facilitation, reciprocity and interoperability

The CBDT-Annex further operationalises CBDF through a set of trust and accountability mechanisms designed to ensure secure and predictable data flow. State Parties must adopt and maintain reasonable and appropriate measures to ensure uninterrupted and secure data flows (CBDT-Annex, Art. 16(3)) and to identify and remove barriers to transfers (CBDT-Annex, Art. 16(5)). Transferred data must receive an equivalent level of protection to that afforded domestically (CBDT-Annex, Art. 17). This reciprocity safeguard anchors cross-border trust without reliance on unilateral adequacy determinations. State Parties are also obliged to publish data-protection measures, notify legal changes, and cooperate in enforcement and capacity-building (CBDT-Annex, Arts. 20 and 21). These obligations reduce regulatory uncertainty and support convergence in practice. The architecture is reinforced by commitments to promote interoperability among legal frameworks to facilitate data flows while protecting personal data, and to permit mutual recognition and reciprocal arrangements (CBDT-Annex, Arts. 15(1) and (2)).

These obligations are complemented by a list of concrete transfer mechanisms that State Parties are encouraged to establish, including regional and local data centres, codes of conduct and certification systems, recognition of regulatory outcomes, and tailored instruments to support micro, small and medium-sized enterprises, women, youth and other underrepresented groups. The Annex also recognises that data is not merely a trade enabler but also a development resource (CBDT-Annex, Art. 22). It thus incorporates a ‘data for development’ orientation, requiring State Parties to facilitate public-benefit data use, data sharing for research, innovation and entrepreneurship, and the development of regional digital infrastructure. In doing so, the Annex frames data governance as a tool of digital industrialisation and inclusion, rather than solely as a matter of privacy protection or trade facilitation.

Finally, the Annex further extends classical trade disciplines into the data flow domain. State Parties must accord national treatment and most-favoured-nation treatment to ‘like data’ (CBDT-Annex, Art. 18), signalling that data itself is treated as a subject of trade discipline rather than merely an input into services or digital products.

Broader facilitation and online trust mechanisms

The CBDT-Annex does not operate in isolation. Broader provisions of the DTP reinforce the practical conditions for CBDF. Part III of the AfCFTA DTP governs digital trade facilitation. Across almost all of its provisions, technological neutrality and interoperability recur as structuring requirements. These are reflected in commitments to mutual recognition of electronic authentication and signatures, paperless trading, electronic invoicing, and customs cooperation, as well as the mutual recognition of digital systems and standards. Part III further obliges State Parties to adopt domestic legal frameworks that enable these forms of cross-border digital interaction (AfCFTA DTP, Arts. 8–19).

Together with the transparency, cooperation and reciprocity safeguards discussed above, these provisions connect technical and regulatory trust. These rules ensure that digital documents, identities and commercial processes are legally valid across borders, thereby reducing transactional friction.

The Protocol further embeds online trust through commitments on cybersecurity, consumer protection and unsolicited communications, as well as limitations on mandatory disclosure of source code subject to public-interest exceptions (AfCFTA DTP, Part V). The Annex on Online Safety and Security introduces additional domestic regulatory obligations, including duties imposed on enterprises and platforms. Collectively, these measures create an enabling ecosystem within which cross-border data flows can function securely and predictably.

Conclusion

The AfCFTA DTP represents the first continent-wide effort to regulate CBDF within a unified regional trade framework. It does not merely replicate existing liberalisation templates. Rather, it combines a general obligation to permit data flows with a structured exception regime, embeds shared continental data-protection standards directly within the transfer architecture, and operationalises trust through reciprocity, interoperability and facilitation mechanisms.

In a global landscape marked by fragmentation and competing digital governance models, the AfCFTA demonstrates that openness can be disciplined, coordinated and development-sensitive. CBDF are neither left to unqualified free flow nor subjected to unilateral adequacy-style control. Instead, they are situated within a rule-based continental framework that narrows discretion, enhances predictability and links digital integration to infrastructure development and inclusive participation.

Whether this model will influence multilateral outcomes depends on implementation and engagement. Its immediate significance, however, lies in demonstrating that regional interoperability among diverse developing economies can be anchored in legal coordination, shared standards and structured regulatory autonomy.

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IV. Openness, Governance and Growth

Internet openness has historically driven productivity, innovation, and trade expansion. Yet, the relationship between openness and economic growth remains under-theorised in digital policy research. This section examines how institutional and regulatory choices influence the economic outcomes of openness, including competitiveness, knowledge diffusion, and entrepreneurship. It examines the political economy of digital closure, exploring both its short-term adaptive strategies and its structural costs for growth and development. It also assesses the role of governance at national, regional, and global levels in safeguarding openness as a prerequisite for sustainable economic progress.

Three contributions approach these questions through specific political contexts. Evrim Görmus examines how commercial spyware circulates between EU foreign policy goals and member-state commercial interests, showing how the EU-GCC relationship exposes the limits of normative commitments to internet freedom when economic and security interests pull in opposite directions. Nori Katagiri interrogates the assumed alignment between democracy and the open internet, arguing that democratic governments have both cooperated with and undermined openness depending on political context. Dechun Zhang introduces the concept of "negotiated openness" to describe how regimes manage digital participation through selective access and conditionality, extending this into the phenomenon of "soft authoritarianism"—governance that constrains the internet without the overt architecture of a firewall.

The final two contributions shift toward political economy. Ildar Daminov's case study of the Belarusian ICT sector examines how firms and connectivity networks adapted to acute political disruption and sanctions pressure in the 2020s, revealing both the resilience of commercial relationships and the structural limits of digital infrastructure when entangled with political crisis. Mallory Knodel argues that the internet commons has been progressively enclosed by "rentierism"—the capture of value through infrastructural bottlenecks rather than productive innovation—and that regulatory tools designed for ex post conduct review are insufficient to address the structural conditions that enable it.

Circulating Control: Spyware and the Struggle for Internet Openness in EU–GCC Relations

Evrim Görmüş

Introduction

Spyware is malicious software that enables unauthorised remote access to an internet-enabled device for surveillance or data extraction (Korzak, 2025). The transnational diffusion of spyware technologies has expanded the scope for their misuse, leading to severe human rights violations and national security risks across authoritarian and democratic regimes. The 2025 report by the Atlantic Council's Cyber Statecraft Initiative identified 561 entities operating in the global spyware market across 46 countries between 1992 and 2024, with 56 % of investors concentrated in Israel, the United States, Italy, and the United Kingdom (Roberts et al., 2024). A complimentary SIPRI mapping exercise identified 43 companies across 18 states. While production is geographically dispersed, 58 % of firms are concentrated in three states—India, Israel, and Italy. Overall, 44 % are based in Europe, 33 % in the Middle East (Israel, Turkey, and the United Arab Emirates), and 21 % in Asia and Oceania (Australia, China, India, and Singapore) (Bromley & Maletta, 2025).

International efforts to govern spyware

States have pursued national, regional, and multilateral initiatives to promote accountability and standardise export controls on spyware technologies. At the multilateral level, the Wassenaar Arrangement on Export Controls for Conventional Arms and Dual-Use Goods and Technologies was established in 1996 to promote "transparency and greater responsibility" in transfers of military and dual-use items and continues to serve as the principal framework for coordinating controls on surveillance and intrusion software among forty-two participating states (Wassenaar Arrangement, 2005).

After the Arab uprisings, evidence linking Western spyware firms to human rights abuses in the Middle East contributed to the inclusion of surveillance tools in the Wassenaar Arrangement's dual-use control list in 2013 (Korzak, 2025). Although the 2013 amendments to the Wassenaar Arrangement's dual-use control list reflected growing concern over the human rights implications of spyware, they have not stopped its proliferation. The continued misuse of spyware technologies by state actors in subsequent years highlights the inherent weaknesses of the Arrangement. Its nonbinding nature and the exclusion of major exporters, including China, Israel, the UAE, Cyprus, and Belarus, make it insufficient to address the growing proliferation and abuse of commercial spyware.

In response to growing concerns over the misuse of commercial spyware, several states have recently supported the Pall Mall Process, a multilateral initiative launched in 2024 and led by the United Kingdom and France. The process aims to address the proliferation and irresponsible use of commercial cyber intrusion capabilities by promoting greater coordination between national authorities, harmonising legal interpretations, and enhancing oversight of their trade and use, with particular attention to challenges posed by cloud-delivered surveillance technologies. While it represents an important step toward international norm-building, the Pall Mall Process remains a voluntary, politically driven initiative that relies on soft-law mechanisms rather than enforceable legal obligations (Ni & Jones, 2024).

At the regional level, the EU adopted several measures to regulate the export of spyware and cyber-surveillance under the recast 2021 Dual-Use Regulation (EUDUR) (Regulation EU 2021/821, 2021). The recast EUDUR introduced cyber-surveillance items as a distinct category and defined them as “dual use items specially designed to enable the covert surveillance of natural persons by monitoring, extracting, collecting or analysing data from information and telecommunication systems” (NATO CCDCOE, 2021). The regulation also introduced, in Article 5, a human-rights-based catch-all clause that enables authorities to require licenses for non-listed items, which could be potentially used for internal repression and/or in violation of human rights and international humanitarian law (Regulation EU 2021/821, 2021).

However, export controls under the EU’s Dual Use Regulation on spyware remain inadequate to prevent the global proliferation of such technologies through opaque trade agreements, which continue to undermine fundamental rights, democracy, and the rule of law. In the absence of binding rules and effective enforcement, the system continues to rely heavily on corporate self-regulation, a mechanism that is inherently unsustainable and bound to fail (Lubin, 2024, p. 843).

Such failure stems from the lack of incentives for businesses to enforce their own standards, the reliance on informed consumers for market oversight, and the ability of businesses to “utilise their technological prowess and superior knowledge to evade detection” (Yeung, 2015).

Beyond these regulatory weaknesses, the effectiveness of the EU’s export-control regime on spyware is structurally constrained by growing economic interdependence with authoritarian states, such as the Gulf Cooperation Council (GCC) countries. The governance of spyware thus reveals broader tensions in governing interdependence, where market integration and human rights protection increasingly collide.

The GCC nexus: spyware and europe’s structural dependencies

The widespread misuse of European-origin spyware technologies in GCC countries, targeting political opponents, journalists, and human rights activists, has sparked worldwide concern following investigations by journalists, civil society organisations, and research groups such as Citizen Lab and Amnesty International since the 2010s. The GCC countries, Bahrain, Kuwait, Oman, Qatar, Saudi Arabia, and the United Arab

Emirates (UAE), have leveraged advanced digital and surveillance technologies as tools for economic diversification, national security, counter terrorism, and smart governance, but also to consolidate authoritarian control and curb dissent. The AI Global Surveillance Index shows that GCC members exhibit exceptionally high levels of digital control through their employment of advanced surveillance tools, including spyware (Vlahos et al., 2022).

The GCC states are not members of the Wassenaar Arrangement and do not maintain formal dual-use export control regimes that regulate surveillance and intrusion software. Although cybercrime and data protection laws have proliferated since 2021, these legislations have mainly been designed to serve domestic objectives such as content control and data localisation, rather than to promote external accountability and transparency. This regulatory approach aligns with the GCC's broader position in international forums, where member states have supported Chinese-led UNGA resolutions that challenge restrictive dual-use export control regimes because they constrain technology transfer and impede innovation (Héau, 2025).

While GCC states have resisted restrictive export control regimes, the EU's regulatory framework on spyware has evolved in response to a series of scandals. The European Parliament's Committee of Inquiry investigating the use of Pegasus and equivalent surveillance spyware (PEGA Committee) issued a report in May 2023 that underscored the illegitimate use of spyware by both national and non-EU governments for political and even criminal purposes. The PEGA report concluded that "Union export rules and their enforcement must be strengthened for the protection of human rights in non-EU countries" (European Parliament, 2023). Following these findings, the European Commission adopted Recommendation (EU) 2024/2659 in October 2024, providing detailed guidance to exporters on human-rights risk assessment and red-flag indicators for potential misuse (European Commission, 2024). In September 2025, the Commission adopted a Delegated Regulation updating the EU dual-use export control list in Annex I of Regulation (EU) 2021/821 to introduce new controls on emerging and disruptive technologies (European Commission, 2025).

While the EU's dual-use export control regime reflects its ambition to integrate human rights into trade governance, its limited enforcement capacity and the economic significance of Gulf markets have constrained meaningful implementation. The result is a persistent gap between the EU's rhetoric of "values-based trade and investment" and the geoeconomic and structural imperatives that shape its engagement with authoritarian partners.

This gap is likely to widen as Gulf sovereign wealth funds continue to invest heavily in European AI and digital infrastructure. In February 2025, the UAE and France announced a €30–50 billion partnership to build a one-gigawatt AI and cloud data centre campus in France, alongside semiconductor and sovereign-cloud cooperation. (France 24, 2025). In parallel, Qatar's sovereign wealth fund has begun joint ventures with European operators to co-develop and co-own data centre assets, marking a shift from passive portfolio investment to strategic co-ownership (Erena Middle East, 2025). In October 2025, the United States approved the export of 500,000 NVIDIA Blackwell GPUs to the

UAE. This deal integrates Gulf actors into the global compute supply chain on which European AI projects depend (Reuters, 2025). These deals, in turn, make the EU both a regulator and a beneficiary of the same transnational circuits of capital and technology that enable digital repression abroad, and limit its capacity to impose effective restrictions on spyware exports.

Conclusion

The global spyware trade demonstrates that internet openness cannot be taken for granted. The EU's export-control reforms represent a pioneering attempt to internalise human rights externalities within trade policy. However, their effectiveness is constrained by deeper contradictions between the Union's normative ambitions and its external economic dependencies, particularly its reliance on foreign capital to finance AI infrastructure essential to asserting digital sovereignty. Addressing these contradictions requires a proactive EU strategy integrating trade, investment, and technology governance to safeguard an open, secure, and rights-respecting internet.

Such a strategy must go beyond regulatory reform to confront the structural realities of digital interdependence. Ultimately, the contest over spyware regulation reveals more than a regulatory shortcoming. It exposes the tensions inherent in governing interdependence within a fragmented international order. The EU's challenge lies not only in managing trade in surveillance technologies, but also in formulating a coherent vision of digital openness resilient to the structural realities of interdependence. The EU's credibility as a global technology regulator will depend on its ability to align market openness with normative coherence, ensuring that economic partnerships do not undermine its human-rights commitments.

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Between Democracy and Open Internet: Trade-off and Collaborators

Nori Katagiri

Introduction

In the history of computing, democracies have been among the strongest advocates of an open internet. Yet the relationship between the two is more complicated than it first appears. It is one of mutual dependence: the open internet and democratic governance rely on each other. This means that democracies may be vulnerable to forces shaping the internet, including abuses such as deepfakes. The internet, in turn, depends on sound governance of physical infrastructure within societies that allows people to use it responsibly. In other words, just as democracies and an open internet stand to gain from the proper condition of each other, they also depend on each other's external environments. This places them in a highly complex cohabitation relationship.

It is a common assumption that democracies govern their digital space by implementing widely accepted principles of pluralism, accountability, and free exchange of ideas (Helbing et al. 2023). They uphold the notion of collective action, hear the voices of people, and defend a normative framework that reinforces freedom of expression, privacy protection, and human rights. Societies also shape technological and institutional policies that enable openness by ensuring fair competition among internet providers and tech firms. However, this depiction is more ideal-typical than real.

The trade-off

In the current geopolitical environment, few democracies worldwide can fully achieve all these goals. Most operate by fulfilling some while falling short on others. For these governments, policy trade-offs are pragmatic even though efforts to achieve freedom, political diversity, and justice are widely considered intrinsically valuable because they align with fundamental human dignity, moral agency, and the creation of a just society (Benkler, 2007). Yet there is only so much that governments can do to maintain this balance through purely liberal-democratic means (Katagiri, 2024). There is also a large group of governments that are not willing to abide by democratic principles. Some have succumbed to incentives to exploit the open internet for political and personal gains. From the Middle East to South Asia to Europe (The Citizen Lab, 2025), elected leaders in many countries have deployed malicious surveillance technologies to spy on their own people under the name of national security.

Given the trade-off, how much openness should we seek? The answer depends on how one perceives the merits and demerits of the internet. Many states have shut it down for various reasons (Recorded Future 2022). The demerits of such a policy are obvious—economic and perhaps legal—making it unpopular overall. However, some governments perceive the merits of such an approach, such as social control through censorship. Authoritarian states believe that there are good reasons to regulate the internet for national security. Some democracies exploit the open internet to spy on their own people in the name of national security. Authoritarian states may also resort to closing the internet for “law and order”, however briefly it may last.

As such, the openness of the internet is a matter of degree that requires balancing between different, sometimes conflicting, policy goals. This is because the internet requires a degree of content moderation to safeguard users against harmful content and protect minors and privacy and autonomy (Gillespie, 2018). An underregulated internet may even enable malicious online activities, including bot attacks and illegal financial transactions, such as money laundering (Haner & Knake, 2021). By default, the internet requires a moderately controlled level of openness.

Democracies and their collaborators

The participation of other actors is critical to ensuring that trade-offs are addressed through democratic governance. Mirroring the widely distributed nature of the internet itself, democracies engage with stakeholders in civil society, the technical community, and the private sector in collective decision-making. Non-governmental stakeholders monitor government actions, lend legitimacy, and contribute resources.

Among the key stakeholders are firms, which own much of the internet infrastructure and exercise outsize influence on digital governance. To be certain, the governments’ relationship with firms is tricky. Democracies confront a variety of issues arising from interactions with private actors, especially tech firms and internet service providers (Schaake 2024, Wong 2023; Katagiri, 2023). They must work with firms reluctant to participate in government activities that carry legal and financial consequences. Of course, there are limits to what governments can do to work in tandem with firms. The private sector is driven by profit, not public interests. But the benefits of working with private actors are enormous; they offer expertise, increase cost efficiency, produce and leverage advanced technology, and provide critical employee training.

In addition to private actors, democracies’ natural partners are found in international institutions. In particular, they are those that embrace the notion of an open internet, such as the Internet Governance Forum and the Internet Corporation for Assigned Names and Numbers (ICANN). These institutions have faced a crisis of their own, with many national leaders openly questioning their worth and legitimacy. In a 2021 survey of close observers of internet governance, for instance, many respondents expressed limited confidence in ICANN (Jongen & Scholte, 2021). Yet it is through the institutions that democracies can promote participatory governance. They are also an important part of international society that keeps internet policies from being monopolised by states or firms.

Conclusions

With limited resources at their disposal, democracies must work together to confront authoritarian states that use the very open internet to undermine it (Maerz, 2026). To defend against it, democracies must strengthen partnerships with international organisations willing to provide credible alternatives to divisive policies and to deliver on their promises. This requires shaping a global environment that fosters hope, reinforces positive engagement, and provides leadership in consolidating gains and advancing effective policy outcomes. They may include enabling secure data flows and rights-based digital policies.

Given the current state of global affairs, there is a small area of opportunity for policy cohesion. An open internet will not sustain itself without governance provided by democratic leaders. Democracies must act together, in partnership with civil society and technology communities, to set standards and practices that ensure the internet remains a space of freedom and innovation.

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Negotiated Openness and the Rise of Soft Authoritarianism

Dechun Zhang

Introduction

Openness has long been a foundational concept in the political and economic imaginaries of the digital age. From early narratives of the internet as a decentralised and emancipatory space to contemporary policy discourses on open markets and cross-border data flows, openness has been closely associated with innovation, growth, and democratic governance. Open systems are assumed to lower barriers to entry, facilitate competition, and accelerate the circulation of knowledge and skills across borders (Benkler, 2006). Within this framework, openness appears both normatively desirable and economically efficient.

In digital societies, openness is increasingly shaped by platforms, algorithms, and data infrastructures rather than emerging organically from market or civic dynamics. As participation, transparency, and connectivity are actively designed and managed, openness functions as a negotiated condition structuring visibility, engagement, and economic value creation. European digital policy should aim to govern openness as an infrastructural condition by strengthening coordination, accountability, and oversight across platform-mediated systems.

Openness as a designed environment

The association between openness and freedom has become increasingly unstable, as in contemporary digital societies, openness no longer operates as a spontaneous or self-regulating condition but is actively produced, managed, and calibrated through platforms, algorithms, and data infrastructures. In China, for example, online participation is widely encouraged while being simultaneously structured through platform regulation, content moderation, and algorithmic visibility, producing openness as a managed environment rather than an autonomous public sphere (Repnikova, 2017). Participation is promoted, transparency ritualised, and connectivity normalised within architectures that shape visibility, relevance, and behavioural possibilities, marking a shift of openness from an ethical ideal to a designed environment.

This shift fundamentally alters the relationship between openness and governance. Rather than standing in opposition to regulation, openness has increasingly become a practical tool for organising and managing digital systems. Digital governance now shapes the environments in which people communicate, share information, and participate online. Techniques such as behavioural nudging and algorithmic regulation—now widely used in liberal democracies—guide user behaviour without overt restrictions, embedding policy goals into platform design and choice architectures. Instead of relying on prohibition, governance takes place through the structuring of interfaces, attention, and participation patterns, reflecting a broader shift toward governing digital spaces through design rather than direct intervention (Zhang, 2025).

Negotiated openness as a strategy

The concept of negotiated openness captures this transformation. Negotiated openness refers to governance arrangements in which participation is actively invited and institutionally channelled, while the terms and limits of openness remain subject to ongoing negotiation. The experience of Singapore illustrates this logic clearly: participatory feedback mechanisms and digital consultations enhance policy responsiveness, while political contestation remains institutionally constrained, making openness a governance strategy rather than a democratic expansion (Rodan, 2013). Openness is not abolished; it is selectively enabled. Citizens and users are encouraged to speak, share, and engage, yet the architectures that determine visibility, amplification, and legitimacy are tightly managed.

Negotiated openness represents a departure from classical authoritarian strategies based on censorship and repression. Instead of silencing voices, contemporary governance increasingly seeks to organise them. In China, this has involved a shift from direct censorship toward platform-based orchestration of participation and sentiment, in which some content is amplified while other content is downplayed, rather than speech being directly prohibited (Noubel, 2025; Repnikova, 2017). Participation becomes a resource rather than a threat. For instance, this dynamic was evident during the COVID-19 pandemic, as seen in controversies such as the Wuhan Red Cross Society scandal over the misallocation of medical supplies. In such cases, authorities did not simply delete posts that criticised the government; they also reduced the visibility of critical content while guiding online discussion toward different acceptable interpretations, such as individual responsibility, accountability, and correction, while still allowing active public engagement. During this process, netizens formed different groups based on their interpretations of the event. Groups aligned with more positive or state-compatible narratives often coordinated to challenge or counter those expressing critical views. As a result, rather than producing large-scale online social movements or escalating into radical nationalist mobilisation, participation remained fragmented. More importantly, it contributed to the emergence of a generally positive discursive environment toward the government and generated forms of peer pressure that discouraged critical expression (Zhang, 2026). The appearance of openness—responsiveness, inclusiveness, engagement—serves to stabilise authority and enhance legitimacy, even as the boundaries of acceptable participation remain structurally constrained.

The shift towards a soft digital authoritarianism

This governance logic underpins the rise of soft digital authoritarianism (Zhang, 2026). Rather than relying primarily on coercion or overt restrictions on freedom, soft authoritarianism operates through affect, communication, and participation. Control is internalised rather than imposed, embedded in everyday practices and infrastructural design rather than enforced through direct intervention. During the COVID-19 pandemic in China, emotional participation—expressed through online performances of gratitude, solidarity, and national pride—was actively encouraged and amplified across digital platforms, contributing to legitimacy-building and behavioural compliance without the need for extensive coercive measures. Participation appeared voluntary and empowering, yet it simultaneously channelled emotion into forms of engagement that reinforced existing power relations.

Crucially, soft digital authoritarianism should not be understood as a regime-specific anomaly. While it is often illustrated through authoritarian contexts, similar techniques are increasingly visible across democratic systems. The South Korea impeachment movement shows how platform-mediated political communication, driven by algorithmic logics, intensifies emotion and immediacy, shaping participation and visibility even in a consolidated democracy, while underscoring the continued centrality of institutional safeguards (Yi & Schüpf, 2025). Algorithmic curation, behavioural nudging, and platform-mediated participation now structure political communication, market behaviour, and policy feedback loops in many liberal democracies as well. This convergence challenges the analytical distinction between democratic openness and authoritarian control.

Political economy of digital capitalism

The rise of negotiated openness is inseparable from the political economy of digital capitalism. Digital participation generates economic value by continuously extracting behavioural and emotional data. In the United States, platform economies rely on sustained user engagement to produce predictive data that fuels market optimisation and emergent forms of governance (Zuboff, 2019). Every act of sharing, liking, searching, or commenting becomes a resource for prediction, optimisation, and governance. Openness is productive precisely because it is measurable and analysable. In this sense, openness is not merely a political arrangement but a core economic mechanism.

This economic logic blurs the boundary between growth and governance. The more open and participatory a system appears, the more data it produces, and the more predictable and manageable it becomes. In India, digital welfare and identification infrastructures increasingly mediate access to public services and rights through data compliance (Naregal, 2025), illustrating how negotiated openness can generate efficiency and inclusion while simultaneously deepening power asymmetries. Growth, legitimacy, and control thus reinforce one another. However, this model of growth is structurally fragile. It depends on sustained engagement, emotional mobilisation, and user compliance rather than on long-term productivity or institutional capacity.

Affective dynamics play a central role in this process. Digital participation increasingly operates through emotion rather than deliberation. Even in European democracies—including Italy, France, Spain, and Germany—affective mobilisation is a powerful driver of political visibility on social media, as emotional engagement tends to amplify the circulation of political content (Gerbaudo et al., 2023). Visibility and relevance are rewarded not for deliberation but for affective intensity. Such affective publics can generate powerful forms of collective identification, but they also lend themselves to governance through emotional alignment rather than institutional accountability (Papacharissi, 2015).

Emotions and authoritarian governance

The emotionalisation of openness further normalises soft authoritarian governance. When participation is framed as empowerment and visibility as inclusion, the governing functions of openness become obscured. Pandemic-related digital mobilisation across both authoritarian and democratic contexts demonstrates how voluntary participation can become intertwined with moral obligation and behavioural conformity (Zhang, 2026), as seen, for example, in the Wuhan Red Cross Society scandal, where online engagement combined emotional expression with peer pressure and normative alignment, often shaped by nationalist framing. Individuals may experience continuous engagement as freedom, even as it deepens surveillance, behavioural steering, and self-regulation. Growth generated under such conditions may appear dynamic and inclusive, yet it rests on the internalisation of governance norms.

These dynamics complicate debates on digital sovereignty, strategic autonomy, and openness, as policymakers often cast openness and control as oppositional despite governance in practice relying on hybrid arrangements that combine selective openness with infrastructural control; comparisons between China’s platform governance and India’s data-driven statecraft show that openness and control are institutionally fused rather than mutually exclusive (Repnikova, 2017; Naregal, 2025), and although such strategies may deliver short-term adaptability and economic resilience, they risk entrenching soft authoritarian logics under the guise of innovation and competitiveness, indicating that openness is no longer a straightforward driver of growth or democracy but a shared technology of governance that reorganises participation, economic value, and political authority without rejecting openness itself.

Priorities for governing openness

If negotiated openness increasingly underpins soft authoritarian governance, the central policy challenge is not how to maximise openness, but how to govern it responsibly. Treating openness as an abstract value obscures its infrastructural, economic, and affective dimensions. Policymakers must instead focus on how openness is designed, institutionalised, and connected to systems of accountability.

First, openness should be approached as a negotiated and infrastructural condition. Policy frameworks must move beyond formal commitments to participation and transparency and examine how digital architectures structure access, visibility, and relevance in practice. Evidence from platform governance and regulatory assessments shows that algorithmic ranking systems, recommender mechanisms, and interface design decisively shape whose voices are amplified, whose content circulates, and whose participation translates into economic or political value (European Commission, 2022; Competition and Markets Authority, 2020). Treating openness as a design issue rather than a purely normative principle is therefore essential for effective digital governance.

Second, the political economy of negotiated openness requires explicit attention. Growth models that rely on continuous participation and data extraction may yield short-term efficiency gains, but they externalise social and political costs. Evidence from competition authorities, labour regulators, and platform governance assessments shows that data-driven platform models prioritise optimisation, prediction, and engagement metrics over long-term innovation, skills development, and institutional trust (European Commission, 2020). Without policy intervention, these dynamics risk locking economies into extractive growth paths that undermine social resilience and democratic legitimacy.

Third, openness must be reconnected to meaningful accountability. Transparency without enforceable accountability can reinforce existing asymmetries, while participation without agency risks depoliticising governance. In data-driven governance systems, openness often widens visibility without strengthening oversight, leaving decision-making authority concentrated while appearing inclusive. Without clear responsibility, audit mechanisms, and effective avenues for redress, openness risks functioning less as democratic empowerment and more as procedural legitimacy for opaque forms of control.

Fourth, the emotional dimension of digital participation should be treated as a core governance issue rather than a side effect. Online participation increasingly operates through affective mobilisation, shaping legitimacy, compliance, and exclusion across political contexts. Emotional engagement can enhance responsiveness and social cohesion, but it can also be strategically amplified to steer behaviour and marginalise dissent. Policymakers should therefore recognise affect as a governance variable and develop safeguards that limit the instrumental use of emotion in digital communication, platform design, and public engagement strategies.

Fifth, approaches to digital sovereignty should avoid equating autonomy with closure. Strategic autonomy does not require restricting openness; rather, it requires strengthening the capacity to govern open systems. Experience from data-driven governance models suggests that resilience is more effectively achieved through diversification, interoperability, and institutional capability than through exclusionary control. Framing sovereignty in infrastructural and regulatory terms allows openness and autonomy to reinforce rather than undermine each other.

Finally, openness should be reclaimed as a relational and ethical practice. Responsible openness does not reject participation or innovation, but resists their automatic conversion into surveillance, optimisation, and behavioural steering. By embedding openness into governance frameworks that prioritise accountability, reflexivity, and long-term public value, policymakers can mitigate the soft authoritarian dynamics increasingly accompanying digital transformation.

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Resilience in the Face of a Polycrisis: How the Belarusian ICT Sector Balanced Business Exit and Connectivity in the 2020s

Ildar Daminov

Introduction

In 2016, the Wall Street Journal referred to Belarus as the "Silicon Valley of Eastern Europe" (Razumovskaya, 2016), with the Belarusian ICT businesses such as EPAM or Wargaming gaining international recognition. Now, a decade later, this highly innovative industry lies in shambles after going through a polycrisis in 2020-2022. Polycrisis is a situation of multiple crises (Lawrence et al., 2024), which, in the case of Belarus in 2020-2022, referred to the interaction among constitutional, geopolitical, public health, and economic crises, among others, resulting in disastrous effects for the industry. Consequently, many ICT businesses relocated abroad—from Kazakhstan to Poland, from Lithuania to the United States. The dramatic rise and fall of the Belarusian ICT sector offers valuable insights into the economic and political costs of authoritarian crackdown on connectivity, going beyond the analyses of state-civil society confrontations. The insights and lessons from this case may be valuable to European Union policymakers.

The rollercoaster of the Belarusian ICT industrial policy (2005-2025)

The story of Belarus's transformation into a regional ICT hub dates to the early 2000s. Back then, the Lukashenka regime, tired of regular economic clashes with Russia and facing a structural economic crisis, caved in to the lobbying efforts of then-Presidential Science and Technology Advisor, Valery Tsapkala, to offer more relaxed ICT regulations. As a result, 2005 saw the opening of the Belarusian Hi-Tech Park (HTP), a special economic zone for IT companies that allowed them to pay no capital taxes, lower social security contributions, and income taxes. Tsapkala and his colleagues saw a strong potential in an emerging tech industry in Belarus, although most of its companies preferred hiding in the shadow economy:

> *Several major business owners complained to us about all the costs they would have to bear. [...] So, very large chunks of the [Belarusian ICT] sector remained in the shadow economy, with some larger businesses often paying salaries under the table to avoid taxes to stay competitive on the international market. Taxes were too high, and social security contributions even higher. Our companies could not compete with India or the others on the market [...] So, we had to get them out into the broad daylight by promising some preferential treatment.*[23]

[23] Interview 1 with a senior ex-official of the Hi-Tech Park.

While initially some businesses were hesitant, proactive efforts to convince companies to register at the HTP paid off.24 In total, Belarusian ICT exports grew 50 times from 2003 to 2014 (ASER & BiK, 2020, p. 12). The regime, upon witnessing progress, offered grants and subsidies in the mid-2010s, in addition to tax benefits, along with further regulatory concessions on blockchain and cryptocurrency (Decree No. 8, 2017; Digital Economy Strategy, 2016). Consequently, even more companies flocked to the HTP with the number of residents exploding more than 5-fold between 2017 and 2021 (HI Tech Park Belarus, 2021b). The exports of ICT services from HTP alone have grown more than 4-fold between 2016 and 2021 – from €800 million to €3,2 billion (HI Tech Park Belarus, 2021a).

The Belarusian ICT industry, as a domestically prominent driver of economic development, has gained greater social and even political weight. It became the most attractive destination for highly qualified specialists by the end of the 2010s, as salaries there were approximately 3 times the national average (BelStat, 2020). Furthermore, the HTP, in addition to being a special economic zone, increasingly took on representative functions for the entire industry. For example, the industry showed its corporate social responsibility during the COVID-19 pandemic by organising humanitarian initiatives. This starkly contrasts with the government's reaction, a mixture of pandemic denial and chaotic regulations (Gerry & Neumann, 2023). Subsequently, when the Lukashenka regime faced a legitimacy crisis due to its mishandling of the pandemic, some sector representatives threw their weight behind the oppositional candidates in the 2020 presidential election. As one interviewee put it, "innovation could not live in the climate of constant fear".25 In the aftermath of the election, some companies launched civic tech initiatives to improve transparency in vote counting, fundraising, and opposition coordination (Navumau et al., 2025). Industry representatives, without the HTP's direct involvement, sent a joint open letter to the authorities calling for an end to the post-election violence.

Unsurprisingly, this entanglement with the 2020 protest movement resulted in a massive backlash from the regime. According to one of the interviewed ex-bureaucrats, Lukashenka saw it as a personal treachery.26 Retaliatory measures followed immediately. The security forces started imprisoning ICT specialists even remotely connected to the protest movement, while demanding financial extortions if the company staff donated any money to civil society organisations in August 2020.27 The repressive measures intensified after late September 2020, when an ICT company staff member shot a KGB officer (Viasna 96, 2021). A tax hike for the HTP residents followed in 2021.

As the crackdown continued into 2021, the situation deteriorated even more once the Lukashenka regime got involved in the 2022 Russian invasion of Ukraine. Several sanctions packages against Belarus were passed not only by the EU and the United States but also by many other nations. Financial sanctions and SWIFT exclusion

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[24] Interview 1.
[25] Interview 2 with a start-up CEO.
[26] Interview 1.
[27] Interview 2; Interview 3, Senior Employee.

paralysed the ICT sector, as most payments could no longer be received. Furthermore, the short-term reputational costs of cooperating with Belarus became all too dangerous (Danilchuk, 2023). A global economic crisis in 2022-2023 in the ICT industry, driven by inflation and spiking energy prices, made the situation even worse.

Resilience in the face of a polycrisis

The 2021-2022 crackdown led to a major economic fallout. As shown in Table 1, the sector's share of GDP shrank from 7.3% in 2021 to 4.5% by 2023. The same period saw a collapse in the HTP exports from 3,2 to 1,8 billion USD, while the FDI rates continued on a downward trajectory from 2021. The job market froze after 2019, and there was a spike in firings in 2022, with more than 15,000 fired compared to approximately 6,000 being hired (Luzgina, 2023). Many ICT specialists fled or were relocated, as a full business exit was not an option that most Belarusian companies preferred between 2020 and 2022.[28] The brain drain from Belarus slowed down only in 2023-2024, but has not fully stopped (Boguslavskaya & Kohl, 2025).

Table 1. Overview of the key indicators of the ICT sector dynamics in Belarus (2013-2024).

Year	GDP Share of the ICT sector	HTP Exports (millions of USD)	FDI in Belarus (billions of USD)
2013	3.1%	447	11,1
2014	3.2%	585	10,2
2015	4.1%	706	7,2
2016	4.9%	821	6,9
2017	5.0%	1025	7,6
2018	5.4%	1414	8,5
2019	6.3%	2195	7,2
2020	7.1%	2735	6,0
2021	7.3%	3250	6,6
2022	6.6%	2500	6,0
2023	4.5%	1800	5,7
2024	4.9%	1800	5,2

Source: Based on (Danilchuk, 2023, p. 11); with complemented data for 2023-2024 from secondary sources 1,2,3.

Having been forced out, many of the industry's representatives persevered despite the polycrisis. They continued with their projects even from abroad, mostly from the EU. According to late 2023 assessments, Poland and Lithuania became the two major destinations for businesses and specialists. This includes up to 220,000 people migrating from Belarus, as well as almost 6,000 companies moving to Poland, and up to 80,000 people and 850 companies moving to Lithuania between late 2020 and early 2023 (Danilchuk, 2023). This correlated with the general migration trend, as the total number

[28] Interviews 2; 3; and 4, Professional Association, Online.

of Belarusians with the EU residence permits reached almost 1 million by the end of 2024, which is more than 10% of Belarus' population (Dev BY, 2025).

The industry's resilience in the face of relocations manifested itself not only in its quick socio-economic integration into new countries but also in the creation of new institutions. Shortly after the first round of relocations, Belarusian entrepreneurs launched diaspora-driven organisations and businesses, including the Association of Belarusian Businesses Abroad, the ZPP Belarus Business Center, and the Imaguru startup association. Not only have those organisations endured despite the pressure from the Lukashenka regime (e.g., ABBA was declared "an extremist organisation"), but they have also expanded the scope of their activities to research, education, and socio-cultural projects. Although they disagree on the extent to which their businesses can be seen as "Belarusian" by now, many emigres agreed that social responsibility and the engagement of relocated businesses at the individual level are also important for maintaining the Belarusian identity (Idea Bank, 2025).

Conclusion

By 2026, neither the repatriation of businesses nor the recovery of the ICT sector in Belarus became very likely without major domestic political transformations. In these conditions, preserving inter-sectoral connectivity abroad and social capital networks is an essential task—not only for economic reasons, but also for civic ones, given the sector's previous engagement with the democratization movement. From this perspective, any internal institutionalisation efforts should be encouraged not only by the Member States but also by the EU institutions. Financial, administrative, and economic support for sector-level initiatives serves not only the interests of these Belarusian businesses but also those of the EU, which can draw substantial economic benefits in the era of tech-driven geoeconomics. In the long run, however, it is exactly these institutionalised platforms that can serve as one of the many platforms of the European re-engagement with Belarus.

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From the Commons to Rentierism: Governance Against a Closed Internet

Mallory Knodel

The open internet has long been associated with productivity and innovation. Interoperable protocols, decentralised governance, and low barriers to entry enabled new forms of entrepreneurship and knowledge diffusion across borders. However, the relationship between openness and economic growth has changed remarkably. Openness, once grounded in interoperability, has been leveraged to entrench platform monopolies. The reconceptualisation of openness has recast economic and social values of technology (Kilic & Knodel, 2025). While openness is often invoked as a normative ideal or technical property, less attention has been paid to the institutional transformations that have altered its economic function.

The current phase of centralisation in digital technology is best understood not simply as a problem of surveillance or extraction, but as a shift toward digital rentierism embedded across multiple layers of infrastructure. In this context, regulatory intervention alone may be insufficient to preserve the open internet. Instead, governance strategies must also address the design of technical protocols and interoperability structures that shape accumulation dynamics upstream of market competition. We must reclaim "open" and a commitment to the commons on social and cultural grounds in the public interest.

From production and appropriation to rent

Early critiques of platform capitalism focused on production. The dominant concern was that users had become the product: their data and attention were monetised by advertising-based platforms (Zuboff, 2019). At the same time, the regulatory response centred on privacy—strengthening data protection rules, enhancing consent mechanisms, and embedding privacy into technical design. Instruments such as the General Data Protection Regulation (GDPR) exemplified this approach.

However, privacy reforms did not significantly alter concentration patterns. If anything, compliance costs and trust dynamics reinforced incumbents, consolidating their position as primary "trust intermediaries." Privacy became a competitive advantage for dominant firms rather than a structural corrective. The production-focused diagnosis, while normatively important, did not sufficiently explain persistent centralisation.

Another critique reframed the problem as appropriation. Here, the emphasis is on extractivism: platforms leveraging network effects and market power to appropriate value generated by users, creators, and complementary innovators (Mazzucato, 2018). The proposed remedy was to create alternatives: public options, regional "stacks," or open-source platforms such as Mastodon. Yet alternatives have struggled to dislodge entrenched incumbents. Network effects remain a formidable barrier, and user exit is constrained by the social and relational infrastructures that platforms monopolise.

This impasse points to a third dimension: rentierism. Digital infrastructures—from submarine cables to cloud services, certificate authorities, DNS resolvers, and dominant social graphs—have become sites where control over essential intermediaries yields ongoing rents with minimal productive contribution. Once entrenched, these actors do not need to innovate or extract more intensively; they simply collect rents from ownership of bottleneck positions (Saito & Sasaki, 2025). Such dynamics extend beyond the application layer into core internet infrastructure. The result is a structural enclosure of what might be termed “human relationality” as a digital common.

The limits of competition and sovereignty

Traditional competition law struggles to address rentier dynamics embedded in layered infrastructures. Antitrust law can discipline specific abuses but rarely restructures the underlying network topology. Even ambitious interventions such as the EU Digital Markets Act (DMA) focus primarily on conduct obligations rather than architectural transformation.

Similarly, digital sovereignty strategies like data localisation mandates, national champions, or regionally bounded service ecosystems may rebalance geopolitical power but do not necessarily restore openness. They can narrow the field of intermediaries without altering the logic of enclosure. Substituting national monopolies for global ones leaves the rent structure intact.

Cross-border data governance frameworks further complicate this picture. Trade agreements seek to liberalise data flows, while rights-based regimes emphasise protection in “digital sovereignty” initiatives that are rebrands of typical networks-versus-services tussles (Clark, 2005). However, neither approach systematically addresses infrastructural concentration. Economic integration may expand markets, but without structural interoperability, gains accrue disproportionately to incumbents controlling key intermediation points.

Weak points in the stack

One indicator of rentier entrenchment is the presence of “patchwork” governance regimes that paper over architectural weaknesses. The certificate authority ecosystem provides a salient example. The CA/Browser Forum is a private governance body coordinating trust anchors for TLS[29] and occupies a highly consequential yet underappreciated position. Decisions within this forum have had geopolitical ramifications, including the removal of certain certificate authorities following Russia’s invasion of Ukraine. The trust model for HTTPS[30] relies on centralised authorities whose

[29] The Transport Layer Security (TLS) protocol is the predominant way client/server applications communicate over the internet in a way that is designed to prevent eavesdropping, tampering, and message forgery.

[30] Hypertext Transfer Protocol (HTTP) is the dominant protocol for distributed, collaborative, hypermedia information systems. HTTPS uses TLS to secure HTTP connections over the internet.

governance is neither fully public nor fully accountable, revealing both concentration and fragility.

Similarly, the DoH31 debate illustrates how privacy-enhancing reforms can inadvertently centralise resolution services. Users prefer a trusted provider for DNS lookups rather than conventional, opportunistic, and more decentralised lookups via their ISP. While encrypting DNS queries mitigates surveillance risks, the shift toward a small number of global DoH providers consolidates trust in a handful of intermediaries. Privacy gains at one layer may therefore reinforce rentier dynamics at another (Knodel, 2023).

Other weak points include the concentration of root server operators, supply chain dependencies in hardware and semiconductors, and the growing dominance of private satellite constellations in space-based internet infrastructure. Each represents a layer where control yields durable rents insulated from competitive churn.

Interoperability as structural intervention

If rentierism is embedded in infrastructural bottlenecks, then preserving openness as a driver of growth requires interventions that alter intermediation structures rather than merely disciplining conduct. Here, interoperability becomes central.

Open social protocols such as ActivityPub, ATProto, and the Decentralised Social Networking Protocol (DSNP) illustrate alternative architectures. By enabling cross-platform communication, these protocols reduce switching costs and weaken network effects that lock users into dominant platforms. However, voluntary adoption has been limited.

At the same time, without regulatory impetus, such as interoperability obligations under the DMA, dominant firms lack incentives to open their networks. Interoperability thus sits at the intersection of standards and regulation. Technical standards define the possibility space; regulation can mandate or incentivise their implementation. This triangular model—standards development, regulatory enforcement, and corporate advocacy—reflects an evolution in governance strategy. Protocol design shapes market structure upstream of competition law. Regulatory frameworks can accelerate adoption. Direct engagement within standards bodies allows public-interest considerations to influence technical architecture before it becomes entrenched.

Importantly, not all technical mandates are appropriate. Expanding the political mandate of standards bodies risks technocratising inherently social disputes. However, carefully targeted interoperability requirements, particularly at higher layers of the stack, may counteract enclosure without fragmenting the global network.

Openness and growth reconsidered

Historically, openness fostered growth by lowering transaction costs, enabling entry, and facilitating knowledge spillovers. Rentier concentration reverses these dynamics. High

[31] DNS over HTTPS (DOH) is a protocol for sending DNS queries and getting DNS responses over HTTPS.

switching costs, data silos, and infrastructural gatekeeping dampen entrepreneurial experimentation and distort capital allocation toward rent-seeking positions. The early internet culture's undying loyalty to openness and the commons has served to stave off rentierism for longer and more durably than earlier regulatory interventions would have.

The digital commons has achieved the fabled balance between innovation and regulation, which is why it has been an ideological and tactical target of Big Tech companies seeking monopoly positions. From a growth perspective, the enclosure of digital commons reduces allocative efficiency and long-term innovation potential. While short-term adaptive strategies, such as national stacks, compliance regimes, or bilateral trade deals, may mitigate geopolitical vulnerabilities, they do not address the structural transformation of openness into controlled intermediation.

Safeguarding openness as a prerequisite for sustainable economic progress, therefore, requires a governance approach attentive to architectural design. Regulatory instruments such as the DMA, GDPR, and cross-border data frameworks are necessary but insufficient. Without interoperable infrastructures that enable meaningful exit and entry, network effects will continue to entrench rent positions beyond the reach of ex post regulation. One way that regulatory action has failed to address equities in a way that would preserve the commons is the Digital Services Act's targeting of Wikipedia as a "very large online platform" (Wikipedia, 2025). In this sense, openness is not merely a normative commitment but an institutional condition for dynamic growth. Its preservation depends on aligning technical standards, regulatory mandates, and market incentives to prevent the irreversible enclosure of digital commons.

If rentier dynamics embedded in infrastructural bottlenecks are the core threat to openness and growth, then interoperability mandates should be treated not as optional competition remedies but as structural economic policy. Requiring dominant platforms and infrastructures to implement open protocols—at social, messaging, and identity layers—would reduce switching costs, restore contestability, and reorient accumulation away from rent extraction toward productive innovation. Absent such measures, regulatory efforts risk managing decline rather than reversing enclosure.

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V. Companies, Markets and Business Models

Private-sector actors have been key architects of the open internet, yet their growing market dominance raises questions about the concentration of control and the erosion of openness. This section explores how business models, data ownership structures, and platform dynamics influence economic openness and innovation. It explores how regulatory interventions—from competition law to data portability—can promote more open, plural, and competitive digital markets.

Guillaume Beaumier opens by contrasting Apple's strategy of curated enclosure with Google's strategy of infrastructural ubiquity, arguing that both converge on concentrated control—and that concentration, more than any particular approach to openness, is the central governance challenge. Matteo Nebbiai traces how geopolitical shifts transmit differently across the logical, physical, and content layers of the internet, finding that the greatest risks to openness are concentrated in the countries—the US, the EU, India—that have historically been most committed to it. Sarah Nicole and Jeb Bell make the case for data cooperatives as a proactive instrument for aligning market incentives with democratic data governance. Camille Françoise and Michele Failla examine extractive pressures on digital knowledge commons through the case of Wikipedia, arguing that European regulation has addressed monopolistic conduct without articulating what a commons-compatible ecosystem should look like.

The chapter closes with three contributions on accountability and institutional design. Tayrine dos Santos Dias and Anita Dorett examine the role of institutional investors and digital rights organisations as accountability actors alongside regulators. Murielle Popa-Fabre argues that civil society evaluation of AI systems is becoming a standard-setting function of structural importance—not a complement to official processes but increasingly a precondition for them. Joshua Tan closes the volume by proposing public AI as a governance category: a model of AI infrastructure designed to serve collective interests and to distribute sovereignty over critical digital systems beyond the small number of firms that currently hold it.

Platform Power and the Future of an Open Internet

Guillaume Beaumier

The dream of an open internet has collided with the rise of a few dominant platforms. Rather than the decentralised network its architects envisioned, today's internet is organised around a small number of companies spanning search engines, social networks, app stores, and online marketplaces (Gorwa, 2019). Beyond their role as service providers, these platforms now act as gatekeepers, sitting between different groups of internet users. This central position chiefly gives them the power to control how individual users and other companies can access and exchange information online (Beaumier & Newman, 2024). How open the internet is, therefore, increasingly depends on the governance choices of a small number of private companies (Haggart, 2020).

Openness in platform ecosystems

Some platforms have moved to close off their ecosystems into what can be dubbed digital walled gardens (Winseck, 2005). Like a private estate, these are curated spaces where the platform controls who enters and on what terms. Apple illustrates this approach clearly. Its users can only download applications from Apple's App Store, restricting the services available to them. More recently, Apple has further tightened its ecosystem by limiting how much data third-party companies can collect from its users (Beaumier & Newman, 2024). The result is an ecosystem in which Apple controls both what users can access and what software and app developers can offer. The company justifies this approach as a way to ensure the security of its ecosystem and the privacy of its users. At the same time, it also uses these controls to gain an advantage over its competitors by promoting its own services. Following the tightening of its privacy controls, Apple's advertising revenue soared as companies turned to Apple's own services to reach their users (McGee, 2021).

In contrast, some platforms have pursued a strategy of openness, allowing third parties to build on and interact with their ecosystems. Rather than curating a closed ecosystem, they aim to open their infrastructure to as many users and developers as possible. Google exemplifies this approach. Google's mobile operating system, Android, is largely open source, meaning any company can use it on its devices. Moreover, Android users can download applications from multiple app stores with few constraints. Google also decided against automatically blocking third-party companies from collecting data from its users (Cohen, 2024). These choices all reflect Google's attempt to establish itself as 'the infrastructure' for the internet. As opposed to Apple's strategy to build a small group of highly valuable users, it aims to attract all possible companies and users onto its platform. Recent estimates indicate that Android and Chrome both accounted for a little over 70% of the mobile phone and desktop browser market (Beaumier & Campbell-Verduyn, 2024; Statcounter, 2026).

Concentrated control as a challenge

While differing in their approach to openness, both cases illustrate how platforms govern their ecosystems in line with their commercial interests. Such interests can sometimes align with the broader public interest, as with Apple's privacy protections, but this alignment is not guaranteed. Even then, protecting the public interest can become a vehicle for private gain, as Apple's data controls ultimately served to capture advertising revenue. Google's openness, meanwhile, follows a similar logic. Its strategy of attracting all users and developers onto its platforms is ultimately a bid for ubiquity and to become the infrastructure through which everyone experiences the internet, rendering the question of openness increasingly moot. This stands in contrast to the decentralised architecture upon which the internet was originally built and which some actors still aim to promote (Musiani, 2015). Openness, therefore, should not be the only variable policymakers consider when assessing the future of the internet. What matters equally is how different infrastructural choices shape which activities are possible, which interactions are enabled, and whose interests are ultimately served, whether the platform is closed or open.

What the Apple and Google cases share, despite their opposing strategies, is an outcome of concentrated control, and it is this concentration, more than any particular approach to openness, that poses the most pressing challenge for internet governance. The internet was originally built as a shared infrastructure, constructed and governed collectively by the community of actors using it (Abbate, 2000, p. 5). What emerged in recent years is an infrastructure dominated by a few platforms that control how it evolves. Beyond shaping how content is distributed, these platforms are increasingly involved in designing other layers of the internet, including the network of submarine cables connecting the world (Gjesvik, 2023). At the same time, most internet users cannot and would not want to contribute to building the internet's infrastructure, as its early architects perhaps dreamed. Moreover, tendencies toward centralisation are deeply embedded and unlikely to fully disappear (Farrell & Newman, 2019). While a return to the internet's early technical community is neither possible nor desirable, policymakers should nonetheless act to rein in platform power and build an internet infrastructure that serves the public interest.

Building the internet for public interest

To build an internet that serves the public interest, policymakers could focus on four strategies: requiring greater transparency from platforms, promoting competition in digital markets, deploying taxation as a check on platform power, and building international coalitions with like-minded countries.

First, an important issue in internet governance nowadays is the lack of transparency of digital platforms. How platforms' decisions affect the lives of all users remains largely hidden under the cloak of business secrecy. Policymakers should require platform companies to disclose more information about their business models and algorithms (Pasquale, 2015). Second, promoting greater competition is essential to prevent platforms from degrading their services once they achieve near-monopoly status (Doctorow, 2025). This should notably take the form of promoting greater interoperability, reducing barriers to entry and supporting open-source solutions. Third, a major source of power for digital platforms is the vast wealth they have accumulated while avoiding paying taxes in most jurisdictions around the world. Countries reasserting their taxation authority would simultaneously help reduce platforms' influence and provide governments with a significant new source of revenue. This argument mirrors similar reasoning in environmental governance, where taxation has been proposed as a tool to reassert public authority over powerful private actors (Green, 2021). Finally, none of these strategies can reach their full potential without international cooperation. While global consensus is difficult to achieve, governments can build coalitions of countries sharing similar values, and by offering access to their digital markets as an incentive, create pressure on platforms and non-participating countries to follow suit (Colgan, 2021, pp. 201-202).

The European Union has made notable progress on several of these fronts, particularly through its Digital Markets Act and Data Act. Both promote greater competition by requiring large digital platforms to support interoperability and share data with users and third-party companies. A growing concern, however, is that European regulators may refrain from enforcing these rules under pressure from the United States to avoid regulating American platforms (Espinoza & Foy, 2025). Such pressure has already led the European Commission to drop its plans for a digital services tax following the re-election of Donald Trump (Sorgi, 2025) and pushed Canada to abandon its own digital taxation plans (Lopez Steven, 2025). It is precisely here that international cooperation becomes essential. Few countries can effectively regulate large platforms on their own. However, a united front would both increase pressure on platforms to change their practices and reduce the risk of bilateral retaliation from the United States.

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Can the Open Internet Survive Geopolitics?

Matteo Nebbiai

Introduction

National states' policies increasingly determine the architecture of the internet. Long gone are the days of John Perry Barlow's "Declaration of the Independence of Cyberspace" (Barlow, 2019), which promoted the internet as a self-governing legal space, independent from national states and governments. On the contrary, moving from an initially loose approach, states have progressively regulated more aspects of the internet, including contract law, intellectual property, and content moderation (Bradford, 2023; Goldsmith & Wu, 2007; Hulvey, 2022).

Despite previous crises and contradictions (Staniland, 2018), the current shift away from the Liberal International Order (LIO), intended as an order whose principles are liberal democratic polity, free trade and multilateralism (Lake et al., 2021), seems more significant due to the aggressive use of economic and military coercion by the global hegemon, the U.S. (Choer Moraes & Wigell, 2022; Herranz-Surrallés et al., 2024) and the increasingly weak position of multilateral international organisations such as the WTO to solve disputes. Could this shift away from the Liberal International Order make the internet less open?

Tracking the impact of geopolitics on internet openness

The potential impact of geopolitical shocks, such as a change in the LIO, on internet openness can be analysed by examining shifts in political preferences, institutions, and internet layers (**Error! Reference source not found.**).

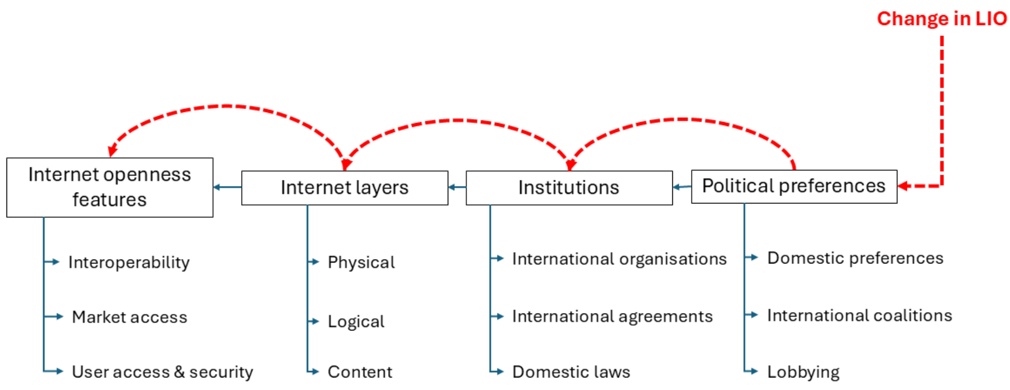
Although internet openness is a contested term that involves multiple elements (West, 2016), it is possible to distinguish three main elements: interoperability, market access and user access and security. Interoperability refers to the technical and legal barriers to connecting systems across organisations and countries (Doctorow, 2023). Market access is the presence of legal and economic institutions that allow organisations to operate across national markets, such as regulations on data flows (Ferracane et al., 2018). User access and security refer to users' capacity to fully access internet services (e.g., the absence of bans and the availability of broadband) (Freedom House, 2025), and the confidentiality, privacy and security granted when doing so (e.g., end-to-end encryption) (Baker et al., 2024).

Each of these features is determined by the technical architecture of the layers composing the internet (Lessig, 2001, 23): a physical layer composed of undersea cables, data centers, satellites, a logical layer composed of protocols such as internet protocol (IP) and the transmission control protocol (TCP), that address and route the delivery of data across services, and a content layer, where user-facing services and applications operate, including web browsing, messaging and platforms.

The architecture of these layers, which in the early days of the internet was shaped by researchers and companies, has increasingly been determined by institutions such as international bodies (e.g., ICANN determining the Domain Name System DNS), agreements (e.g., data flows provisions in Free Trade Agreements) and national laws (e.g., content moderation regulation).

Finally, these institutions are shaped by the political preferences of the states, both in domestic politics (what parties and citizens want), international politics (what coalitions states form), and lobbying (the influence of interest groups). The weakening of the LIO primarily affects governments' political preferences, with cascading repercussions for institutions, the layers' architecture, and ultimately the openness of the internet. [32] However, the length of this chain shows how a variety of factors might ultimately mediate this shock.

Figure 1: A framework to analyse the impact of international politics on internet openness.



Interoperability

At the *logical and physical layers*, interoperability is based on protocols governed by multistakeholder technical institutions and governance bodies, such as the Internet Engineering Task Force (IETF) and the Internet Corporation for Assigned Names and Numbers (ICANN) (DeNardis, 2009). In these bodies, the most common coalitions are formed around two poles: the G7 states and Russia and China. This has clearly manifested since the 2000s in the contestation by China and Russia, who perceived the internet as disproportionately governed by US entities and attempted to shift governance towards United Nations bodies, such as the International Telecommunication Union

[32] Political preferences are not the only drivers. Preferences of firms also play a role here, i.e., under the form of engineers participating to highly technical groups such as the Internet Engineering Task Force (IETF). However, I see as implausible that geopolitical events might significantly change the preferences of these actors without intervention of political actors.

(DeNardis, 2009, 167-168; Negro, 2020).[33] However, these moves never attempted to "decouple" the logical layer, as this would have enormous costs on countries with strong economic interdependency with foreign businesses like China (Jia & Winseck, 2018). At the moment, these bodies remain highly technical, and China does not seem interested in politicising them, as its technological catch-up has allowed it to shift from being a "norm taker" to a "norm maker" in some standards, such as 5G (Negro, 2023).

The interoperability in the *content layer*—like social media and messaging platforms—is more contested. Service interoperability is constrained by a thicket of regulations, including those governing trade secrets, intellectual property (IP), and data sharing (Doctorow, 2023). One of the biggest limitations to interoperability is anticircumvention law, which makes any retro-engineering of copyrighted technology illegal (Doctorow, 2025). Most governments worldwide have adopted these policies under pressure from U.S. diplomats during free trade agreement (FTA) negotiations (Brown, 2006). While a declining LIO might weaken the intellectual property system's legitimacy and countries' commitment to FTAs, companies retain significant powers to pressure countries into compliance: they can threaten exit from national markets (Culpepper & Thelen, 2020), and venue shopping: due to the multinational nature of companies, FTA provisions can be enforced even in countries that did not sign them (Gray, 2018). Additionally, a decreasing LIO might translate into increased access for U.S. big tech companies to lobby the government for greater pressure on foreign governments (Goddard & Newman, 2025; Wallach, 2025).

Some countries have attempted to increase interoperability through domestic law, such as data-sharing policies (e.g., UK Data (Use and Access) Act 2025) and service interoperability (e.g., mobile messaging interoperability in the Digital Markets Act). US administrations have long opposed policies of this sort because they are perceived as discriminatory against U.S. tech companies (U.S. Trade Representative, 2022). The second Trump administration's escalation of threats and sanctions (Le Monde, 2025; Munich Security Conference, 2025) might push countries to adopt a more cautious approach to internet interoperability regulation, especially when, at the domestic level, the governing parties politically align with the U.S. (Gkritsi, 2026).

Overall, the weakening of LIO seems unlikely to *reduce* current internet interoperability, given the logical layer's low political salience (Antoine, 2022) and the already low level of interoperability in the service layer. However, it might effectively block or chill attempts to increase it. In particular, middle-power governments might be disincentivised from proposing laws that increase data sharing and interoperability in the content layer out of fear of U.S. coercion.

[33] Some UN-driven inititatives, such as the Working Group on Internet Governance, stressed more the role of national government's sovereingty relatively to technical multi-stakeholder groups (DeNardis, 2009).

Market access

The weakening of LIO is increasingly prompting policymakers to think in terms of "weaponised interdependence" (Farrell & Newman, 2019) or "de-risking"(Gabor, 2023): dependence on resources or infrastructure controlled by a rival country might be used for surveillance or coercion. This will make a global market for the internet infrastructure and services increasingly unlikely. At the *physical layer*, physical internet infrastructure, such as data centres and telecom components, has increasingly been classified as "critical". This increases the likelihood that policymakers will privilege security concerns over economic efficiency (Walter & Trampusch, 2024), for instance, by banning certain firms or countries from providing services (e.g., 5G Chinese telecom equipment). In the *content layer*, policymakers might become more likely to privilege security reasons over free speech concerns: Sputnik and Russia Today operations were suspended by the EU immediately after the Russian invasion of Ukraine (Council of the EU, 2022), and the U.S. (temporarily) banned TikTok (Rozenshtein, 2025). This is a significant policy change only for countries open to international markets; countries already enforcing barriers to foreign firms' internet services, such as China, Egypt, and Russia, will continue to use or expand them (Freedom House, 2025).

There are, however, opposing forces. Between economically interdependent countries, business-minded policymakers resist regulations that hinder market access and foreign investment. The U.S. government and multinational corporations (MNCs) are opposing foreign data localisation or data sovereignty policies (Satter et al., 2026; Wallach, 2025). Internally, the EU seems split on an "industrial policy" approach that privileges European-produced products (Politico, 2026a), due to economic (Bauer et al., 2016) and political costs.

Overall, the weakening of LIO has different effects across layers. The content layer was already globally fragmented, mainly due to China and Russia's isolated ecosystems, so the impact is not great. The physical layer became significantly decoupled in some markets (e.g., telecommunications), but remains entangled in others (e.g., semiconductors). Finally, the fragmentation of the G7 market remains a key variable of future transatlantic relations and domestic political preferences.

User access and security

The weakening of the LIO seems to reduce the relative value of soft power vis-à-vis forms of economic coercion: this might lead to the reduction of funds for political initiatives to fund the open internet around the world, for instance, the slashed USAID budget contained various initiatives on internet Freedom Promotion (German Council on Foreign Relations, 2025).

On safety, an increasing sense of insecurity might translate into a lower priority for individual rights (e.g., data protection, encryption, consumer rights) when seen as in contrast with national security or competitiveness (Moens & Tamma, 2025). Moreover, the increasing aggressiveness against digital policies from the U.S. (Le Monde, 2025; Munich

Security Conference, 2025) introduces a trade-off between regulation and security: is it worth losing U.S. military protection to introduce digital rights? The "Brussels Effect", famously showing how regulation from big markets is more likely to be exported, might disappear (Ylönen, 2025).

Still, less regulation does not necessarily mean a less open internet. The type or regulation that is (not) adopted is crucial, and domestic party politics crucially drives it. As a result of the LIO weakening, right-wing parties could increasingly pick and choose between Trump's libertarian rhetoric against government intervention in social media and Artificial Intelligence (Lewis et al., 2026) and its litigation against the media and universities deemed too partisan (Barr, 2025). In response to the U.S. aggressive positions, left-wing and centrist parties will increasingly consider regulating the online environment as a political response to U.S. coercion (Politico, 2026b). However, the impact of measures such as social media bans for children (Malik, 2026) and cryptography exemptions to combat child pornography (Politico, 2025) on the openness of the internet is debatable.

Conclusion

The weakening of the Liberal International Order will not have a direct and immediate impact on internet openness, because such a shock is mediated by various actors (e.g. political parties and business lobbyists) and institutions (e.g. technical bodies and extant trade agreements). The biggest potential negative impact on internet openness is concentrated in countries with relatively open internet environments, such as the U.S., the EU, and India. In contrast, countries with already closed environments, such as China and Russia, will maintain their policies. Finally, the impact varies depending on the internet layer: the logical layer is mostly technical and depoliticised, and should therefore be more shielded from partisan and security-focused politics, while remaining dominated by firms' expertise ("quiet politics", Culpepper, 2010). The physical layer is most clearly impacted by weaponisation fears, leading to higher geoeconomic sensitivity: this means strong new market barriers against clear rivals (e.g., U.S. semiconductor sanctions against China) and ambiguity between potential future rivals (EU vs US). Finally, the content layer mixes fears of dependency with partisan domestic politics, especially regarding the trade-offs between free speech and online safety, and liberalised markets and industrial policy.

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Beyond Ex-post Regulation: Data Cooperatives for Aligned Markets and Governance

Sarah Nicole and Jeb Bell

Introduction

Governments played a prominent role in the development of the early internet and helped establish the underlying infrastructure of today's data economy. Today, however, they are largely sidelined, as private actors rapidly release increasingly sophisticated artificial intelligence (AI) systems. The consequence is not only an absence of government oversight and know-how, but an evolving market structure that looks likely to further concentrate profits and data in the hands of a few.

A 360-degree response is necessary to build a world where data agency is the norm: this requires governance and business models to align market incentives with individual agency and societal prosperity. Beyond ex post punitive regulations, which often face limits in application, data cooperatives, member-governed structures built on democratic control and federated growth, offer a proven alternative that aligns market incentives with public interest. By leveraging the existing cooperative model and market-shaping tools like interoperability standards and regulatory sandboxes, governments can decluster innovation and create conditions where ethical alternatives thrive.

The limits of reactive governance

The emerging policy challenge concerns the role and tools governments have at their disposal to shift the course of the growing generative and agentic AI economy. A traditional response is ex post regulation. Governments have long relied on punitive regulatory approaches to mitigate negative externalities or correct business practices that harm citizens or communities. But this type of regulation has several limitations.

The pace of technological innovation is one. It is spurring the release of new AI systems at a rate much faster than conventional timelines for developing and implementing new punitive regulations. Therefore, governments need to act with an eye toward the future, not the past. Already, there are indications that, thanks to AI, the future economy will look nothing like the one we experience today (Taylor, 2026).

At the same time, punitive, ex post regulation is not well-suited to address the deep structural conditions that define the market, such as data rights. By focusing on enforcement *after* damage has occurred, governments missed an opportunity to intervene earlier in the innovation cycle, where both standards and incentives could drive more fundamental changes in the marketplace.

The effectiveness of ex post regulation, even at the surface level of prohibited or harmful business practices or market behaviours, is also questionable. Europe, for one, is witnessing the limited impact of punitive regulation. Fines and penalties seem to be baked into the budgets of many tech companies as the cost of doing business in Europe. Hyperscalers, in particular, seem largely undeterred by punitive measures (Proton, 2025). The focus on "bad actors" in the ex post regulation also distracts governments from the needs of "good actors" – those building alternative technologies, open-source solutions, and innovative business models.

These challenges suggest that regulation is not a sufficient government strategy, especially in the era of rapidly evolving AI systems. What is needed is a proactive approach that shapes markets toward desirable outcomes from the start, creating conditions where ethical innovation becomes the default rather than the exception. Data ownership and control—core to an AI-driven economy—are prime areas where governments could take steps to encourage a more competitive, dynamic marketplace that benefits citizens and communities as much as companies.

The current extractive business model between concentration and innovation

There is no question that some of the most successful AI companies today have relied on mass data extraction to train and expand the generative and reasoning capacities of their AI systems. This approach favours incumbent hyperscalers and those first-to-market. It also concentrates innovation in the hands of a few tech giants who control vast data repositories and the computational infrastructure to process them. Smaller companies, researchers, and community-based organisations are locked out, unable to access the data resources necessary to develop contextualised solutions for specific populations or problems. This concentration stifles the diversity of innovation that emerges when a broader range of actors can participate in technological development (Acemoglu et al. 2019).

Most hyper-scaled platforms that rely on the mass extraction of data to train frontier AI models struggle with quality and relevance when training large language models (Sajith, Srinivasan, 2024), making them poorly suited to community-specific needs. The dominant advertising-driven business model treats data as a commodity to be indiscriminately harvested, prioritising volume over value. This approach not only undermines individual privacy and agency but also proves increasingly inefficient for genuine innovation.

Besides, this model is reaching its practical limits. Having scraped virtually all available web data, leading AI companies now increasingly rely on synthetic data. This creates a closed loop, posing a risk of degrading model quality over time. It further entrenches the advantages of incumbent players who control the infrastructure for generating and processing such data. Alternatively, put more simply, the system built on extracting ever more human data now finds itself cannibalising its own outputs because it has exhausted genuinely novel human-generated content. The rise of agentic AI systems only compounds this problem further.

Beyond inefficiency, the extractive model actively harms society (Schaake, 2024). It fuels manipulative advertising practices, algorithmic discrimination, and the erosion of democratic discourse through attention-capturing content that prioritises engagement over truth. The business incentive to maximise data extraction and user engagement creates platforms optimised for addiction and polarisation rather than human flourishing.

A market-shaping alternative: the data cooperative model

The opportunity lies in investing in alternative solutions that fundamentally shift the incentive structures powering the attention economy, the advertising model that drives companies to extract personal data for profit. Governments can act as market shapers rather than mere regulators. Something they already do with tools such as procurement policies, tax incentives, and infrastructure investments. These instruments have historically catalysed entire industries by creating demand for alternative business models and rewarding practices aligned with public interest (Mazzucato, et al. 2025).

Data cooperatives offer a structural match for this alternative vision. Their principles, including democratic control, shared ownership, and member benefit, align closely with the nature of data and the need for more equitable data governance. The value of data is contextual and relational, not intrinsic or fixed; the same dataset can support multiple, even conflicting, uses (Nicole et al., 2025). Data cooperatives represent an emerging yet proven model. Indeed, data cooperatives are a type of data intermediary that uses long-established cooperative models and global regulatory frameworks to manage data with or on behalf of members (data producers) for the benefit of members or for trade with external queriers (data users) (Flink, 2024).

How cooperatives scale offers distinct advantages for managing and governing data. Unlike conventional corporations, which grow through centralisation and vertical integration, cooperatives expand through federation, building networks of autonomous, member-driven organisations that collaborate while preserving local governance. The U.S. rural electric cooperative network illustrates this well: it covers more than half the country's landmass yet comprises over 900 independent entities. This federated approach allows cooperatives to remain responsive to and accountable within their local communities, while still benefiting from shared infrastructure, pooled data resources, and coordinated strategy. That kind of distributed growth naturally reflects the decentralised, context-dependent, and interconnected nature of data itself.

Many governments already have legal frameworks for cooperatives, from Switzerland, Brazil, France, the UK, and the U.S. This existing infrastructure provides a significant advantage amid the legal uncertainty surrounding AI and emerging technologies. However, supporting frameworks and policies are still lacking and would benefit the growth of the data cooperative ecosystem.

Proposed policy pathways towards proactive market shaping

While the cooperative legal structure exists, adapting it to data governance requires addressing novel challenges around data rights, portability, interoperability, and collective bargaining power in digital markets. The lack of clear legal definitions, the complexities of establishing data fiduciaries from data cooperatives, data trusts, etc., and the absence of supportive policies, especially at the local level where co-ops usually operate, create significant barriers.

Governments must move beyond reactive regulation toward proactive market shaping. This requires a multifaceted approach that combines the following:

1. Create legal pathways between data cooperatives and other legal models. This would decluster innovation from a single organisational archetype by removing barriers that prevent data cooperatives from embracing innovation the way legal models like startups often do. Governments should explore hybrid organisational structures that allow different entities to adopt principles from multiple models while accessing the tools and resources typically reserved for each.
2. Support interoperability standards by demonstrating that cooperative models inherently offer stronger data protection than extractive platforms, not through ex-post enforcement that has repeatedly failed, but through structural design built on open standards and member control. This shifts the narrative from punishing bad actors to empowering trustworthy alternatives.
3. Establish regulatory sandboxes and experimental zones, helping test and refine cooperative models before scaling them broadly. These controlled environments allow innovators to navigate legal uncertainties while providing regulators with real-world evidence of what works and which barriers need to be addressed.

The transition from punitive ex-post regulation to proactive market shaping will not happen overnight. But the costs of inaction are clear: continued concentration of data power, erosion of individual agency, and the foreclosure of alternative paths for technological development. Data cooperatives offer a proven organisational form adapted to the unique challenges of data governance, one that aligns economic incentives with democratic values and distributes both the benefits and control of data among those who generate it. The question is not whether such alternatives are possible, but whether governments will act decisively to create the conditions for them to flourish.

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Open Knowledge in the Age of Extractive Digital Ecosystem

Camille Françoise and Michele Failla

The evolution of economic models in a digital ecosystem

Before the development of the internet, business models for digital ecosystems were mostly closed. Organisations or companies would develop a product or service to sell at a lower price than their competitors. The first digital encyclopaedias, such as Encarta, are an example. The past decades have brought a drastic change in the business model promoted by the US digital companies, whose strategy shifted from the service-for-a-price model to offering a service “for free”. They enacted two aggressive strategies to capture the market: (1) forcing competitors out of the market and, once a monopoly was established, demanding payment while preventing competitors from accessing the market and therefore sustaining lock-in mechanisms; (2) using user data as payment to facilitate the resale of this data to external parties.

Competition laws in Europe have struggled to keep pace with these practices and address the challenges they pose to legal frameworks, either through the exploitation of grey areas or through legislative gaps. The lack of effective tax mechanisms to address the competitive advantage of global tech companies operating in the European Union (EU) over EU-established companies further complicates matters. The European Commission has partially addressed these challenges with the adoption of the Digital Markets Act (DMA) and the Digital Services Act (DSA). It has been actively working to regulate large platforms, with the objectives of limiting monopolies, enforcing existing laws, and striking a balance between content moderation and fairer revenue streams.

However, in its ambition to establish a digital ecosystem that is fairer and more respectful of people's ability to exercise their own self-determination online, the European legislators have focused on limiting the impacts of monopolistic enterprises, without proposing what a desirable future of digital platforms and commerce should look like to foster a thriving digital ecosystem in the EU. In other words, it did not address the compatibility of certain business models with European values enshrined in Article 2 of the Treaty on European Union (TEU). Policymakers seem to be forgetting that there are infrastructures such as the Wikimedia projects, of which Wikipedia is the most famous, that have a unique model; a model that supports desirable digital infrastructures: open source, transparent, community-driven, and privacy-focused. This peculiar model represents the most unique democratic collaboration system within a digital ecosystem. The unique visibility offered by Wikipedia often allows policymakers to carve out spaces for this model to continue legally. But what about other Digital Commons, such as OpenStreetMap? Wikimedia is part of the Digital Commons ecosystem, which aims to create this desirable digital future, and deserves more carefully designed policies.

Open data, open content: fuelling big tech or open democratic societies?

Information enables empowerment. Wikimedia projects contribute to gathering knowledge to share it freely and openly. They enable citizens in their daily lives by providing access to neutral, verifiable information, supporting education, autonomy, empowerment, informed decision-making, democratic participation, accountability, innovation, economic development, social cohesion, and resilience for all people, wherever they live. The model provides equal access to all through the information available within the open ecosystem: an academic in Argentina, a farmer in Belgium, a civil servant in Thailand, a CEO of a medium-sized enterprise in Kenya, or a large tech company in the United States or Europe.

The interstices of this model created a condition under which big tech companies can exploit existing laws to extract value from data and content. The black box syndrome challenges the protection of personal data under the GDPR: a lack of transparency, a lack of human rights enforcement, and the extraction of value at unprecedented scale, against the intentions of the creators, occur without fair remuneration, increasing wealth inequities and social developments. These models challenge the notion that information is equally accessible and reusable by everyone, which aligns with the idea of equity.

This raises a question whether, in democratic societies, restricting access to information because of the inequitable and extractive use by a few companies is a justified response. Such restrictions would have a major impact on people, including the progress towards the United Nations Sustainable Development Goals (SDGs). However, maintaining the status quo is not a viable option either. Finding solutions that will allow for fair and equitable remuneration mechanisms, ensure the visibility of sources, including human contributions, and facilitate transparent, accountable infrastructural designs for the digital commons ecosystem are urgently needed.

Economic case for protecting common goods in a predatory digital ecosystem

One main aspect of the difficulties that digital commons face is not only the question of the materiality and digitality of the infrastructures, but also the financial extraction of the value contained in data, which is turned into financial flows to the benefit of a few concentrated global powers. This process diminishes the power of other infrastructures and communities while tilting the balance of power in favour of a few big tech companies.

In the era of the attention economy and data extraction in exchange for access to monopolistic digital infrastructures, the Wikimedia Movement and Projects remain one of the last bastions promoting the values of the open internet and net neutrality. Monopolistic digital infrastructures prioritise short-term commercial gains by enclosing public spaces and failing to respect and promote the fundamental rights of individuals and communities.

Wikimedia Projects do not pursue profit and are oriented towards the public good: providing free knowledge and neutral, verifiable information to everyone. They do not sell information or collect or sell user data. They do not exploit attention-economy ecosystems, such as addictive designs, to keep people in the infrastructure. Algorithms are not used to give readers what reinforces their own beliefs and convictions based on profiling methods. Respect and promotion of people's privacy serve to protect them and their ability to self-determination by fostering critical thinking. Equal access to information, the same facts, and multiple perspectives are the crucial enablers for forming opinions and meaningfully participating in democratic life.

Supporting communities and infrastructure, whether physical or digital, requires financial streams. This raises the question of how to fairly redistribute the value back to the people who created it. The goal is to continue developing a more equitable, fair, and inclusive society, while protecting the commons and digital public goods.

One solution adopted by the Wikimedia Foundation is the launch of a commercial service, Wikimedia Enterprise, to ensure that the value extracted by a few powerful big tech companies is at least partially returned to the people who created it: the Wikimedia Community. Such a solution, however, is not definitive given the unique features of Wikimedia projects and the undesirability of a one-size-fits-all approach. Other NGOs and Commons may have different business models, which may prevent them from effectively engaging in negotiations with big companies. Born a quarter of a century ago, Wikipedia serves as a reminder that the principles of net neutrality and an open internet are under attack by a few monopolistic enterprises, which replicate and reinforce inequalities at every level of society.

Internet Openness by and for Whom? The Role of the Digital Rights Community and Investors

Tayrine Dias and Anita Dorett

Introduction

internet openness can be described in terms of its collaborative, modular and scalable architecture and infrastructure, open protocols and standards, distributed networks, unequal but widespread accessibility across countries and collectives, and the legal and regulatory frameworks that organise how information circulates online. Considering that access to the internet is widely understood as a civic or human right (Reglitz, 2024; United Nations, 2021), internet *openness* gains further importance, especially when it comes to the expansion of the spectrum of openness to include the layers of corporate accountability and multistakeholder governance of the internet and online services. Therefore, the calls for a more open internet should take into account *who* participates in the private and public governance of the internet and its infrastructure, and *how* these actors do it beyond the more paradigmatic notions of openness. Digital rights organisations and activists, civil society, and the community of responsible investors all play crucial roles and offer unique perspectives in reconstructing a more inclusive and resilient practice of online openness.

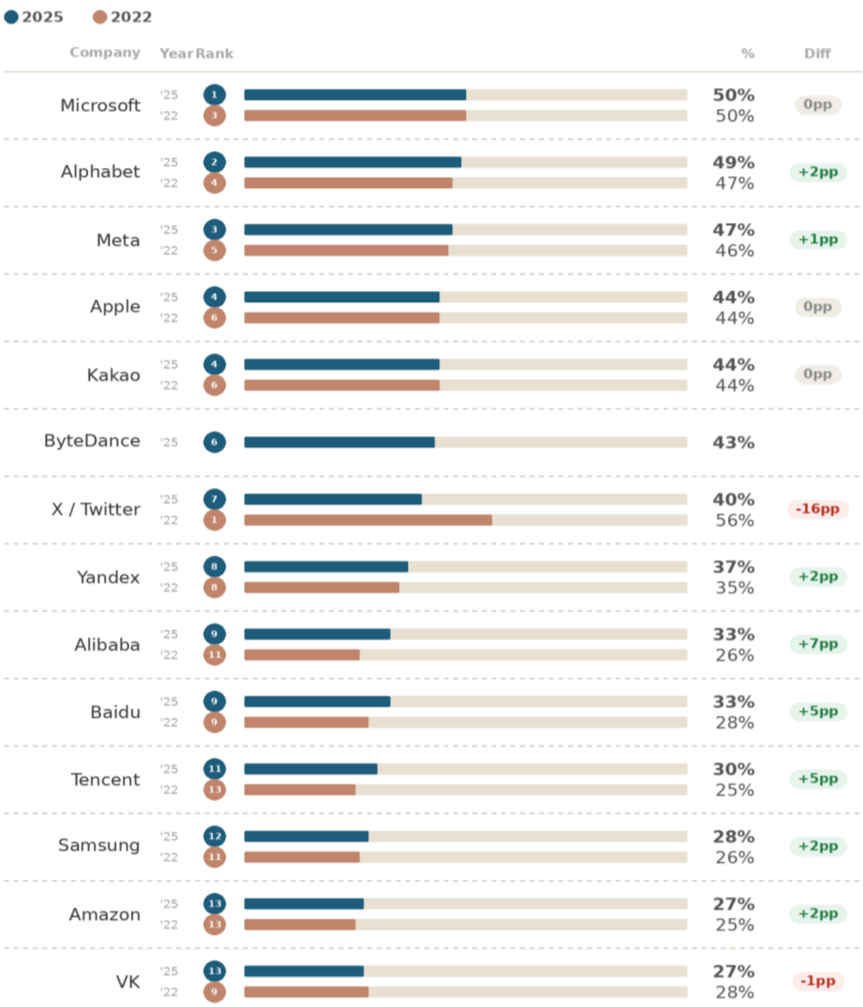
Digital sovereignty and multistakeholder governance

In the current context of intense geopolitical competition and growing insularity, a fiercer dispute among a few national players to control the tech value chain encompasses everything from chip production to shadowbanning and censoring activists online. In this context, the idea of digital sovereignty has gained greater traction among other governments seeking to sustain or regain their positions amid the growing concentration of technopolitical power. While in Africa (African Union Commission, 2022) the debate around digital sovereignty strongly leans into the risks of data extractivism, surveillance, and exploitation, in Europe and Canada these discussions have largely focused on tackling US dependence when it comes to compute power, cloud infrastructure and platforms, with little effort to recognize and tackle their own role in producing those negative impacts in the Global Majority.

Most of the debate around digital sovereignty has understressed other stakeholders, such as civil society, investors, and the field of 'smaller' tech, particularly those representing the Global Majority, who are necessary to counter the tendency toward closed systems that stems from the primacy of national or regional interests in the Global North. The recent episodes of shutdowns, censorship and political crackdowns in Russia and Iran or the deployment of ICE's technosurveillance apparatus in the US, highlight the dangers of the unchecked implementation of digital sovereignty and show how fragile the spectrum of openness can be when power concentrates either in governments or a handful of companies constituting quasi-monopolies.

This challenging political and economic scenario also threatens to drive a further decline in internet openness. The network and its governance are squeezed among geopolitical tensions, "enshittified"[34] business models (Doctorow, 2025) and big tech pressure for further deregulation, more so as the generative and agentic AI gold rush pushes ahead at full steam. In 2025, Ranking Digital Rights reported that tech giants are moving slowly and not addressing well-documented issues related to transparency, governance, and the implementation of mechanisms to protect and respect digital rights.

Figure 1: RDR Index: Big Tech Edition—2022 and 2025 comparison



Source: Total scores for companies assessed in the 2022 and 2025 RDR Index: Big Tech Edition, based on the scores across the categories of governance, freedom of expression and privacy.

[34] Cory Doctorow coined the term in 2022, to describe "the sudden on-set platform collapse" (Doctorow, 2025, p. 3), explaining the way an online service got worse, how the worsening unfolds and the contagion effects observed (p. 4)

Companies have stagnated in building multistakeholder governance processes, or in protecting the freedom of expression and privacy of their users; among the 14 companies assessed, the biggest gain from 2022 to 2025 was 7 percentage points.

To resist threats to openness in the form of deregulation, “enshittification” and national or region-centric response that may entrench inequalities and digital rights violations, we pose the following questions: what is the role of digital rights organizations and civil society in reshaping practices, advocating for policy change, and dismantling big tech power concentration and what role can investors play in supporting and enabling these efforts? Which strategies and paths have these groups designed and implemented, or which ones should they consider to resist these threats?

Advocacy efforts: new paths of resistance

In the last few decades, a significant part of the digital rights advocacy effort has been directed towards increasing transparency to promote a race to the top in the field, particularly through the publication of benchmarks and reports. However, more organisations, activists, digital rights advocates, investors and other stakeholders have been calling for more substantive actions, pushing companies to move from declarations and commitments to enforcement mechanisms that translate policies into practices with measurable impact benefiting users.

From a civil society perspective, advocacy efforts have then shifted to redefining approaches in the tech ecosystem, reducing the burden on users to choose and switch between platforms, and strengthening spaces that institutionalise corporate accountability. Some of the more relevant strategies include the following.

Interoperability across online platforms

Open common standards[35] enable communication and the flow of information across platforms, giving users more control over data portability, platform interactions and choosing which services they want to use and pay for. This strategy is at the cross-platform infrastructure level, with potential benefits for users and the whole ecosystem by promoting platform and service decentralisation.

Supporting alternative business models and platforms

Small Tech platforms are explicitly designed as alternatives to big tech, which may enact responsible tech governance more rapidly, driven by the avoidance of extractive business models, greater proximity to users, and simpler decision-making structures. Nebula, Qobuz, DuckDuckGo, Mastodon, Proton, and Bluesky are notable examples of small tech. These platforms either commit to or organise their services around greater user freedom over content, higher privacy standards or more content creator power. For instance, Mastodon's federated architecture distributes moderation authority rather than concentrating it, Proton's legal domicile and technical architecture make mass

[35] See more on interoperability at the Fediverse webpage: https://socialwebfoundation.org/what-is-the-fediverse/

surveillance structurally difficult, and Nebula is owned in part by the creators it hosts. A more open, decentralised and inclusive digital ecosystem depends not only on pressure towards big tech, either by civil society initiatives or regulatory efforts, but also on support for alternatives with policy (via public procurement strategies, for example) and broader stakeholder alliances that shape a multistakeholder governance approach focused on respect for digital rights.

Multistakeholder spaces with accountability processes

Participation in digital rights initiatives such as the Global Network Initiative (GNI) is an advocacy strategy that brings together practitioners, activists, academics, investors and tech companies, providing a framework and space to share best practices, empower policy and promote accountability. One challenge of these fora is that they tend to be dominated by the larger tech companies from the Global North. At the same time, Global Majority actors are less represented due to resource constraints, therefore failing to enable and sustain a shift or sharing of power in the hands of stakeholders other than companies, especially the largest platforms. This strategy potentially improves openness at the cross-platform governance level, benefiting users.

Evidence-based campaigning and alliance-making

Conducting cross-topic and cross-regional independent research has been crucial to supporting all the strategies mentioned above. The production of benchmarks, standards, recommendations and guidance helps civil society mobilise and helps investors engage and support efforts by regulators and companies to conduct business activities that account for societal expectations. As important as this strategy may be, without resistance paths to ensure and enforce more competent corporate practices, processes, governance, and accountability, this strategy risks losing its relevance and influence. Alliances between policymakers, investors and activists are a form to strengthen this strategy's impact. This approach is at the platform and ecosystem governance level, with benefits for users and the whole ecosystem's accountability.

The approaches above combine a path to create spaces for digital rights organisations, activists and affected communities to have a say, supporting data ownership and portability to reduce the ecosystem centralisation, enabling alternative services and approaches, challenging the black box around big tech services and products, and questioning the ubiquitousness of their business models.

Where such spaces are created for more transparency, decentralised decision-making and more robust accountability, investors are also better equipped and informed to be advocates and engage with their portfolio tech companies for a more open internet that more equitably serves and respects its users' rights, thereby increasing corporate value with more sustainable, inclusive and resilient business models that secure long-term investment returns.

Financial leaders and investors: strategies and angles to promote internet openness

The internet's early design was based on principles of openness, underpinned by a public interest ethos. As power concentration grows, it has been reconfigured as a collection of closed and tightly controlled platforms that support surveillance business models built on centralisation, extractive data practices and data monetisation. In parallel, the astronomical rise and gargantuan growth of the tech sector, dominated by a few large platform companies that have flourished in a regulatory void, have been the investment darling of the capital markets in the last decade. With the long-standing status as some of the most valuable companies in the world by market capitalisation, big tech is virtually in every investment portfolio. In this scenario, what role can investors play, alongside digital rights organisations, to support internet openness?

Institutional investors, as business actors, have a responsibility to respect human rights in line with the UN Guiding Principles on Business and Human Rights (UNGPs). Through every stage of investment decision-making, investors are expected to manage human rights risk by conducting human rights due diligence, which includes understanding risks to impacted rightsholders through collaboration with civil society and engaging with their portfolio companies to address these risks. Where portfolio companies fail to manage human rights risks, this will also result in significant operational, financial, legal, and reputational risks—hence adversely impacting the value of their investment.

Most investors are looking to maximise the long-term value of their investments to secure sizable retirement benefits and returns for themselves, their families, and future generations. To achieve this goal, we argue that investors must focus on the *resiliency* of their investee companies to sustain long-term growth and value creation, which is built on equitable and robust business models that respect broader social and environmental interests through responsible and transparent business practices. We claim that a more open internet ecosystem is more resilient *because* it is more decentralised and more permeable to power checks that can enshrine rights in line with societal expectations. In contrast, as per the Ranking Digital Rights data presented above, the stagnation of big tech's transparency, governance, and implementation of mechanisms to protect and respect digital rights tells a different story—one that reflects the universal decline in global public trust (Indah & Sukresna, 2025) in large platform companies and their failure in upholding openness principles.

To respond to this challenge, some approaches and strategies for investors to re-evaluate, consider and aspire to enabling a more open internet include:

1. **Reaffirming principles of sustainability in investment due diligence and decision-making** that direct investments to companies that prioritise responsible business practices that support long-term value creation and reduce systemic risks, enabling a more resilient, equitable, and inclusive economy that benefits all stakeholders. By insisting on sustainable investing with durable and continued returns, investors can drive transformative change in corporate behaviour and enhance overall market integrity.

2. **Addressing risks that can affect the long-term viability and profitability of tech investments.** The “enshittified” business models where users’ behaviours, preferences and personal data are commodified, exploited and manipulated with little to no regard for their safety, mental health, informed consent, free choice or their digital rights; driven by the pursuit of short-term profits at the expense of long-term sustainability and public interest, is a core risk that must be addressed. Investors know well that tech companies put profit before people, and this has played out in the first bellwether case in the landmark social media trials in California, where the verdict concluded that the defendants—Meta and Google—deliberately built their platforms to be addictive, and the companies' executives knew this and failed to protect their youngest users. (Guynn, Snider & Brown, 2021; Chmielewski, Rozen, & Godoy, 2026). The outcome of this bellwether case is tied to about 2,000 other pending social media addiction lawsuits brought by parents and school districts. Failing to respect users’ digital rights, hiding and designing manipulative and addictive algorithmic systems behind proprietary walls and treating people and collectives as data resources to extract—investors need to address the resultant persistent and massive reputational, financial, business, legal and statutory risks and losses—and more significantly the losses and risks to human lives, safety and dignity. Under the UNGPs, if potential or actual harms are identified and have not been prevented or mitigated, investors must consider steps to end investment relationships responsibly.

3. **Strong corporate accountability governance structures**. Publicly listed tech companies have enshrined a dual-class share structure that guarantees founders and management an outsized voting power compared to ordinary shareholders. Despite strong views and support from most ordinary shareholders to address a company’s accountability gaps, founders and management can control voting outcomes at shareholder meetings, thus silencing investor voice and inhibiting participation in corporate governance. Investors need to continue advocacy efforts to end dual and multi-class share structures in public capital markets, which have enabled corporate actors to operate with impunity and consolidate power while engaging in rights-abusing business practices.

4. **Diversifying investments** to support tech alternative business models and platforms with the explicit intent of cultivating open and responsible tech governance. As discussed in the previous section, these would be small tech and new models of ownership and distributed governance committed to a more open, decentralised and inclusive digital ecosystem, that organise their services around user freedom and privacy. One of the hurdles to implementing this strategy is that many of the alternative small tech players have avoided looking to the capital markets for funding, choosing instead to “billionaire-proof” their services, protecting their user-centred mission and principles rather than enriching shareholders. A remaining question is how investors, who are also consumers, can help support alternative business models and platforms while avoiding the

traps of requiring unsustainable profit-on-profit growth that has led to the entrenchment of today's dominant big tech paradigms?

5. **Alternative investment structures and approaches** that provide capital to address pressing global challenges, operating, for example, at the intersection of digital needs of the global community for open, equitable and inclusive access to the internet *and* strategic investing. Investing in businesses that enshrine long-term mission and value creation at the core of their governance and operations through structures like benefits corporations or perpetual purpose trusts that take into consideration all stakeholders that are materially affected by that company's decision-making, like workers, customers, local communities, wider society, investors and the environment. Such structures can ensure that profits are directed toward the company's long-term mission, effectively insulating its mission from potential market pressures or changes in ownership that could divert its purpose. Impact investing[36] is another approach to generate positive social and/or environmental outcomes alongside financial returns.

6. **New frontiers in tech investments** where investors are incentivised to channel capital towards tech research funding and innovation to drive a more open internet. Through government-supported investment programs, government bonds and other incentives that guarantee annual returns and, as such, are seen as a solid and safe investment option, this can drive much-needed capital towards serving public interest in digital infrastructure and services that are free, open and interoperable to protect user choice, information access and civil liberties or human rights. One of the challenges in this approach is protecting public investment and the products or services created from being converted into proprietary assets, instead of upholding principles of openness and transparency.

Conclusion

The internet's openness is facing a triple threat: aggressive deregulation, the "enshittification" of extractive business models, and the rise of national or region-centric agendas that risk entrenching inequalities and digital rights violations. Resisting these converging pressures requires a multistakeholder alliance, which includes the digital rights community and responsible investors. These two collectives have collaborated and exchanged mutual support in a myriad of governance spaces and campaigns.

On the advocacy side, civil society organisations are pursuing interoperability standards to reduce platform lock-in, building coalitions for evidence-based campaigning, championing small tech alternatives with more decentralised, user- or creator-centric governance, and institutionalising corporate accountability through multistakeholder

[36] Impact investments are investments that deploy capital towards companies, funds or projects dedicated to addressing social or environmental issues, while still aiming at obtaining a financial return. See more in: https://thegiin.org/publication/post/about-impact-investing/

frameworks like GNI, in which investors can also play an active role. These efforts are essential to support investors in making informed and rights-respecting investment decisions.

On the investment side, responsible investors can reinforce these advocacy gains by embedding sustainability principles into decision-making, actively addressing the reputational, legal, and financial risks posed by rights-violating business models, diversifying portfolios toward more open and user-centred platforms, and channelling capital into public-interest digital infrastructure.

Digital rights organisations and responsible investors together represent a meaningful counterweight to concentrated technopolitical power. Their collaboration is crucial to building a more resilient, inclusive, and genuinely open internet that serves people before profit, creating and enhancing both economic and social benefit and value.

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AI Evaluations: Civil Society as the Standard Setter in the AI Era

Murielle Popa-Fabre

Introduction

Artificial intelligence is reshaping the meaning of internet openness. Where openness once referred primarily to open standards, interoperability and the free flow of information, it is now increasingly defined by who controls the accumulation of data, the opacity of algorithms, and the computational resources that determine who builds, owns and ultimately governs the AI stack. This calls for governance models that can sustain openness, transparency, and accountability in current AI ecosystems, while recognising the crucial role AI evaluation can play in both AI governance and competitiveness.

Openness in the age of AI is about AI observability

The modern AI stack can be understood across four interdependent layers: compute and foundation models at the base (Layer 1); data processing, pipelines and databases (Layer 2); deployment tools including agent frameworks, prompt management and orchestration (Layer 3); and at the top, a layer dedicated to observability, evaluation and security (Layer 4) (Tully, et al., 2024).

Observability refers to this uppermost layer that tracks, logs and monitors all actions of an AI system, and in particular all inputs, outputs and decision pathways of Large Language Model-based systems and agentic workflows. Tools such as Truera or Langfuse are already populating this layer commercially, making the statistical response patterns of generative AI visible and auditable in real time. What was, until recently, a niche concern of responsible AI practitioners is rapidly entering the mainstream in the business and enterprise world. Yet observability remains largely absent from governance discussions.

Consulting firms such as McKinsey QuantumBlack are now confirming what governance researchers have long argued: without systematic evaluation and monitoring, trust in AI systems cannot be built or sustained at scale (Yee et al., 2025). Investing in evaluations means following a clear technical path: track and verify every step of agentic workflows by building evaluation and monitoring directly into the pipeline. As a concrete illustration, a legal services provider using an agentic document review workflow that experiences a sudden drop in accuracy can, if observability tools are in place, identify the root causes in real time and course-correct rapidly, rather than discovering the failure weeks later through manual review.

Why observability is a governance imperative

The case for making observability a core component of AI governance frameworks rests on three arguments. First, it enables accountability by making visible what is otherwise undetectable to individual end-users: the statistical patterns of generated content and their reliability, including scores for completeness, factual accuracy, faithfulness, and relevance. Second, it enables oversight at the algorithmic and population level, capturing systemic response patterns that only become apparent at scale. Third, and critically, it is a tool for fighting lock-in. Without the ability to easily evaluate an AI application built on a given base model, organisations become trapped with their initial Large Language Model (LLM) provider, as manually verifying whether an internal AI system produces reliable outputs can take a company several months. Observability transforms an opaque technological dependency into a governable, auditable and interchangeable one. Moreover, doing so becomes a precondition for any meaningful form of AI openness.

Governance on device: the new frontier

The rise of Small Language Models (SLMs) running directly on devices—from Apple's M5 chip, explicitly positioned to "kill cloud AI", to Google quietly releasing an app in May 2025 to download and run AI models locally—is not merely a technical development. It is a governance rupture. The SLM ecosystem has dramatically expanded since 2022, with models such as Phi-3, SmolLM or Qwen 2.5 already capable of running fully on personal devices since late 2024 (Lu et al., 2024). As AI moves from the cloud to personal devices, an entirely new and urgent question emerges: What does "governance on device" look like? The question is no longer only who has the technical capacity to audit AI systems, but what an AI model does inside a personal smartphone or laptop, unobserved continuously and at scale. An observability layer at the device level becomes even more critical than in cloud-based settings. This is key for policy action, as one cannot govern what one cannot measure.

Measuring linguistic, cultural and political leanings: A democratic imperative

Concretely, measuring the cultural, linguistic, implicit values or political biases embedded in conversational AI is a complex and pressing task, now that AI agents are deployed and adopted at scale, including on personal devices (Popa-Fabre, 2026). The capacity to assess AI systematically requires carefully designed evaluation settings that account for the advanced levels of memorised data that shape interaction patterns with end-users.

Two concrete examples illustrate what is at stake: political biases in chatbots during peri-electoral periods, and culturally grounded biases that affect the competitiveness of AI global products.

One of the most sensitive and underexplored dimensions of AI evaluation and impact assessment concerns the implicit political leanings encoded in mainstream LLMs. Static evaluation tests already show that mainstream LLMs occupy positions on the political spectrum that are measurably different (Rozado, 2024), but a striking illustration comes from a recent study that reproduced political surveys with LLMs (Cao et al., 2025; Pollished, 2025). Following methodologies developed in commercial market research, the study assessed how different models predict candidates' scores in the French presidential election, revealing strong divergences across models: mainstream models span virtually the entire political spectrum, from Mistral on the far left to Gemini on the far right, with Meta, Anthropic, xAI and others clustered in between. Without measurement, citizens have no way of knowing which political orientation silently shapes the information they receive.

The governance implications are profound. Empirical studies have begun documenting that LLM-based systems can measurably sway political views (Hackenburg et al., 2025; Lin et al., 2025) and opinion shifts in interaction with conversational AI are increasingly well documented (Rogiers et al., 2024; Potter et al., 2024; Chandra et al., 2026). The models under scrutiny were not experimental, but rather the systems that millions of citizens interact with daily to seek information, navigate public debates, or form opinions. The fact that their implicit political orientations vary so dramatically and remain invisible to the average user makes AI evaluation and impact assessment a democratic necessity.

AI benchmarks: A venue for co-designing?

Peri-electoral periods in particular represent a critical window, where, without systematic, independent measurement of political orientations and their impacts on users, neither citizens nor regulators can know which political lens is shaping the information landscape. Political bias evaluation and the continuous monitoring of how political persuasion changes as models evolve should therefore become a standard and a publicly available component of AI accountability frameworks. Furthermore, civil society, not just developers, must have a seat at the table in designing them.

The proliferating, fragmented landscape of AI benchmarks makes this co-design question even more urgent. Critical gaps in accountability and comparability remain (Reuel et al., 2025), particularly in capturing the interactional and temporal dynamics of AI impacts such as persuasion, and in informing risk assessment and policy responses as new models are deployed. AI benchmarks, currently designed and controlled by a narrow set of actors, could and should be reimagined as participatory venues that bring together technical experts, policymakers, and civil society.

AI evaluation as a competitive asset

As LLMs become the bedrock of AI products and everyday applications deployed globally, the inability to establish a corpus of benchmarks to evaluate cultural and linguistic biases across languages and regions makes it not only impossible to assess discrimination but also to anticipate commercial underperformance.

India offers a striking case study of an English-speaking but culturally distinct country. Indian users were shown to require significantly more corrections to ChatGPT English outputs than their American counterparts, because language and culture appear deeply entangled in LLM training (Agarwal et al., 2024). Indian participants were not only less efficiently served by AI writing assistance, but were subtly pushed toward adopting Western writing styles, a homogenising effect that would be invisible without evaluation.

A recent multilingual study reinforces these findings (Jain & Chopra, 2026). Users were systematically disadvantaged by the language they used to interact with AI models, across 20 everyday questions on health, investment, career, and education in 6 languages. Translating 750 factual questions from the CulturalBench benchmark into six languages, the research team found that an LLM queried in English on Indian cultural subjects outperforms the same model queried in Hindi on the same subject. Thus, confirming the governance insight that bias evaluation must be multilingual to meaningfully serve the fairness and accountability (Popa-Fabre, 2026).

The implication for any company deploying AI globally is direct: cultural bias is not only an ethical problem but also a competitive liability for the global deployment of AI products. In a world where trust in AI systems is becoming scarce or contested, the ability to demonstrate measurable, evidence-based trustworthiness across languages, cultures and value systems is a genuine go-to-market advantage. For Europe in particular, positioning itself as a trust barycenter in the global AI landscape is a value-based industrial strategy that turns democratic accountability into a differentiator in the global AI race.

Conclusion

As the latest wave of AI fast becomes an interface through which citizens access information, form opinions and engage with the world in everyday applications, openness is no longer simply a question of open standards. It is now a question of who gets to observe, measure and ultimately shape the values embedded in AI systems and who can evaluate their impact.

AI evaluation is the mechanism through which that openness becomes meaningful and effective; it allows civil society to have a concrete say on the linguistic, cultural, political and epistemic biases or deceptive design patterns carried by mainstream conversational AI.

The stakes could not be higher. What is at risk is information integrity, one of the foundations of democratic pluralism. Evaluation frameworks are not purely technical or company instruments; they are normative. Whoever designs the benchmarks and impact assessments shapes the values against which the technology is measured.

The AI standard-setting role should therefore be exercised through shared governance between civil society, technology companies, research institutions and policymakers. Civil society has the legitimacy to claim a seat at that table and to inject human agency into the incremental improvement of these systems, opening a path towards the resilience technology needs in this defining moment of the AI policy.

Building multicultural, multilingual, and value-aware AI evaluations and impact assessment frameworks is complex work that requires participatory governance to approximate pluralism in systems that increasingly shape public discourse. This is the new multilateralism and the practical contribution of techplomacy can make: forging an incremental, evidence-based path toward an AI future that reflects the full breadth of human cultures, languages and democratic values.

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Public AI and the Architecture of Sovereignty

Joshua Tan

Introduction

Artificial intelligence is becoming the core infrastructure of the digital economy, and Europe is building it on borrowed ground. The compute is American. The foundation models are American and Chinese. The data pipelines route through American servers, governed by American terms of service, shaped by the priorities of the current American administration. Europe has significant AI research talent, regulatory ambition, and a genuine tradition of treating public goods as public goods. What it has not produced is the infrastructure it controls.

This is not for want of trying. The European AI strategy has pursued two parallel paths: regulation and openness. The AI Act establishes risk-tiered obligations for AI systems deployed in Europe. And open-source AI—models like Mistral and Swiss AI's Apertus, released with public weights and permissive licenses—has been embraced as Europe's competitive answer to GPT and Claude. Both instincts are defensible. Neither is sufficient (Kak, 2024).

Regulation is a conduct remedy, not a structural one: it polices behaviour within an existing market without altering the market's underlying architecture. A Europe that regulates American AI is still a Europe that depends on American AI. Open source, meanwhile, has a complementary assets problem that its proponents have been slow to reckon with (Tan *et al*., 2025).

Why open source is not enough

Foundation models are not like traditional software. A Linux kernel, once released, can be deployed on commodity hardware by anyone with a laptop. A frontier language model requires persistent inference infrastructure, continuous fine-tuning pipelines, evaluation ecosystems, and governance institutions capable of maintaining alignment with changing requirements. Open weights without these complementary assets are like publishing architectural blueprints without providing land, materials, or builders (Tan *et al*., 2025). The challenges are structural, and they run three layers deep.

The first is resources. Modern models are trained on thousands of Graphics Processing Units (GPUs) over weeks or months, requiring capital and engineering infrastructure available only to well-funded corporations or state-supported institutions. But the resource problem doesn't end at training. Post-training—the fine-tuning, alignment work, and feedback loops that make models actually useful in practice—is typically kept closed. And inference at scale demands ongoing GPU access and cost management that most public institutions cannot sustain independently. Open weights without inference are inert: unusable to all but a small elite with the capital, compute, and engineering to deploy them.

The second layer is licensing. The layman's understanding of "open source" AI is more accurately termed open-weight AI—and even that framing is generous. The license of LlaMa—a family of open-source large language models (LLMs) released by Meta AI—contains restrictive terms and explicit revocability that the open source community has widely criticised as incompatible with established open source standards. Companies like Meta can stop releasing models or add restrictions to future licenses at any time. More fundamentally, releasing weights alone does not provide the transparency that makes open source software auditable and reproducible. With traditional open source, access to the code means the ability to understand, reproduce, and verify. With open weights, researchers typically lack access to training data, curation decisions, alignment procedures, and compute configurations. The "source" in open source once meant access to the complete blueprint; open weights provide only the finished artifact.

The third layer is governance. Open does not mean safe: openly released models are often less safe than their closed counterparts, because safety work—red-teaming, alignment tuning, monitoring for emergent harms—requires sustained investment that volunteer communities cannot reliably provide. And there is a deeper structural irony: community-contributed evaluations, tooling, datasets, and fine-tuning techniques often accrue value to large firms whose commitment to openness is tenuous at best. open-source contributors imagine they are building shared infrastructure; in practice, they may be fueling a pipeline that concentrates power. What looks like a commons is increasingly functioning as a subsidised input to proprietary systems.

France's Mistral illustrates the trajectory. Having released open-weight models as a calling card, it has subsequently pursued proprietary commercial models, Microsoft investment, and enterprise API services. This is a rational business decision; it is not a public infrastructure strategy. The same arc (open weights as marketing, proprietary systems as the actual business) is visible across the industry. The result is that as AI systems grow more complex, the gap between "available weights" and "usable systems" widens. What began as a token prediction from weights that fit on a local GPU is morphing into AI assistants that blend multimodal reasoning, access to proprietary tools, and complex orchestration layers. Open source does not face a failure of values. It faces a failure of structure. The open-source model, which flourished in an earlier era of low-cost computation and interoperable standards, is no longer sufficient on its own.

Public AI

Public AI begins from a different premise: that AI infrastructure should be governed as a public good, not merely that AI outputs should be made publicly available. The distinction is fundamental (Sieker *et al.,* 2025; Marda *et al.,* 2025; Vincent, 2023).

In practice, this means shared compute infrastructure operated under public mandate and open governance—not owned by a single company or a single government, but accountable to multiple stakeholders across jurisdictions. It means commons-governed datasets that create a data flywheel feeding back into shared rather than proprietary capability. It means open foundation models hosted on durable, non-proprietary infrastructure, available as utility services with transparent pricing, neutral governance, and no lock-in (Sieker *et al.,* 2025; Marda *et al.,* 2025).

Several institutions are already working in this direction. The Swiss National AI Initiative (ETH Zurich, EPFL, and the Swiss National Supercomputing Centre), is training the Apertus model on publicly funded supercomputing infrastructure, accessible to Swiss research institutions under a public mandate. The Barcelona Supercomputing Centre contributes sovereign compute to projects including multilingual models that serve Catalan, Spanish, and other European linguistic communities that would otherwise be underserved by English-centric incumbents. These are not ideological projects; they are practical experiments in what public infrastructure provision looks like at the model layer (Sieker *et al.,* 2025).

The Public AI Inference Utility serves many of these institutions directly, providing a public-access point for sovereign models, allowing users across jurisdictions to interact with publicly governed AI through a single interface without routing through proprietary cloud providers. The architecture is deliberately modular: compute contributions from different national institutions, model governance distributed across stakeholders, and access policies set by the hosting jurisdiction.

European strategy

Europe's strategic situation makes public AI unusually well-suited as a policy instrument for three reasons.

First, Europe already has the institutional infrastructure. CSCS, the Barcelona Supercomputing Centre, Jülich Supercomputing Centre, and similar facilities represent substantial publicly funded compute capacity. What has been missing is the governance and distribution layer to coordinate them for AI inference at scale—not more hardware, but an architecture for shared use.

Second, Europe's linguistic and cultural diversity is a failure mode for incumbents and an opportunity for public alternatives. ChatGPT is excellent at English but still inconsistent in many European languages, such as Catalan, Basque, Swiss German, Welsh, and others with smaller but politically significant user communities. Incumbents have limited commercial incentive to serve these communities well. Public AI institutions, whose mandates include cultural and linguistic representation, have both the incentive and, with coordinated compute, the capacity to do so.

Third, the European regulatory tradition already understands infrastructure as a public good. Water, electricity, rail, and telecommunications have all, at various points, been organised as public utilities, natural monopoly frameworks, or heavily regulated common carriers. The argument that AI's intelligence layer warrants similar treatment is not a radical departure from European political economy. It is a logical extension of it. The BBC, conceived as a public broadcast utility operating under democratic accountability rather than commercial interest, is a useful analogy: not state-controlled propaganda, but publicly governed infrastructure with a mandate to serve plural publics (Vincent, 2023).

The time is now

The deepest challenge for public AI is timing. Infrastructure markets exhibit strong path dependence. Once enterprise systems integrate with proprietary APIs, once workflows are trained around specific model behaviours, once switching costs accumulate, market structures become difficult to reverse. The AI stack is not yet locked. Foundation models are still being trained, infrastructure choices are still being made, and governance institutions are still being designed.

The European Commission's AI various factories initiatives, the proposed OpenEuroLLM project, and the emerging network of national AI institutes represent genuine steps toward public infrastructure provision. The gap is coherence: these efforts remain fragmented across member states, insufficiently coordinated at the European level, and without a clear governance model for shared public infrastructure that crosses jurisdictions (Kak, 2024; Sieker *et al.,* 2025).

Public AI offers one such model—not a single European state AI project, but a federated utility architecture in which member states compute contributions, national model development efforts, and distributed governance institutions are coordinated around shared access principles and open, interoperable infrastructure. The Airbus analogy is imperfect but instructive: a consortium of European states, none of which could credibly challenge American aerospace dominance alone, pooled resources under shared governance to create a durable alternative (Tan *et al.,* 2025).

The architecture of AI remains malleable. The choice is not between markets and public governance. It is between infrastructure that concentrates and infrastructure that creates the conditions for plural competition. Europe has both the institutional tradition and the strategic incentive to make that choice deliberately. The opportunity is finite.

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